

- 21.** In a Young's double-slit experiment with light of wavelength 520 nm, the fringe width is 0.52 mm. What will be the fringe width if the entire apparatus is immersed in water (refractive index $\frac{4}{3}$) ?

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$$\beta = \frac{D\lambda}{d}$$

$$\frac{\beta'}{\beta} = \frac{\lambda'}{\lambda}$$

$$= \frac{v/v}{c/v} = \frac{1}{n}$$

$$\beta' = \frac{\beta}{n}$$

$$\beta' = \frac{0.52}{\frac{4}{3}}$$

$$= 0.39 \text{ mm}$$

Alternative:

$$\beta' = \frac{\beta}{n}$$

$$= \frac{0.52}{\frac{4}{3}}$$

$$= 0.39 \text{ mm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

- (b) State Huygens principle. Using it, draw a diagram and discuss the case of refraction of plane wave of light from a rarer medium to a denser medium at their plane interface. Hence derive Snell's law. 5

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Huygen's Principle:

Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets emanating from the wavefront are usually referred to as secondary wavelets and if we draw a common tangent to all these spheres, we obtain the new position of the wavefront at a later time.

Discussion : In time τ distance BC travelled by the incident plane wavefront with the velocity v_1 is $v_1 \tau$. To determine the shape of the refracted wave wavefront , draw a sphere of radius $v_2 \tau (=AE)$ from point A and draw a tangent from C to E as shown in the diagram.

From the above diagram

ΔABC and ΔAEC

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$$

$$\text{and } \sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$$

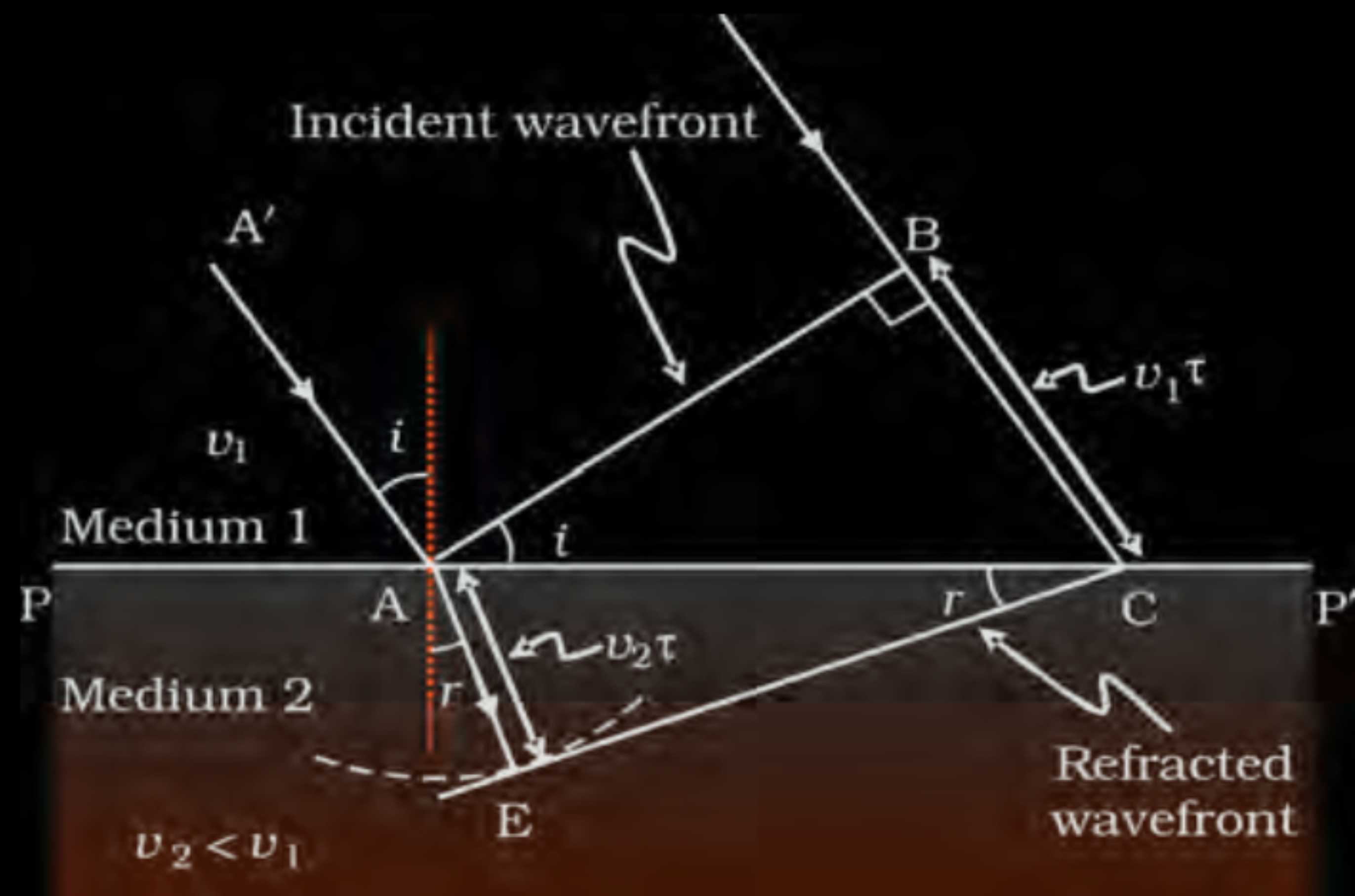
$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} \dots\dots\dots(1)$$

$$\text{As } n_1 = \frac{c}{v_1} \text{ and } n_2 = \frac{c}{v_2}$$

$$\therefore \frac{v_1}{v_2} = \frac{n_2}{n_1} \dots\dots\dots(2)$$

From equations (1) and (2)

$$n_1 \sin i = n_2 \sin r$$



1

1/2

1/2

1/2

1/2

1/2

- 21.** In a Young's double-slit experiment with monochromatic light of wavelength 600 nm, and slit separation of 2 mm, fringes are observed on a screen at some distance. Calculate the change in fringe width if the screen is moved towards the slits by 10 cm.

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$\beta = \frac{\lambda D}{d}$	$\frac{1}{2}$
$\beta' = \frac{\lambda D'}{d}$	
$\Delta\beta = \beta - \beta' = \frac{\lambda}{d}[D - D']$	$\frac{1}{2}$
$\Delta\beta = \frac{\lambda}{d}[D - (D - 0.1)]$	$\frac{1}{2}$
$\Delta\beta = \frac{600 \times 10^{-9}}{2 \times 10^{-3}} \times 0.1$	
$= 3 \times 10^{-5} \text{ m}$	$\frac{1}{2}$

- 21.** Light of wavelength 640 nm incident on the slits in a Young's double-slit experiment produces an interference pattern of fringe width 7.2 mm. Calculate the wavelength of light which will produce a pattern of fringe width 8.1 mm.

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$$\beta = \frac{D\lambda}{d}$$

$$\frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2}$$

$$\lambda_2 = \frac{\beta_1}{\beta_2} \times \lambda_1$$

$$\lambda_2 = \frac{8.1}{7.2} \times 640$$

$$\lambda_2 = 720 \text{ nm}$$

1/2

1/2

1/2

1/2

15. *Assertion (A) :* In a Young's double-slit experiment, interference pattern is obtained by using yellow light. If yellow light is replaced by red light, the fringe width increases.

Reason (R) : Fringe width is inversely proportional to the distance of the screen from the slits.

15. *Assertion (A) :* In a Young's double-slit experiment, interference pattern is obtained by using yellow light. If yellow light is replaced by red light, the fringe width increases.

Reason (R) : Fringe width is inversely proportional to the distance of the screen from the slits.

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(C) Assertion (A) is true, but Reason (R) is false.

- 25.** (a) Explain the diffraction of light due to a single slit.
- (b) Light of wavelength 540 nm is incident on a 0.09 cm wide single slit and forms a diffraction pattern on the screen 1.5 cm away from the slit. Find the width of the central maximum.

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- 25.** (a) Explain the diffraction of light due to a single slit.
 (b) Light of wavelength 540 nm is incident on a 0.09 cm wide single slit and forms a diffraction pattern on the screen 1.5 cm away from the slit. Find the width of the central maximum.

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(a) When a single narrow slit is illuminated by a monochromatic source. Waves coming from different part of slit superimpose then a broad pattern with a central bright region is seen. On both sides with alternate dark and bright regions, the intensity becoming weakening away from the center. This is called diffraction of light.	1
(b) $\beta_o = \frac{2D\lambda}{a}$	$\frac{1}{2}$
$= \frac{2 \times 1.5 \times 10^{-2} \times 540 \times 10^{-9}}{0.09 \times 10^{-2}}$	1
$= 1.8 \times 10^{-7} \text{ m}$	$\frac{1}{2}$

30. A monochromatic laser beam is incident perpendicularly to the wall of a water-filled glass container. The laser beam passes through two slits in a block kept in the container. An interference pattern is formed on a screen inside the container. The water is then replaced with a transparent fluid of greater refractive index than that of water. Again interference pattern is formed on the screen inside the liquid. CBSE 25 C

(i) How will the speed and the wavelength of light change in the liquid ?

- | | |
|--------------------------|--------------------------|
| (A) decreases, decreases | (B) decreases, increases |
| (C) increases, increases | (D) increases, decreases |

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|--------------------------|--------------------------|
| (A) decreases, decreases | (B) decreases, increases |
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(A) decreases, decreases

- (ii) The angular separation of the bright fringes in liquid will : 1
- (A) increase
 - (B) decrease
 - (C) remain the same
 - (D) may increase or decrease depending upon the refractive index of liquid

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(B) decrease

(iii) (a) In the above experiment (with water), if the separation between the slits is halved and distance between the slits and the screen is doubled, the fringe width is :

(A) doubled

(B) quadrupled

(C) halved

(D) unchanged

(iii) (a) In the above experiment (with water), if the separation between the slits is halved and distance between the slits and the screen is doubled, the fringe width is :

1

(A) doubled

(B) quadrupled

(C) halved

(D) unchanged

(B) quadrupled

(iii) (b) The two slits are 1 mm apart and the screen is placed 1 m away from the slits. If light of wavelength 500 nm is used, then fringe separation in water ($\mu = \frac{4}{3}$) is : 1

- (A) 0.375 mm (B) 0.5 mm
(C) 0.75 mm (D) 1.0 mm

(iii) (b) The two slits are 1 mm apart and the screen is placed 1 m away from the slits. If light of wavelength 500 nm is used, then fringe separation in water ($\mu = \frac{4}{3}$) is : 1

- (A) 0.375 mm (B) 0.5 mm
(C) 0.75 mm (D) 1.0 mm

(A) 0.375 mm

- (iv) The intensity at the dark fringes in the interference pattern can be zero when the two interfering waves have : 1
- (A) same frequency, same intensity and stable phase difference π .
 - (B) same frequency, same intensity and stable phase difference zero.
 - (C) different frequency, different intensity and stable phase difference π .
 - (D) different frequency, different intensity and stable phase difference zero.

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 - (D) different frequency, different intensity and stable phase difference zero.

(A) same frequency, same intensity and stable phase difference π .

- (b) A beam of light consisting of two wavelengths 400 nm and 500 nm is used to obtain interference fringes in a double slit experiment. The distance between the slits is 1.5 mm and the distance of the screen from the plane of the slits is 1.5 m.
- (i) (I) Find the distance of the fifth bright fringe from the centre of the screen for the wavelength 500 nm.
- (II) Find the distance of 3rd dark fringe from the centre of the screen for the wavelength 400 nm.
- (ii) What is the least distance from the centre of the screen where the bright fringes due to both wavelengths coincide ?

(i) $\lambda_1 = 400 \text{ nm}$, $\lambda_2 = 500 \text{ nm}$	
Distance of 5 th bright fringe	
$y_n^{\text{max}} = \frac{n\lambda D}{d}$	$\frac{1}{2}$
$y_5^{\text{max}} = \frac{5\lambda_2 D}{d}$	
$y_5^{\text{max}} = \frac{5 \times 500 \times 10^{-9} \times 1.5}{1.5 \times 10^{-3}}$	$\frac{1}{2}$
$y_5^{\text{max}} = 2.5 \times 10^{-3} \text{ m} = 2.5 \text{ mm}$	$\frac{1}{2}$
$y_n^{\text{min}} = \frac{(2n-1)\lambda D}{2d}$	$\frac{1}{2}$
Distance of 3 rd dark fringe	
$y_3^{\text{min}} = \frac{5 \times 400 \times 10^{-9} \times 1.5}{2 \times 1.5 \times 10^{-3}}$	$\frac{1}{2}$
$y_3^{\text{min}} = 1 \text{ mm}$	$\frac{1}{2}$

(ii) $n\lambda_2 = (n+1)\lambda_1$	$\frac{1}{2}$
$\frac{n+1}{n} = \frac{\lambda_2}{\lambda_1} = \frac{500}{400}$	
$n = 4$	$\frac{1}{2}$
Minimum distance	$\frac{1}{2}$
$y_4 = \frac{n\lambda_2 D}{d}$	
$= \frac{4 \times 500 \times 10^{-9} \times 1.5}{1.5 \times 10^{-3}}$	$\frac{1}{2}$
$= 2 \text{ mm}$	

18. In a Young's double-slit experiment, the intensity at the central maximum in the interference pattern on the screen is I_0 . Find the intensity at a point on the screen where the path difference between the interfering waves is $\frac{\lambda}{6}$.

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$$I_0 = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$$

$$= I + I + 2I\cos 0^\circ$$

$$= 2I + 2I = 4I$$

$$I = I_0 / 4$$

Intensity at point P

$$I' = I + I + 2I\cos\pi/3$$

$$= 3I$$

$$= \frac{3I_0}{4}$$

Alternatively:

$$I = I_0\cos^2\phi/2$$

$$I = I_0\cos^2\pi/6$$

$$= \frac{3I_0}{4}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (b) (i) Two coherent light waves, each of intensity I_0 superpose each other and produce interference pattern on a screen. Obtain the expression for the resultant intensity at a point where the phase difference between the waves is ϕ . Write its maximum and minimum possible values.
- (ii) In a single slit diffraction experiment, the aperture of the slit is 3 mm and the separation between the slit and the screen is 1.5 m. A monochromatic light of wavelength 600 nm is normally incident on the slit. Calculate the distance of (I) first order minimum, and (II) second order maximum, from the centre of the screen.

- 19.** The ratio of the intensities at maxima to minima in Young's double-slit experiment is 25 : 9. Calculate the ratio of intensities of the interfering waves.

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$$\text{Given, } \frac{I_{\max}}{I_{\min}} = \frac{25}{9}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{25}{9}$$

$$3(a_1 + a_2) = 5(a_1 - a_2)$$

$$\frac{a_1}{a_2} = \frac{4}{1}$$

$$\frac{I_1}{I_2} = \frac{16}{1}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

19. Write two differences in the patterns of double-slit interference experiment and single-slit diffraction experiment.

Light waves from two pinholes illuminated by two sodium lamps do not produce interference patterns. Explain why.

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Interference pattern	Diffraction pattern	
1. The interference pattern has a number of equally spaced bright and dark bands. 2. The interference pattern is obtained by superposing two waves originating from the two narrow slits. 3. The maximas are of same intensity.	1. The diffraction pattern has a central bright maximum which is twice as wide as the other maxima. 2. The diffraction pattern is obtained by superposition of a continuous family of waves originating from each point on a single slit. 3. The intensity falls as we go to successive maxima away from the centre, on either side.	1
(Any two differences) The light wave coming out of two independent source of light will not have constant phase with time. Alternatively: Two sodium lamps are incoherent source of light.		1

2. Two coherent waves, each of intensity I_0 , produce interference pattern on a screen. The average intensity of light on the screen is :

(A) zero

(B) I_0

(C) $2I_0$

(D) $4I_0$

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(C) $2I_0$

- 19.** (a) Find the intensity at a point on the screen in Young's double slit experiment, at which the interfering waves of intensity I_0 each, have a path difference of (i) $\frac{\lambda}{3}$, and (ii) $\frac{\lambda}{2}$.

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19. (a) Find the intensity at a point on the screen in Young's double slit experiment, at which the interfering waves of intensity I_0 each, have a path difference of (i) $\frac{\lambda}{3}$, and (ii) $\frac{\lambda}{2}$.

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(i)

$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x$$

$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = 4I_0 \cos^2 \frac{\phi}{2}$$

$$I = 4I_0 \cos^2 \frac{\pi}{3}$$

$$I = I_0$$

(ii) $\Delta\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} = \pi$

$$I = 4I_0 \cos^2 \frac{\pi}{2}$$

$$I = 0$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

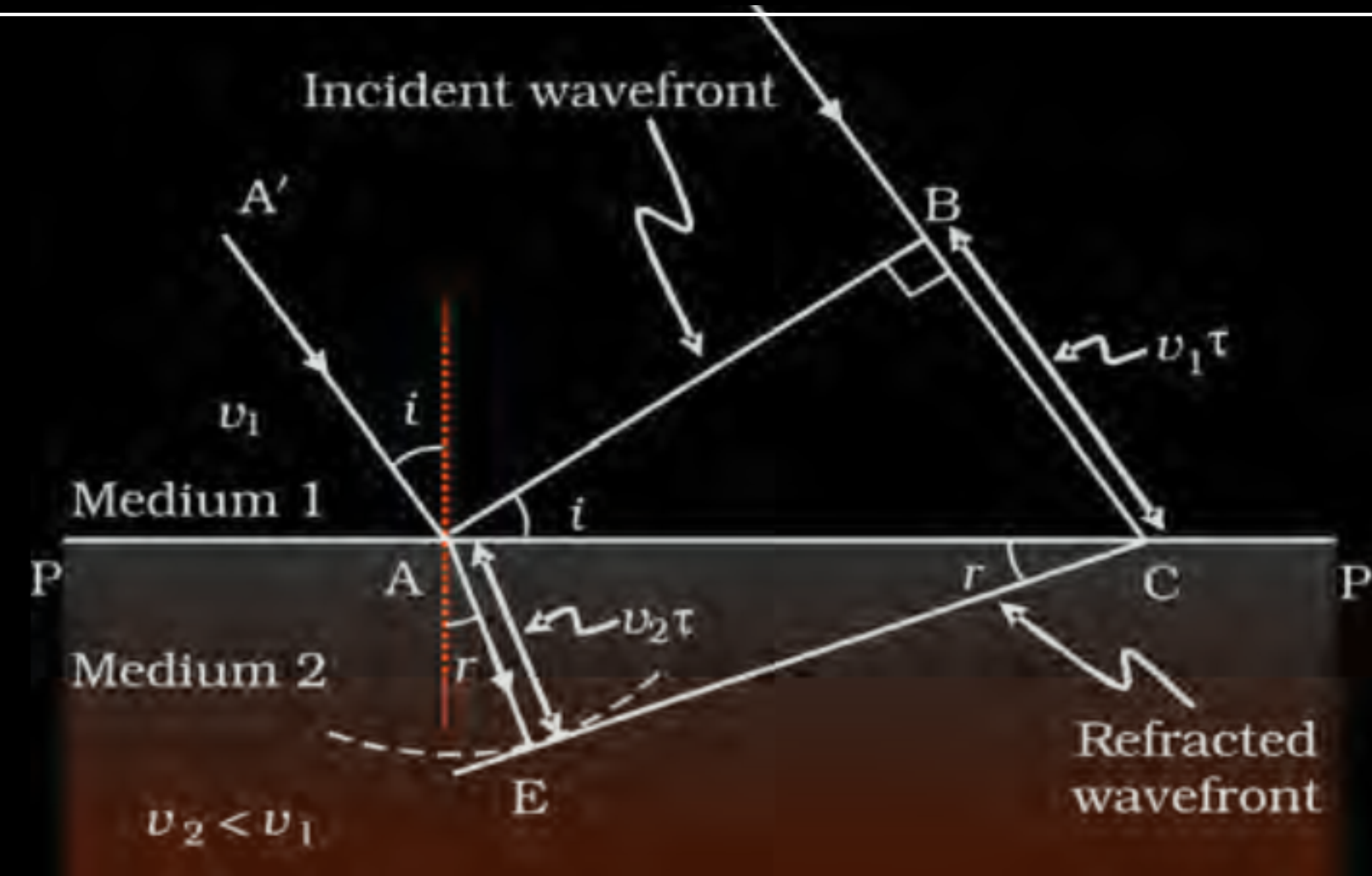
$\frac{1}{2}$

- (b) (i) Using Huygens' principle, explain the refraction of a plane wavefront, propagating in air, at a plane interface between air and glass. Hence verify Snell's law.

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$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$	1/2
$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$	1/2
$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$	1
$\frac{\sin i}{\sin r} = \frac{c/n_1}{c/n_2}$	1/2
$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$ or $n_1 \sin i = n_2 \sin r$	1/2



15. **Assertion (A)** : When two coherent sources in Young's double-slit experiment (YDSE) are infinitely close to each other, interference pattern is observed on a screen.

Reason (R) : The fringe width in YDSE does not depend on separation between the two slits.

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15. **Assertion (A)** : When two coherent sources in Young's double-slit experiment (YDSE) are infinitely close to each other, interference pattern is observed on a screen.

Reason (R) : The fringe width in YDSE does not depend on separation between the two slits. CBSE 25

(D) Both Assertion (A) and reason (R) are false

19. Light of wavelength 700 nm is incident normally on a slit of width 0.2 mm. Find the angular width of the central maximum on a screen placed at a distance of 1 m from the slit. CBSE 25

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19. Light of wavelength 700 nm is incident normally on a slit of width 0.2 mm. Find the angular width of the central maximum on a screen placed at a distance of 1 m from the slit. CBSE 25 **2**

Angular width of central maxima= $\frac{2\lambda}{a}$	1
$= \frac{2 \times 700 \times 10^{-9}}{2 \times 10^{-4}}$ radian	$\frac{1}{2}$
$= 0.007$ radian	$\frac{1}{2}$

33. (a) (A) In Young's double slit experiment, the two slits are 0.15 mm apart. The slits are illuminated by a monochromatic light and an interference pattern is obtained a screen kept 3.0 m away from the slits. The distance between first and eighth minima is 8.0 cm. Find the wavelength of the light used.
- (B) Two coherent waves of amplitude A_1 and A_2 are used in an interference experiment. Derive expression for the ratio of minimum and maximum intensities of light obtained in the interference pattern.

(A) The distance of n^{th} minima

$$Y_n = \frac{(2n+1)\lambda D}{2d}, \quad n = 0, 1, 2, \dots$$

$$Y_8 - Y_1 = \frac{15\lambda D}{2d} - \frac{\lambda D}{2d} = \frac{7\lambda D}{d}$$

$$\lambda = \frac{(Y_8 - Y_1) \times d}{7D}$$

$$\lambda = \frac{8 \times 10^{-2} \times 1.5 \times 2 \times 10^{-4}}{7 \times 3} \text{ m}$$

$$\lambda = 5.71 \times 10^{-7} \text{ m} = 571 \text{ nm}$$

(B) Intensity of resultant wave

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$I \propto A^2$ (where A is amplitude of resultant wave)

$$A = \sqrt{A_1^2 + A_1^2 + 2A_1A_2 \cos \varphi}$$

$$A_{\text{max}} = (A_1 + A_2)^2$$

$$A_{\text{min}} = (A_1 - A_2)^2$$

$$\frac{I_{\text{min}}}{I_{\text{max}}} = \frac{(A_1 - A_2)^2}{(A_1 + A_2)^2}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

12. Two coherent light waves, each having amplitude 'a', superpose to produce an interference pattern on a screen. The intensity of light as seen on the screen varies between :

(A) 0 and $2a^2$

(B) 0 and $4a^2$

(C) a^2 and $2a^2$

(D) $2a^2$ and $4a^2$

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12. Two coherent light waves, each having amplitude 'a', superpose to produce an interference pattern on a screen. The intensity of light as seen on the screen varies between :

(A) 0 and $2a^2$

(B) 0 and $4a^2$

(C) a^2 and $2a^2$

(D) $2a^2$ and $4a^2$

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(B) 0 and $4a^2$

- (b) In Young's double slit experimental set-up, the intensity of the central maximum is I_0 . Calculate the intensity at a point where the path difference between two interfering waves is $\lambda/3$. CBSE 25 2

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$$\phi = \frac{2\pi}{\lambda} \times \Delta x$$

$$= \frac{2\pi}{\lambda} \times \frac{\lambda}{3}$$

$$= \frac{2\pi}{3}$$

$$I' = 4I \cos^2 \frac{\phi}{2} \quad \text{Given } 4I = I_0$$

$$= I_0 \cos^2 \frac{2\pi}{6}$$

$$= \frac{I_0}{4}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Note: If a student attempt by using $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$, award full credit for correct answer.

24. In Young's double slit experiment, the separation between the two slits is 1.0 mm and the screen is 1.0 m away from the slits. A beam of light consisting of two wavelengths 500 nm and 600 nm is used to obtain interference fringes. Calculate :

3

- (a) the distance between the first maxima for the two wavelengths.
- (b) the least distance from the central maximum, where the bright fringes due to both the wavelengths coincide.

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24. In Young's double slit experiment, the separation between the two slits is 1.0 mm and the screen is 1.0 m away from the slits. A beam of light consisting of two wavelengths 500 nm and 600 nm is used to obtain interference fringes. Calculate :

3

- (a) the distance between the first maxima for the two wavelengths.
- (b) the least distance from the central maximum, where the bright fringes due to both the wavelengths coincide.

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a) Distance = $\frac{n\lambda_1 D}{d} - \frac{n\lambda_2 D}{d}$

For n=1

$$\text{Distance} = \frac{(600 - 500) \times 10^{-9} \times 1}{10^{-3}} = 10^{-4} \text{ m}$$

b) $n\lambda_1 \frac{D}{d} = (n+1)\lambda_2 \frac{D}{d}$

$$n \times 600 \times 10^{-9} = (n+1) \times 500 \times 10^{-9}$$

$$n = 5$$

$$x = 5 \times \frac{\lambda_1 D}{d}$$

$$= \frac{5 \times 600 \times 10^{-9} \times 1}{10^{-3}}$$

$$= 3 \text{ mm}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

Alternatively

$$n_1 \lambda_1 = n_2 \lambda_2$$

$$\frac{n_1}{n_2} = \frac{5}{6}$$

therefore n = 5

Position of 5th bright for λ_1 (600 nm) $x = 5 \times \frac{\lambda_1 D}{d} = 3 \text{ mm}$

13. *Assertion (A) :* In double slit experiment if one slit is closed, diffraction pattern due to the other slit will appear on the screen.

Reason (R) : For interference, at least two waves are required.

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(C) Assertion is true, but Reason (R) is false.

18. In Young's double slit experiment, the screen is moved 30 cm towards the slits. As a consequence, the fringe width of the pattern changes by 0.09 mm. If the slits separation used is 2 mm, calculate the wavelength of light used in the experiment.

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$\beta_1 = \frac{D\lambda}{d}$	$\frac{1}{2}$
$\beta_2 = \frac{D_2\lambda}{d}$	
$\beta_1 - \beta_2 = \frac{\lambda}{d}(D_1 - D_2)$	$\frac{1}{2}$
$= \frac{\lambda}{d}[D - (D - 30 \times 10^{-2})]$	
$\lambda = \frac{(\beta_1 - \beta_2)d}{30 \times 10^{-2}} = \frac{9 \times 10^{-5} \times 2 \times 10^{-3}}{30 \times 10^{-2}}$	$\frac{1}{2}$
$\lambda = 6 \times 10^{-7} \text{ m}$	$\frac{1}{2}$

- 26.** (a) Write the conditions under which two light waves originating from two coherent sources can interfere each other (i) constructively, and (ii) destructively, in terms of wavelength. Can these be applied for two lights originating from two sodium lamps ? Give reason.
- (b) Monochromatic light of green colour is used in Young's double slit experiment and an interference pattern is observed on a screen. If the green light is replaced by red monochromatic light of the same intensity, how will the fringe width of interference pattern be affected ? Justify your answer.

- 26.** (a) Write the conditions under which two light waves originating from two coherent sources can interfere each other (i) constructively, and (ii) destructively, in terms of wavelength. Can these be applied for two lights originating from two sodium lamps ? Give reason.
- (b) Monochromatic light of green colour is used in Young's double slit experiment and an interference pattern is observed on a screen. If the green light is replaced by red monochromatic light of the same intensity, how will the fringe width of interference pattern be affected ? Justify your answer.

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(a)	
(i) Conditions for constructive interference	
$\Delta = n\lambda$	$\frac{1}{2}$
(ii) Conditions for destructive interference	
$\Delta = (n + \frac{1}{2})\lambda$	$\frac{1}{2}$
$n = 0, 1, 2, \dots$	$\frac{1}{2}$
• No	
Two independent monochromatic source of light can not be coherent	$\frac{1}{2}$
(b) Fringe width $\beta = \frac{\lambda D}{d}$	$\frac{1}{2}$
$\lambda_{\text{red}} > \lambda_{\text{green}}$	
$\beta_{\text{red}} > \beta_{\text{green}}$	$\frac{1}{2}$

- (b) (i) A physics teacher wants to demonstrate interference with the help of double slit experiment using a laser beam of 633 nm wavelength. Since the hall is large enough, interference pattern is formed on the wall 5.0 m from the slits. For clear and comfortable view by all the students they want the fringe width 5 mm.
- (I) Find the slit separation for obtaining the desired interference pattern.
- (II) How far will the first minimum be from the central maximum ?
- (ii) A parallel beam of light of wavelength 650 nm passes through a slit of width 0.6 mm. The diffraction pattern is obtained on a screen kept 60 cm away from the slit. Find the distance between first order minima on both sides of the central maximum.

(i)
 (I) Slit separation

$$\beta = \frac{D\lambda}{d}$$

$$d = \frac{D\lambda}{\beta}$$

$$= \frac{633 \times 10^{-9} \times 5}{5 \times 10^{-3}}$$

$$= 633 \times 10^{-6} m$$

$$= 633 \mu m$$
 (II) Distance of first minimum from central maximum

$$x_n = \frac{(2n-1)\lambda D}{2d}$$

$$n = 1$$

$$x = \frac{633 \times 10^{-9} \times 5}{2 \times 5 \times 10^{-3}}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$x = 316.5 \times 10^{-6} m$$

$$x = 316.5 \mu m$$

(ii) Distance between first order minima on both the side

$$W = \frac{2D\lambda}{d}$$

$$= \frac{2 \times 650 \times 10^{-9}}{0.6 \times 10^{-3}} \times 60 \times 10^{-2}$$

$$= 1.3 \times 10^{-3} m$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

18. In a double slit experiment, the two slits are 1.5 mm apart. The slits are illuminated by a mixture of lights of wavelengths of 600 nm and 400 nm and the interference pattern is observed on a screen 1.5 m away from the slits. Find the minimum distance of the point from the central maximum at which bright fringes of the interference patterns of the two wavelengths coincide.

18. In a double slit experiment, the two slits are 1.5 mm apart. The slits are illuminated by a mixture of lights of wavelengths of 600 nm and 400 nm and the interference pattern is observed on a screen 1.5 m away from the slits. Find the minimum distance of the point from the central maximum at which bright fringes of the interference patterns of the two wavelengths coincide.

CBSE 25

2

$$n\lambda_1 = (n+1)\lambda_2$$

$$n \times 600 = (n+1) \times 400$$

$$\therefore n = 2$$

$$\therefore x = \frac{2\lambda_1 D}{d}$$

$$= \frac{2 \times 600 \times 10^{-9} \times 1.5}{1.5 \times 10^{-3}}$$

$$x = 1.2 \times 10^{-3} \text{ m}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

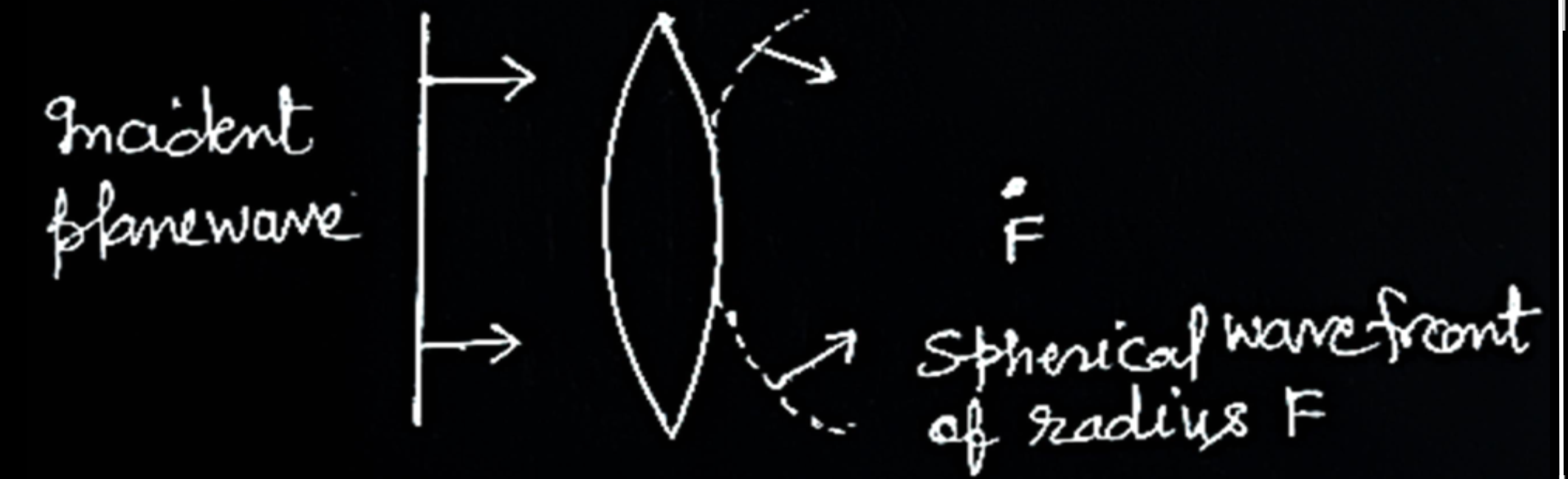
- 26.** (a) Define 'wavefront' of a light wave. A plane wavefront is refracted from a convex lens. Draw the shape of the refracted wavefront.
- (b) A plane wave travelling in a medium is incident on a plane surface separating this medium from a rarer medium. Draw a diagram to show refraction of the wave. Hence, verify Snell's law.

CBSE 25

26. (a) Define 'wavefront' of a light wave. A plane wavefront is refracted from a convex lens. Draw the shape of the refracted wavefront.
- (b) A plane wave travelling in a medium is incident on a plane surface separating this medium from a rarer medium. Draw a diagram to show refraction of the wave. Hence, verify Snell's law. CBSE 25

3

(a) Wave front- It is a continuous locus at every point on it, all the particles of medium are vibrating in same phase.



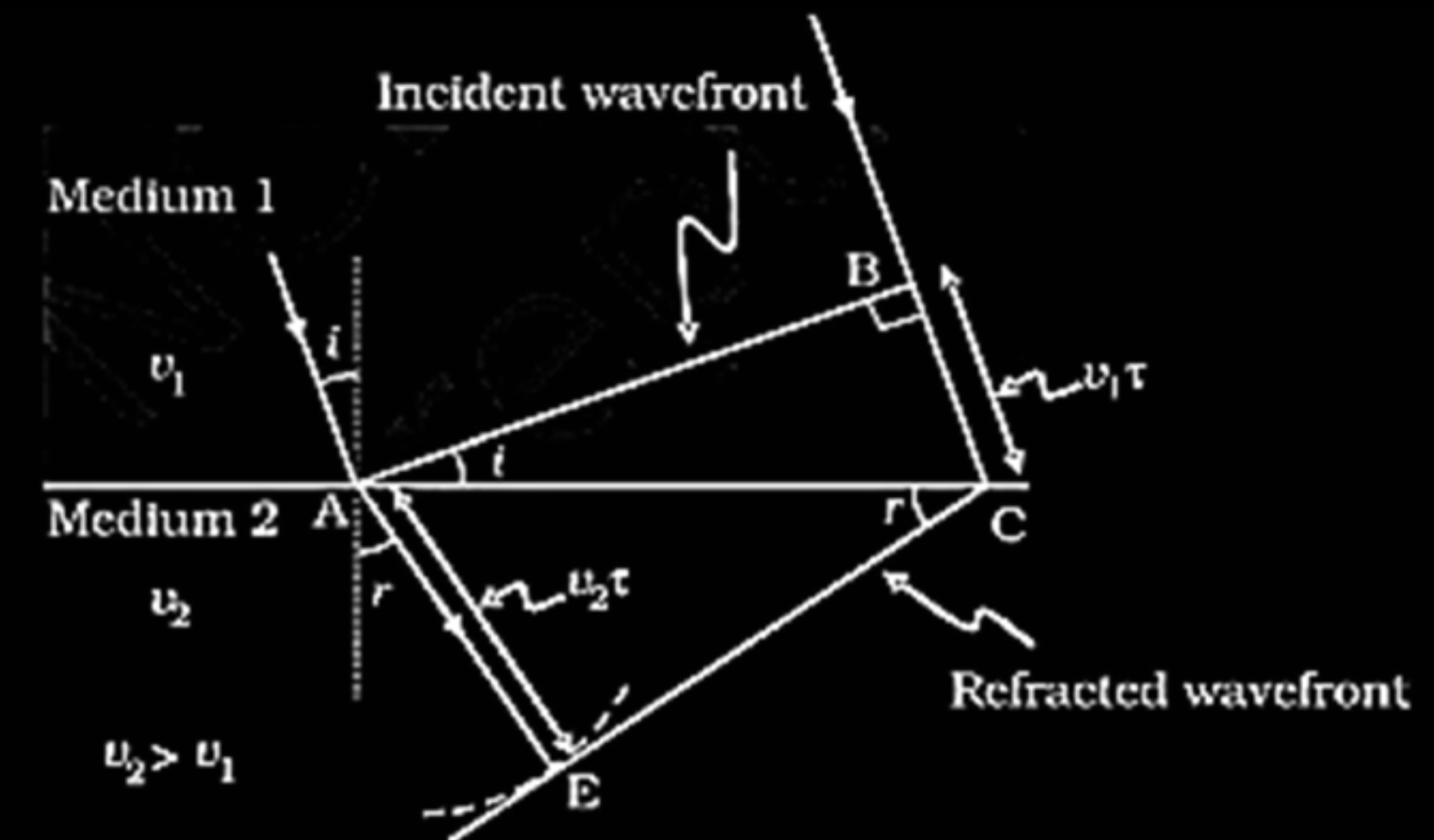
Considering triangle ABC and AEC

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC} \text{ -----(1)}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC} \text{ -----(2)}$$

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

$$n_1 \sin i = n_2 \sin r$$



18. Find the angle of diffraction (in degrees) for first secondary maximum of the pattern due to diffraction at a single slit. The width of the slit and wavelength of light used are 0.55 mm and 550 nm, respectively. 2

CBSE 25

18. Find the angle of diffraction (in degrees) for first secondary maximum of the pattern due to diffraction at a single slit. The width of the slit and wavelength of light used are 0.55 mm and 550 nm, respectively. 2

CBSE 25

$$\theta = (2n+1) \frac{\lambda}{2a}$$

For first secondary maxima $n=1$

$$\theta = \frac{3\lambda}{2a}$$

$$\theta = \frac{3 \times 550 \times 10^{-9}}{2 \times 0.55 \times 10^{-3}}$$

$$\theta = 1.5 \times 10^{-3} \text{ radian} = 0.086 \text{ degree}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

26. A double slit set-up was initially placed in a tank filled with water and the interference pattern was obtained using a laser light. When water is replaced by a transparent liquid of refractive index $n > n_{\text{water}}$, what will be the effect on the following ?

3

- (a) Speed, frequency and wavelength of the light of laser beam.
- (b) The fringe width, shape of interference fringes and shift in the position of central maximum.

CBSE 25

26. A double slit set-up was initially placed in a tank filled with water and the interference pattern was obtained using a laser light. When water is replaced by a transparent liquid of refractive index $n > n_{\text{water}}$, what will be the effect on the following ?

3

- (a) Speed, frequency and wavelength of the light of laser beam.
- (b) The fringe width, shape of interference fringes and shift in the position of central maximum.

CBSE 25

(a)	• Speed of light will decrease • Frequency remains unaffected • Wavelength decreases	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
(b)	• Fringe width decreases • Shapes does not change • Position of central maxima does not change.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

14. **Assertion (A)** : Out of Infrared and radio waves, the radio waves show more diffraction effect.

1

Reason (R) : Radio waves have greater frequency than infrared waves.

CBSE 25

14. **Assertion (A)** : Out of Infrared and radio waves, the radio waves show more diffraction effect. **1**
Reason (R) : Radio waves have greater frequency than infrared waves.

CBSE 25

(C) If Assertion (A) is true but Reason (R) is false.

18. In a double slit experiment, it is observed that the angular width of one fringe formed on the screen is 0.2° . The wavelength of light used in the experiment is 500 nm. Calculate the separation of the two slits.

2

CBSE 25

18. In a double slit experiment, it is observed that the angular width of one fringe formed on the screen is 0.2° . The wavelength of light used in the experiment is 500 nm. Calculate the separation of the two slits.

2

CBSE 25

$$\begin{aligned}\text{Angular width } (\theta) &= \frac{\lambda}{d} \\ d &= \frac{\lambda}{\theta} \\ &= \frac{500 \times 10^{-9}}{0.2 \times \frac{\pi}{180}} \\ d &= \frac{45}{\pi} \times 10^{-5} \text{ m} \\ &= 0.14 \text{ mm}\end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

25. (a) The amplitude of a light wave becomes n times. This results in intensity of the wave becoming m times. What is the relation between n and m ?
- (b) White light is incident on three identical surfaces – a black surface, a yellow surface and a white surface, one by one. For which surface, the pressure exerted on the surface by the incident light will be (i) maximum (ii) minimum ? Justify your answer.

CBSE 25

25. (a) The amplitude of a light wave becomes n times. This results in intensity of the wave becoming m times. What is the relation between n and m ?
- (b) White light is incident on three identical surfaces – a black surface, a yellow surface and a white surface, one by one. For which surface, the pressure exerted on the surface by the incident light will be (i) maximum (ii) minimum ? Justify your answer.

CBSE 25

(a) Intensity (I) $\propto$ (amplitude) $\Rightarrow m = n^2$	1
(b) (i) White surface	$\frac{1}{2}$
As white surface will reflect maximum light falling on it, the change in momentum of light will be maximum and as a consequence the pressure exerted by the light will be maximum.	$\frac{1}{2}$
(ii) Black surface	$\frac{1}{2}$
As black surface absorbs maximum light falling on it, the change in momentum of light will be minimum and as a consequence the pressure exerted by the light will be minimum.	$\frac{1}{2}$

- (b) (i) Light consisting of two wavelengths 600 nm and 480 nm is used to obtain interference fringes in a double slit experiment. The screen is placed 1.0 m away from slits which are 1.0 mm apart.
- (1) Calculate the distance of the third bright fringe on the screen from the central maximum for wavelength 600 nm.
 - (2) Find the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.
- (ii) (1) Draw the variation of intensity with angle of diffraction in single slit diffraction pattern. Write the expression for value of angle corresponding to zero intensity locations.
- (2) In what way diffraction of light waves differs from diffraction of sound waves ?

(i)

(1) Distance of the nth bright fringe from the central maximum(x_n) = $\frac{n\lambda D}{d}$

For n = 3

$$x_3 = \frac{3 \times 600 \times 10^{-9} \times 1}{1 \times 10^{-9}}$$

=1800m

(2)

$$n_1 \lambda_1 = n_2 \lambda_2$$

$$n_1 \times 600 = n_2 \times 480$$

$$\frac{n_1}{n_2} = \frac{480}{600}$$

$$\frac{n_1}{n_2} = \frac{4}{5}$$

Position of the 4th bright fringe of 600 nm = 4 x 600 = 2400 m

Alternatively: -

Position of the 5th bright fringe of 480 nm = 5 x 480 = 2400 m

Alternatively: -

$$(n-1)\lambda_1 = n\lambda_2$$

$$(n-1) \times 600 = n \times 450$$

on solving n = 4

Position of the 4th bright fringe of 600 nm = 4 x 600 = 2400 m

Position of the 5th bright fringe of 480 nm = 5 x 480 = 2400 m

1/2

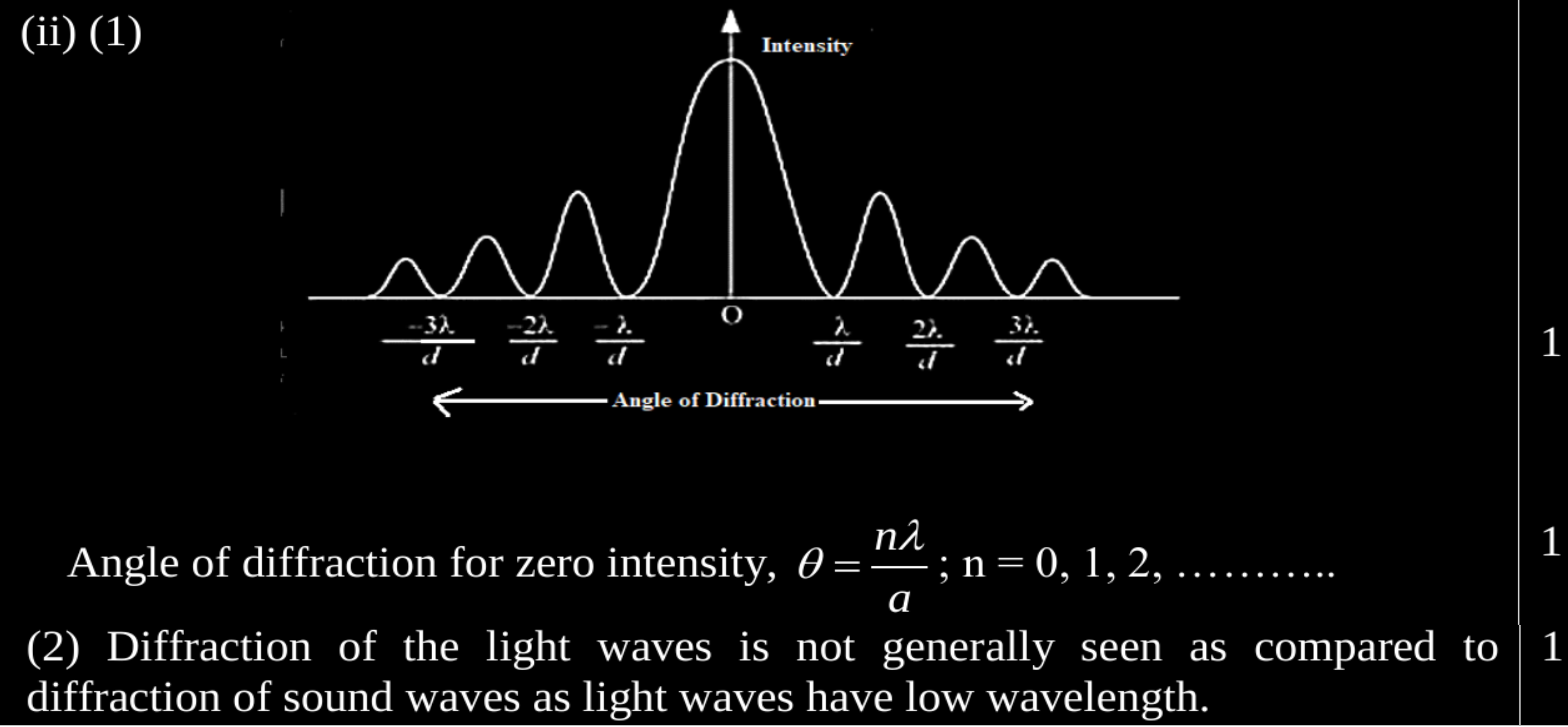
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1

1

1

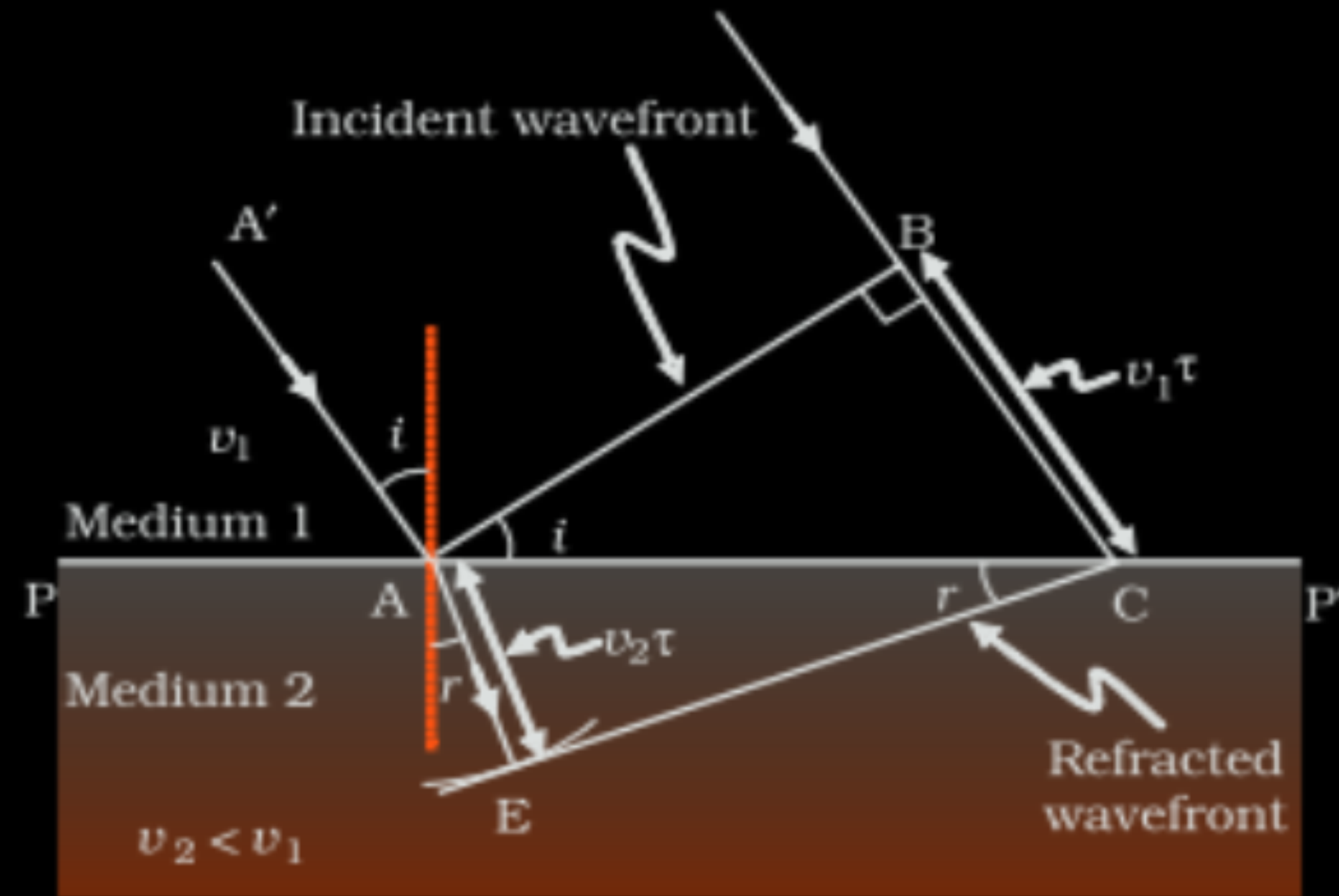
18. Show the refraction of light wave at a plane interface using Huygens' principle and prove Snell's law.

CBSE 25

18. Show the refraction of light wave at a plane interface using Huygens' principle and prove Snell's law.

CBSE 25

2



1

Consider the triangles ABC and AEC,

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC} \dots \dots \dots (i)$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC} \dots \dots \dots (ii)$$

Divide eq. (i) & (ii): -

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} \dots \dots \dots (iii)$$

Refractive index of medium 1,

$$n_1 = \frac{c}{v_1} \dots \dots \dots (iv)$$

Refractive index of medium 2,

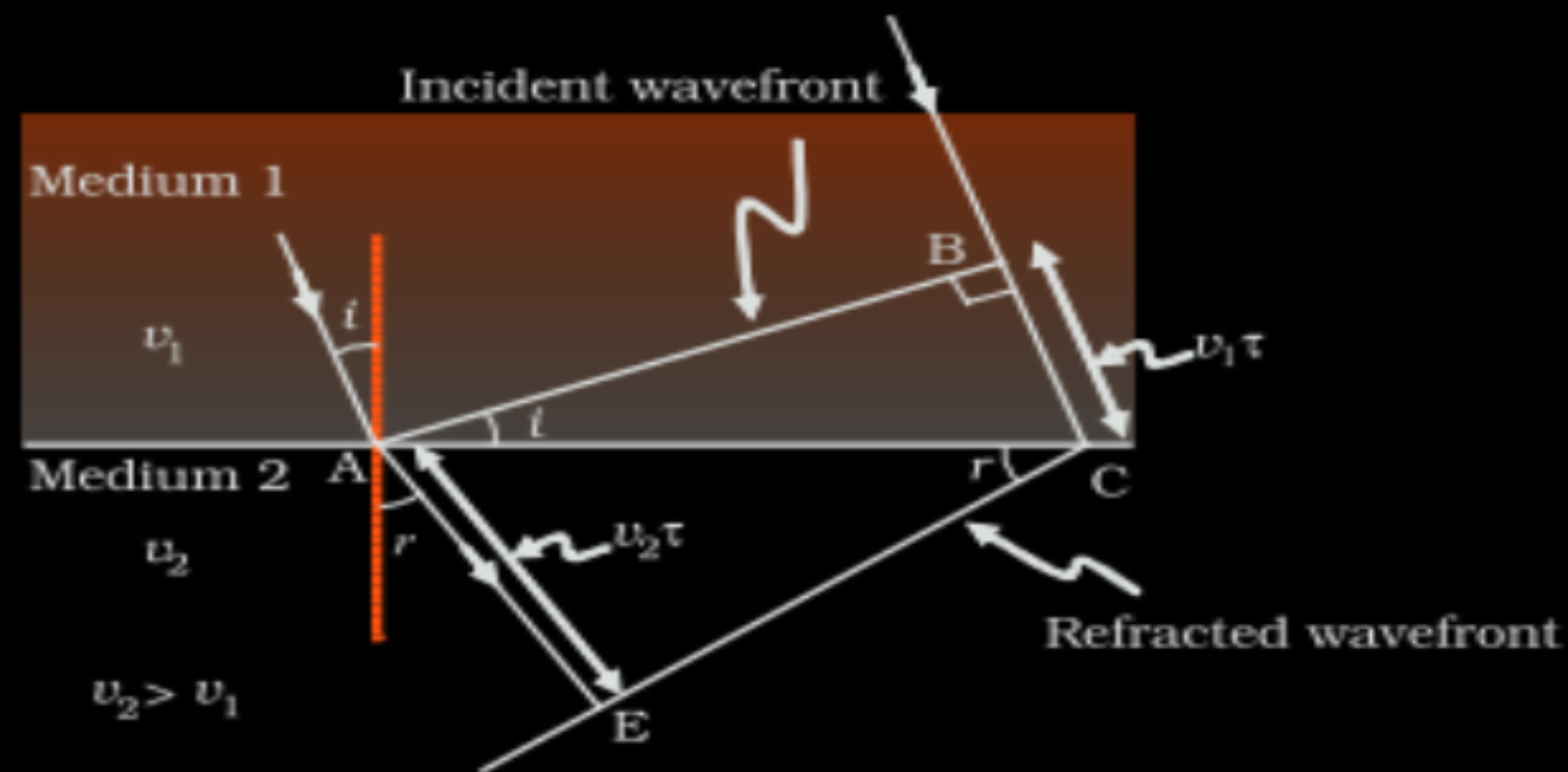
$$n_2 = \frac{c}{v_2} \dots \dots \dots (v)$$

From eq. (iii); (iv) & (v): -

$$\frac{n_2}{n_1} = \frac{\sin i}{\sin r}$$

1

Alternatively: -



18. In a double-slit experiment, 6th dark fringe is observed at a certain point of the screen. A transparent sheet of thickness t and refractive index n is now introduced in the path of one of the two interfering waves to increase its phase by $2\pi (n - 1) t/\lambda$. The pattern is shifted and 8th bright fringe is observed at the same point. Find the relation for thickness t in terms of n and λ . CBSE 25

18. In a double-slit experiment, 6th dark fringe is observed at a certain point of the screen. A transparent sheet of thickness t and refractive index n is now introduced in the path of one of the two interfering waves to increase its phase by $2\pi (n - 1) t/\lambda$. The pattern is shifted and 8th bright fringe is observed at the same point. Find the relation for thickness t in terms of n and λ .

CBSE 25

2

Phase difference for 6th dark fringe = 11π
 Phase difference for 8th bright fringe = 16π

$$\Delta\phi + \phi_6 = \phi_8$$

$$2\pi(n-1)\frac{t}{\lambda} + 11\pi = 16\pi$$

$$t = \frac{5\lambda}{2(n-1)}$$

Alternatively: -

Path difference for 6th dark fringe (x_6) = $\frac{11\lambda}{2}$

Path difference for 8th bright fringe (x_8) = 8λ

Increase in path difference after introducing the sheet (Δx)

$$= \frac{\lambda}{2\pi} \times 2\pi(n-1)\frac{t}{\lambda} = (n-1)t$$

$$\therefore x_6 + \Delta x = x_8$$

$$\frac{11\lambda}{2} + (n-1)t = 8\lambda$$

On solving:-

$$t = \frac{5\lambda}{2(n-1)}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

18. (a) In a diffraction experiment, the slit is illuminated by light of wavelength 600 nm. The first minimum of the pattern falls at $\theta = 30^\circ$. Calculate the width of the slit.

CBSE 25

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CBSE 25

2

Condition for Minima $a \sin \theta = n\lambda$ For First Minima $n=1$ $a \sin 30^\circ = 600 \times 10^{-9} \text{ m}$ $a \times \frac{1}{2} = 600 \times 10^{-9} \text{ m}$ $a = 1200 \times 10^{-9} \text{ m}$ $= 1.2 \times 10^{-6} \text{ m}$	1 $\frac{1}{2}$ $\frac{1}{2}$
--	---

- (b) In a Young's double-slit experiment, two light waves, each of intensity I_0 , interfere at a point, having a path difference $\frac{\lambda}{8}$ on the screen. Find the intensity at this point.

31. (a) (i) (1) What are coherent sources ? Why are they necessary for observing a sustained interference pattern ? 5
(2) Lights from two independent sources are not coherent. Explain.
(ii) Two slits 0.1 mm apart are arranged 1.20 m from a screen. Light of wavelength 600 nm from a distant source is incident on the slits.
(1) How far apart will adjacent bright interference fringes be on the screen ?
(2) Find the angular width (in degree) of the first bright fringe.

i) 1) If the phase difference between the displacement produced by each of the wave from two sources does not change with time then two sources are said to be coherent.

Alternatively

Two sources are said to be coherent if they emit light continuously of same frequency / wavelength and having zero or constant phase difference.

Coherent sources are required to get constant phase difference.

2) Two independent sources will never be coherent because phase difference between them will not be constant.

ii) 1) Distance between adjacent bright fringe = fringe width

$$\beta = \frac{\lambda D}{d}$$

$$= \frac{600 \times 10^{-9} \times 1.2}{0.1 \times 10^{-3}} = 7.2 \text{ mm}$$

2) $\theta = \frac{\lambda}{d}$

$$= \frac{600 \times 10^{-9}}{0.1 \times 10^{-3}} = 6 \times 10^{-3} \text{ rad} = 0.34^\circ$$

Give full marks if the student writes the answer in radians only.

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

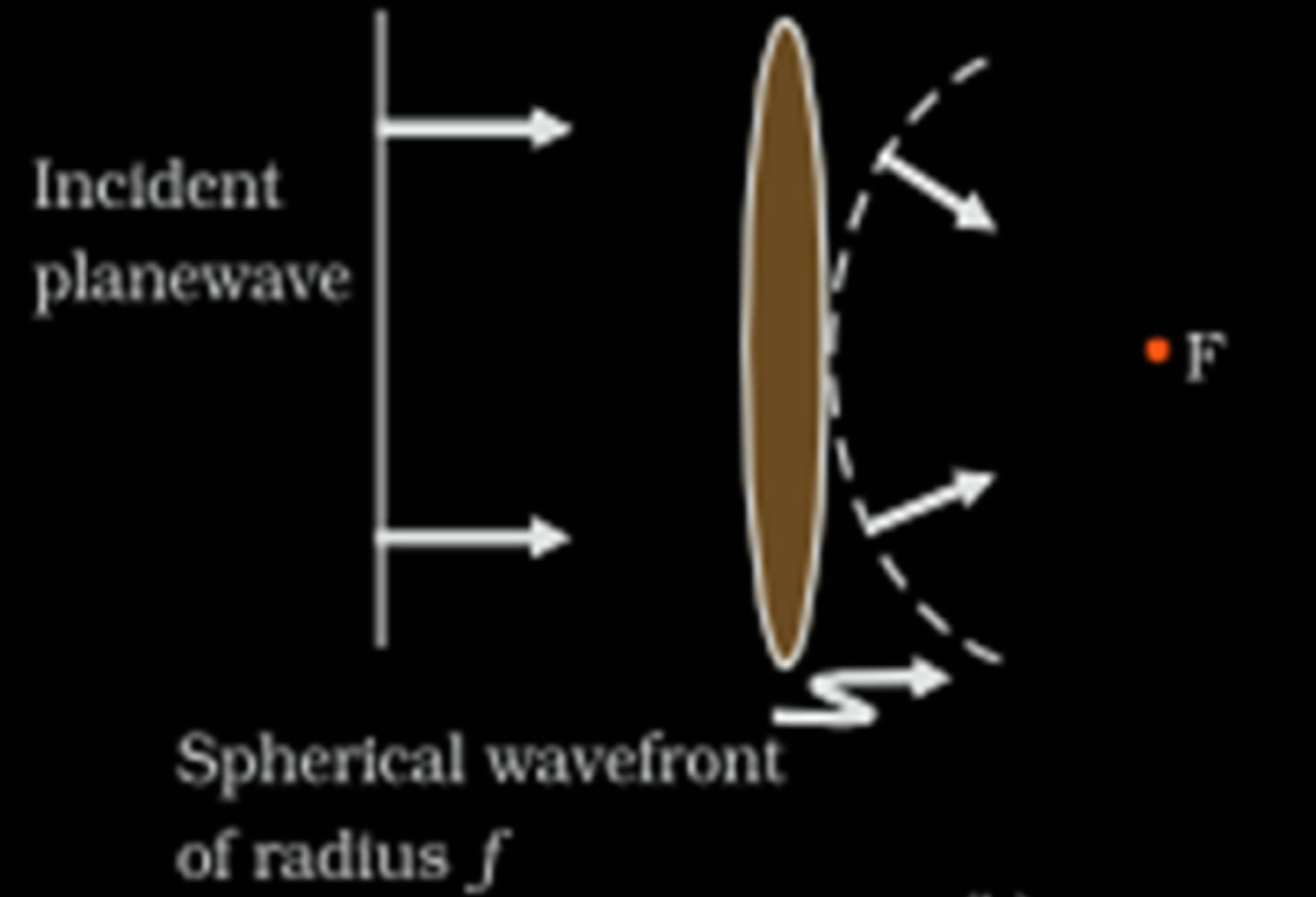
$\frac{1}{2}$

- (b) (i) Define a wavefront. An incident plane wave falls on a convex lens and gets refracted through it. Draw a diagram to show the incident and refracted wavefront.

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CBSE 25

i) A wavefront is a locus of all the points which oscillate in phase.



1

$\frac{1}{2} + \frac{1}{2}$

- 21.** State Huygens principle. Using it draw a diagram showing the details of passage of a plane wave from a denser into a rarer medium.

CBSE 24 C

21. State Huygens principle. Using it draw a diagram showing the details of passage of a plane wave from a denser into a rarer medium.

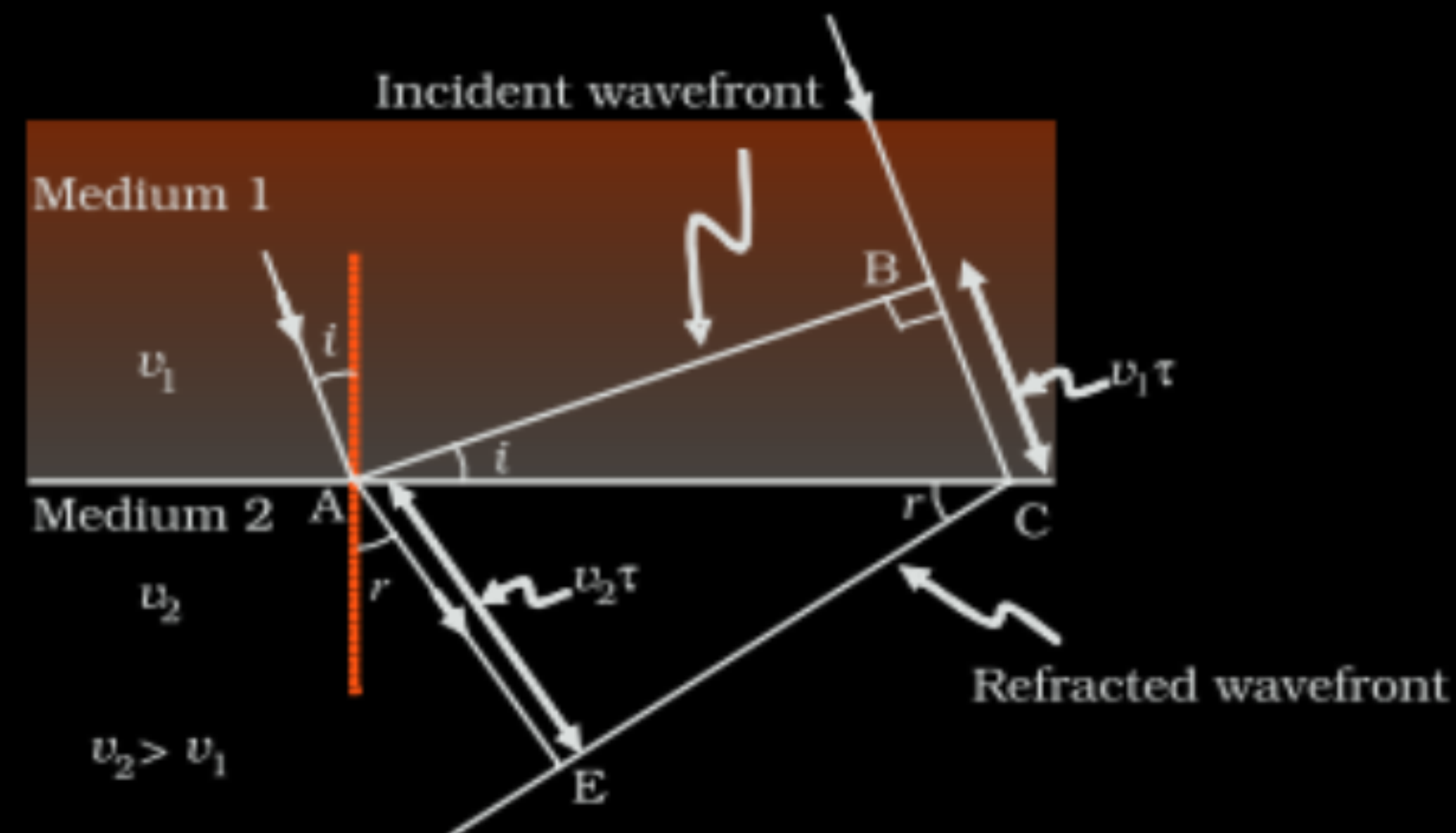
CBSE 24 C

2

Huygens Principle :

Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets emanating from the wavefront are usually referred to as secondary wavelets and if we draw a common tangent to all these spheres, we obtain the new position of the wavefront at a later time.

1



1

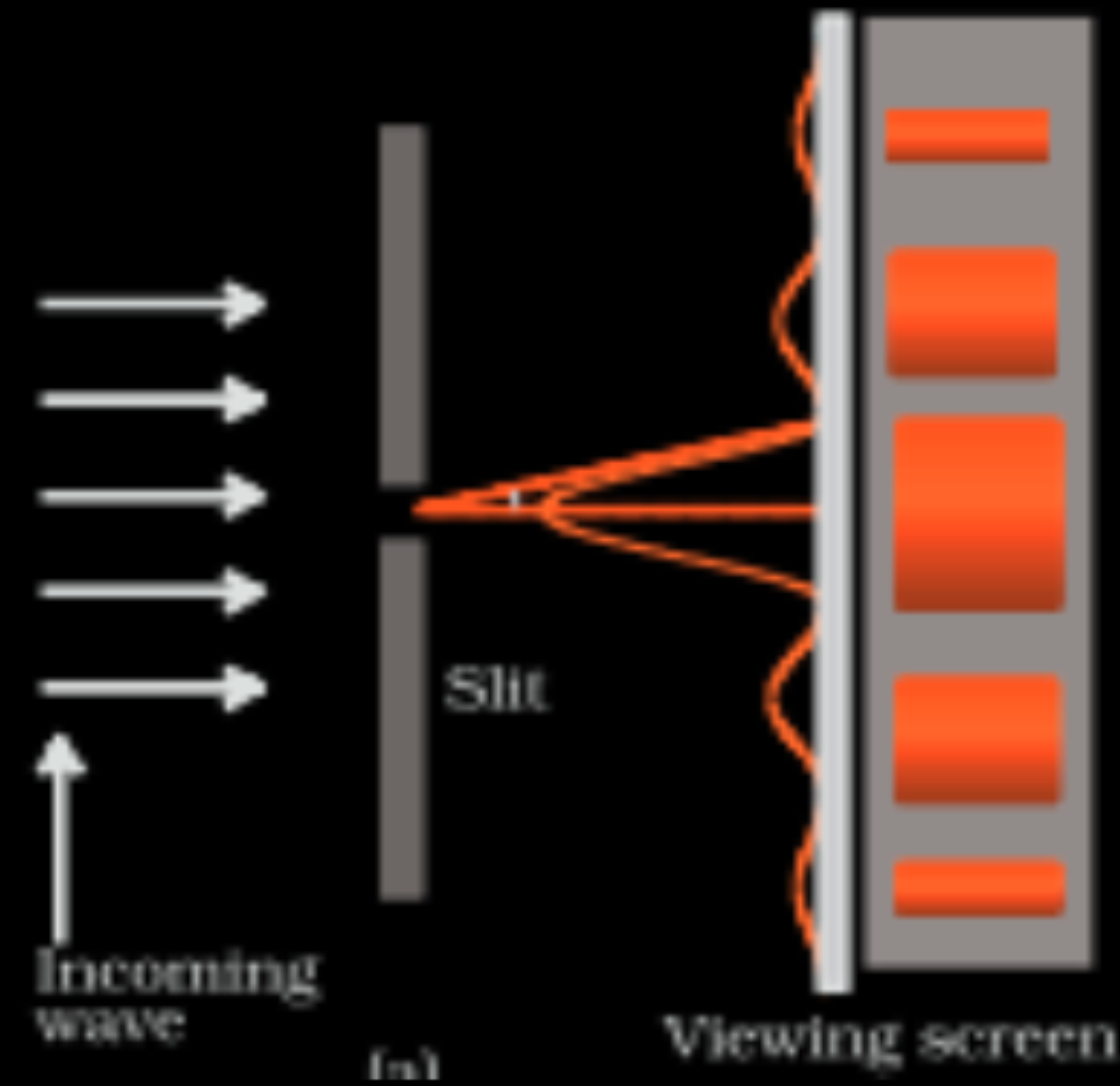
28. (a) Draw the shape of intensity distribution curve of the fringes due to diffraction at a single slit.

CBSE 24 C

28. (a) Draw the shape of intensity distribution curve of the fringes due to diffraction at a single slit.

CBSE 24 C

(a)



- (b) (i) Describe a simple activity to observe diffraction pattern due to a single slit.

CBSE 24 C

- (b) (i) Describe a simple activity to observe diffraction pattern due to a single slit.

CBSE 24 C

(i)	We hold two razor blades in such a way that their edges are parallel and with a narrow slit in between. Keep the slit parallel to the filament of electric bulb, right in front of the eye. A diffraction is seen with its bright and dark bands.	2
-----	---	---

- (b) The intensity of the two interfering waves in Young's double slit experiment is I_0 each. Find the intensity at a point on the screen where path difference between the interfering waves is (i) $\frac{\lambda}{2}$, and (ii) $\frac{\lambda}{3}$.

3

- (b) The intensity of the two interfering waves in Young's double slit experiment is I_0 each. Find the intensity at a point on the screen where path difference between the interfering waves is (i) $\frac{\lambda}{2}$, and (ii) $\frac{\lambda}{3}$.

3

CBSE 24 C

(b)

$$I = 4I_0 \cos^2 \frac{\phi}{2}$$

(i) $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$

$$I = 4I_0 \left(\cos \frac{\pi}{2} \right)^2 = 0$$

(ii) $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$

$$I = 4I_0 \left(\cos \frac{\pi}{3} \right)^2$$

$$I = I_0$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

8. The phase difference between any two points on a wavefront of a wave is :

- | | |
|-----------|---------------------|
| (A) π | (B) $\frac{\pi}{2}$ |
| (C) Zero | (D) $\frac{\pi}{6}$ |

8. The phase difference between any two points on a wavefront of a wave is :

(A) π

(B) $\frac{\pi}{2}$

(C) Zero

(D) $\frac{\pi}{6}$

CBSE 24 C

(C) Zero

19. A monochromatic light beam is incident on a slit of width 0.2 mm. The diffraction pattern is formed on a screen placed 1.5 m away from the slits. The first maximum is formed at a distance of 3.0 mm from the centre of the screen. Calculate the wavelength of light used.

CBSE 24 C

19. A monochromatic light beam is incident on a slit of width 0.2 mm. The diffraction pattern is formed on a screen placed 1.5 m away from the slits. The first maximum is formed at a distance of 3.0 mm from the centre of the screen. Calculate the wavelength of light used.

2

CBSE 24 C

Distance of nth maxima from centre of screen

$$x_{nth} = \frac{(2n+1)\lambda D}{2a}$$

For n=1

$$x = \frac{3\lambda D}{2a}$$

$$\lambda = \frac{2ax}{3D}$$

$$\lambda = \frac{2 \times 0.2 \times 10^{-3} \times 3 \times 10^{-3}}{3 \times 1.5}$$

$$\lambda = 2.67 \times 10^{-7} \text{ m}$$

1/2

1/2

1/2

1/2

- (b) (i) Give reasons for the following :
- (1) Why is an objective lens of large focal length and large aperture used in a telescope ?
 - (2) Two sodium lamps (independent) cannot emit coherent lights.
 - (3) The resultant intensity at any point on the screen varies between zero and four times the intensity, due to one slit, in Young's double slit experiment.
- (ii) In Young's double slit experiment, fringes are obtained on a screen using two slits separated by 2 mm. When the screen is moved 20 cm towards the slit, the change in fringe width is found to be 6×10^{-5} m. Calculate the wavelength of the light used.

(1) The objective lens of a telescope should have large focal length and large aperture so that it is able to gather more light and hence has better resolving power.	1
(2) Two sodium lamps which are independent cannot emit coherent lights as two independent sources cannot produce waves which have no phase difference or have constant phase difference between them.	1
(3) The resultant intensity $I = 4I_o \cos^2 \frac{\phi}{2}$ I is maximum for $\phi = 0, 2\pi, 4\pi, \dots \Rightarrow I_{\max} = 4I_o$ I is minimum for $\phi = \pi, 3\pi, \dots \Rightarrow I_{\min} = 0$	1
(ii) Fringe width $(\beta) = \frac{\lambda D}{d}$ ----- (1) When screen is moved 20 cm towards the slit; fringe width	$\frac{1}{2}$
$(\beta') = \frac{\lambda(D-20)}{d}$ ----- (2) Subtracting equation (1) and (2)	$\frac{1}{2}$
$(\beta - \beta') = \frac{\lambda \times 20 \times 10^{-2}}{d}$ On solving $\lambda = 6 \times 10^{-7} \text{ m} = 600 \text{ nm}$	1

13. **Assertion (A)** : In a Young's double-slit experiment, interference pattern is not observed when two coherent sources are infinitely close to each other.

Reason (R) : The fringe width is proportional to the separation between the two sources.

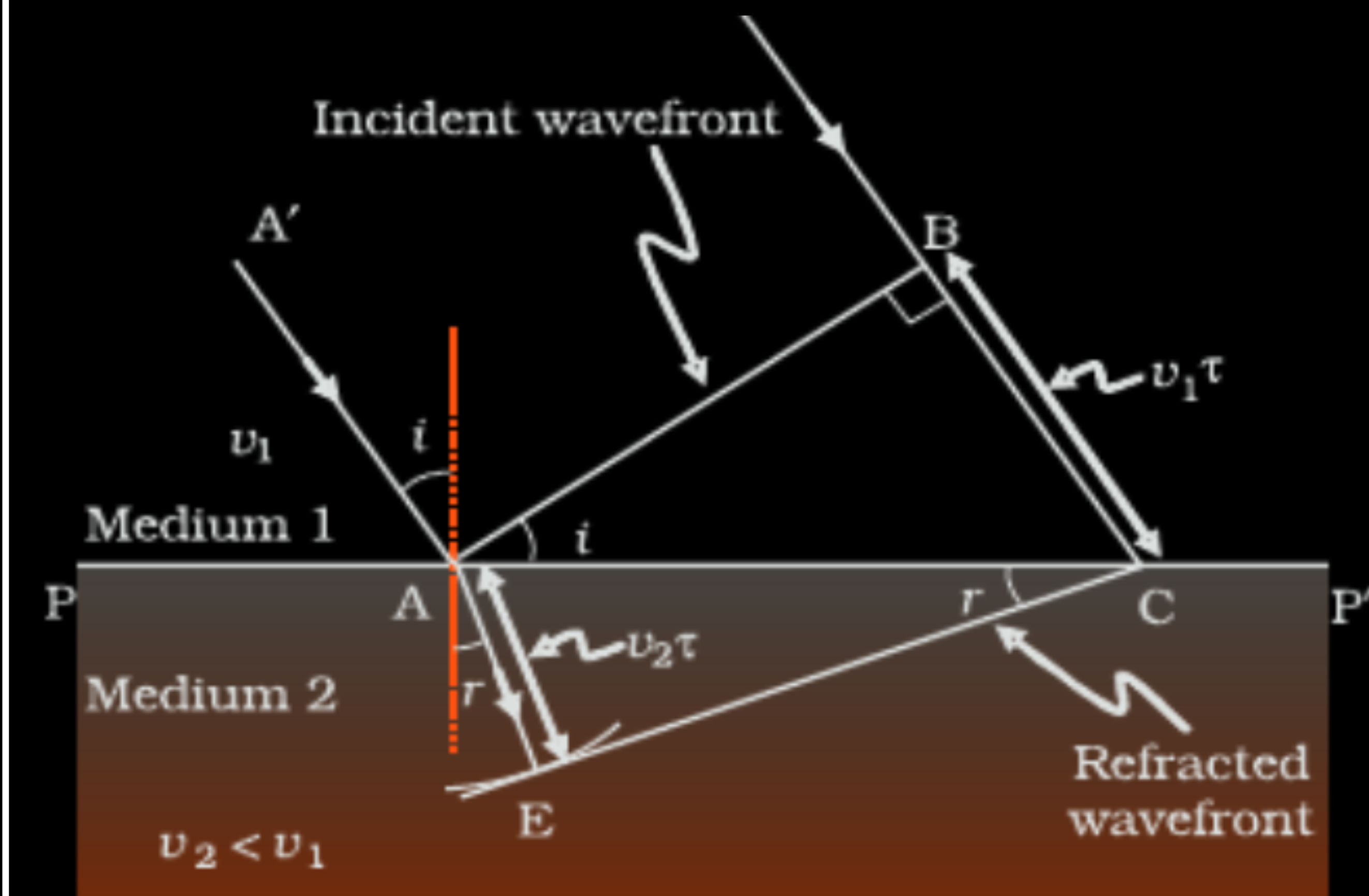
13. **Assertion (A)** : In a Young's double-slit experiment, interference pattern is not observed when two coherent sources are infinitely close to each other.

Reason (R) : The fringe width is proportional to the separation between the two sources.

CBSE 24

(C) Assertion (A) is true but Reason (R) is false.

32. (a) (i) A plane light wave propagating from a rarer into a denser medium, is incident at an angle i on the surface separating two media. Using Huygen's principle, draw the refracted wave and hence verify Snell's law of refraction.
- (ii) In a Young's double slit experiment, the slits are separated by 0.30 mm and the screen is kept 1.5 m away. The wavelength of light used is 600 nm. Calculate the distance between the central bright fringe and the 4th dark fringe.



Considering triangles ABC and AEC

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC} \quad \text{and} \quad \text{-----}(1)$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC} \quad \text{-----}(2)$$

From equation (1) and equation (2)

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} \quad \text{-----}(3)$$

If c represents the speed of light in vacuum, then

$$n_1 = \frac{c}{v_1} \quad \text{and} \quad n_2 = \frac{c}{v_2}$$

In terms of refractive indices

$$n_1 \sin i = n_2 \sin r$$

which is Snell's law of refraction.

(ii)

$$X_4 = \frac{(2n-1)\lambda D}{2d}$$

$$X_4 = \frac{(2 \times 4 - 1) \times 600 \times 10^{-9} \times 1.5}{2 \times 0.3 \times 10^{-3}}$$

$$= 1.05 \times 10^{-2} \text{ m}$$

1/2

1

1/2

1/2

1/2

1/2

1

1/2

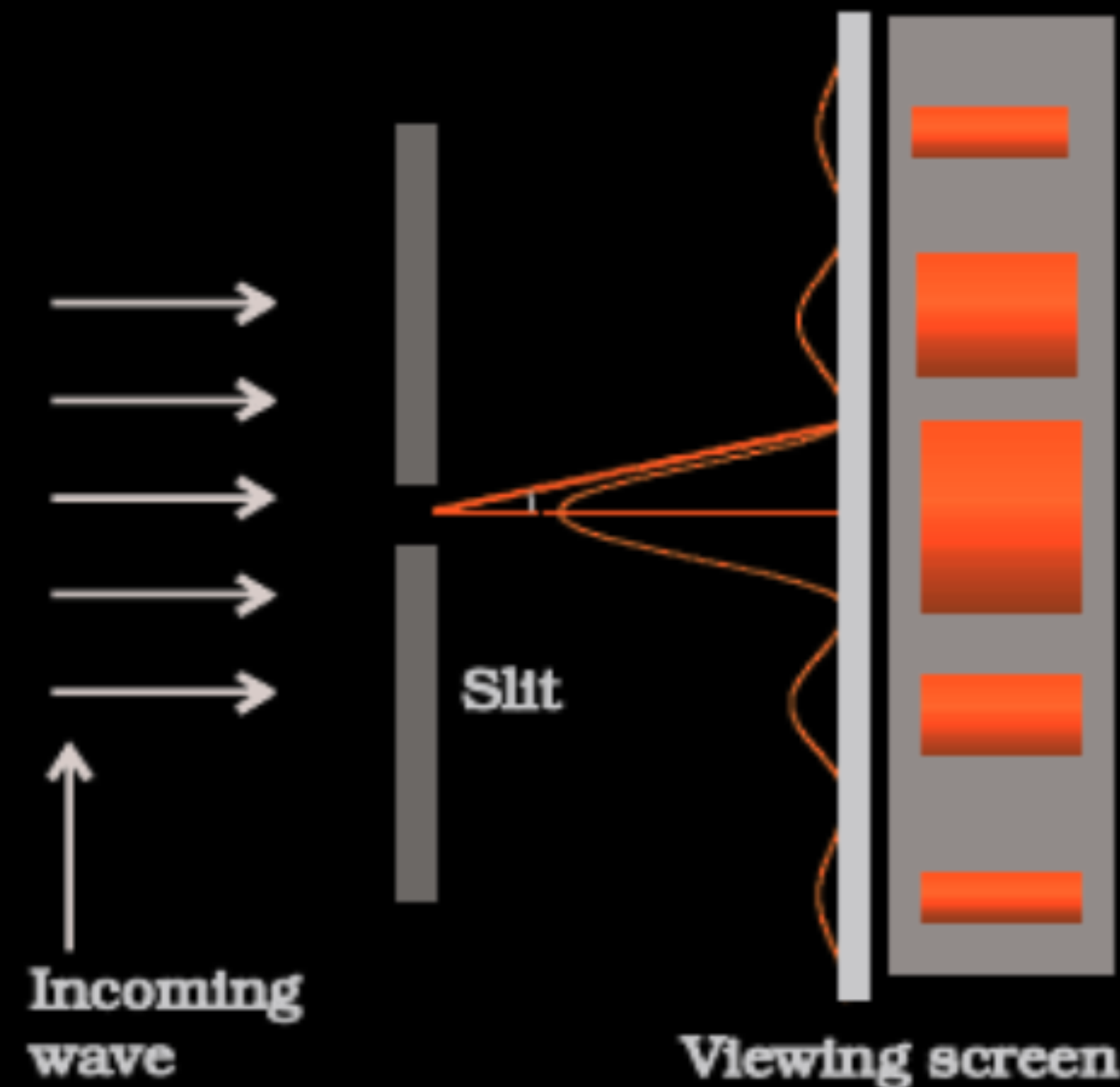
32. (b) (i) Discuss briefly diffraction of light from a single slit and draw the shape of the diffraction pattern.

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CBSE 24

(i) A beam of light falls normally on a single slit and bends around its corners. This phenomenon is called diffraction.

When a beam of light falls normally on a narrow single slit, then diffracted light goes on to meet a screen. It is observed that at the center of the screen intensity is maximum and goes on decreasing as one move away from the center on either side of screen.



1

1

1

8. In a Young's double-slit experiment in air, the fringe width is found to be 0.44 mm. If the entire setup is immersed in water $\left(n = \frac{4}{3}\right)$, the fringe width will be

(A) 0.88 mm

(B) 0.59 mm

(C) 0.33 mm

(D) 0.44 mm

CBSE 24

8. In a Young's double-slit experiment in air, the fringe width is found to be 0.44 mm. If the entire setup is immersed in water $\left(n = \frac{4}{3}\right)$, the fringe width will be

(A) 0.88 mm

(B) 0.59 mm

(C) 0.33 mm

(D) 0.44 mm

CBSE 24

(C) 0.33 mm

10. A Young's double-slit experimental set up is kept in a medium of refractive index $\left(\frac{4}{3}\right)$. Which maximum in this case will coincide with the 6th maximum obtained if the medium is replaced by air ?

(A) 4th

(B) 6th

(C) 8th

(D) 10th

10. A Young's double-slit experimental set up is kept in a medium of refractive index $\left(\frac{4}{3}\right)$. Which maximum in this case will coincide with the 6th maximum obtained if the medium is replaced by air ?

(A) 4th

(B) 6th

(C) 8th

(D) 10th

CBSE 24

(C) 8th

14. **Assertion (A)** : In interference and diffraction of light, light energy reduces in one region producing a dark fringe. It increases in another region and produces a bright fringe.

Reason (R) : This happens because energy is not conserved in the phenomena of interference and diffraction.

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14. **Assertion (A)** : In interference and diffraction of light, light energy reduces in one region producing a dark fringe. It increases in another region and produces a bright fringe.

Reason (R) : This happens because energy is not conserved in the phenomena of interference and diffraction.

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1

(C) Assertion (A) is true and Reason (R) is false.

- (b) (i) Differentiate between a wavefront and a ray.
- (ii) State Huygen's principle and verify laws of reflection using suitable diagram.
- (iii) In Young's double slit experiment, the slits S_1 and S_2 are 3 mm apart and the screen is placed 1.0 m away from the slits. It is observed that the fourth bright fringe is at a distance of 5 mm from the second dark fringe. Find the wavelength of light used.

5

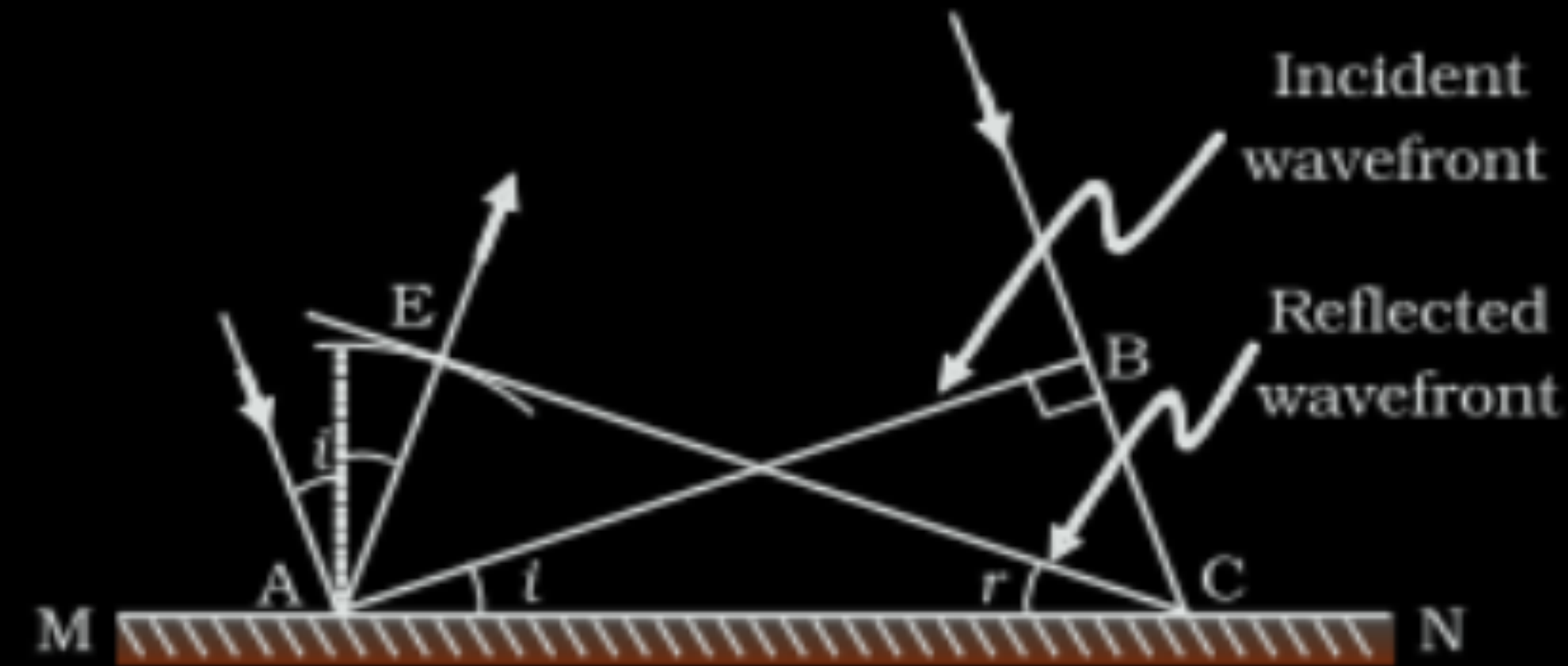
i) Wavefront is a surface of constant phase.

Alternatively Locus of points, which oscillate in phase

Ray - The straight line path along which light travels (or energy propagates).

Alternatively – Ray is normal to wave front.

ii) Huygens' Principle Each point of the wave front is the source of secondary disturbance and the wavelets emanating from the points spread out in all directions with speed of wave. The wavelets emanating from wave front are usually referred to as secondary wavelets. A common tangent to all these spheres gives the new position of the wave front at a later time.



Triangles EAC and BAC are congruent therefore $\angle i = \angle r$

iii) Position of 4th bright fringe

$$x_{4(\text{bright})} = 4 \frac{D\lambda}{d}$$

Position of 2nd dark fringe

$$x_{2(\text{dark})} = \frac{3}{2} \frac{D\lambda}{d}$$

$$x_{4(\text{bright})} - x_{2(\text{dark})} = 5\text{mm}$$

$$4 \frac{D\lambda}{d} - \frac{3}{2} \frac{D\lambda}{d} = 5 \times 10^{-3}$$

$$\lambda = 6 \times 10^{-6} \text{ m}$$

19. What is a sustained or stable interference pattern ? What are the conditions for obtaining such an interference pattern ?

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19. What is a sustained or stable interference pattern ? What are the conditions for obtaining such an interference pattern ?

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- | | |
|--|----------|
| ❖ When position of maxima and minima is not changing with time, interference pattern is called sustained or stable interference. | 1 |
| ❖ Light sources must be coherent | 1 |

30. Diffraction and interference are closely related phenomena that occur together. Diffraction is the phenomenon of bending of light around the edges of the obstacle, while interference is the combination of waves that results in a new wave pattern. In order to get interference, there must be at least two waves that are diffracting. So while diffraction can occur without interference, interference cannot occur without diffraction.

Two slits of width $2\text{ }\mu\text{m}$ each in an opaque material are separated by a distance of $6\text{ }\mu\text{m}$. Monochromatic light of wavelength 450 nm is incident normally on the slits. One finds a combined interference and diffraction pattern on the screen.

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(i) The number of peaks of the interference fringes formed within the central peak of the envelope of the diffraction pattern will be :

1

(A) 2

(B) 3

(C) 4

(D) 6

30. Diffraction and interference are closely related phenomena that occur together. Diffraction is the phenomenon of bending of light around the edges of the obstacle, while interference is the combination of waves that results in a new wave pattern. In order to get interference, there must be at least two waves that are diffracting. So while diffraction can occur without interference, interference cannot occur without diffraction.

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(i) The number of peaks of the interference fringes formed within the central peak of the envelope of the diffraction pattern will be :

1

(A) 2

(B) 3

(C) 4

(D) 6

(D) 6

- (ii) The number of peaks of the interference formed if the slit width is doubled while keeping the distance between the slits same will be : *1*
- (A) 1 (B) 2
- (C) 3 (D) 4

- (ii) The number of peaks of the interference formed if the slit width is doubled while keeping the distance between the slits same will be : *1*
- (A) 1 (B) 2
(C) 3 (D) 4

(ii) (C) 3

(iii) (a) If instead of 450 nm light, another light of wavelength 680 nm is used, number of peaks of the interference formed in the central peak of the envelope of the diffraction pattern will be :

(A) 2

(B) 4

(C) 6

(D) 9

(iii) (a) If instead of 450 nm light, another light of wavelength 680 nm is used, number of peaks of the interference formed in the central peak of the envelope of the diffraction pattern will be :

(A) 2

(B) 4

(C) 6

(D) 9

(C)6

- (b) Consider the diffraction of light by a single slit described in this case study. The first minimum falls at an angle θ equal to :

1

(A) $\sin^{-1}(0.12)$

(B) $\sin^{-1}(0.225)$

(C) $\sin^{-1}(0.32)$

(D) $\sin^{-1}(0.45)$

- (b) Consider the diffraction of light by a single slit described in this case study. The first minimum falls at an angle θ equal to :

1

(A) $\sin^{-1}(0.12)$

(B) $\sin^{-1}(0.225)$

(C) $\sin^{-1}(0.32)$

(D) $\sin^{-1}(0.45)$

(B) $\sin^{-1}(0.225)$

(iv) The number of bright fringes formed due to interference on 1 m of screen placed at $\frac{4}{3}$ m away from the slits is : *1*

(A) 2

(B) 3

(C) 6

(D) 10

(iv) The number of bright fringes formed due to interference on 1 m of screen placed at $\frac{4}{3}$ m away from the slits is : *1*

(A) 2

(B) 3

(C) 6

(D) 10

(D) 10

- (b) (i) The displacement of two light waves, each of amplitude 'a' and frequency ω , emanating from two coherent sources of light, are given by $y_1 = a \cos \omega t$ and $y_2 = a \cos (\omega t + \phi)$. ϕ is the phase difference between the two waves. These light waves superpose at a point. Obtain the expression for the resultant intensity at that point.
- (ii) In Young's double slit experiment, find the ratio of intensities at two points on a screen when waves emanating from two slits reaching these points have path differences (i) $\frac{\lambda}{6}$ and (ii) $\frac{\lambda}{12}$.

$$(i) \quad y_1 = a \cos \omega t$$

$$y_2 = a \cos(\omega t + \phi)$$

According to the principle of superposition

$$y = y_1 + y_2$$

$$y = a \cos \omega t + a \cos(\omega t + \phi)$$

$$y = a \cos \omega t + a \cos \omega t \cos \phi - a \sin \omega t \sin \phi$$

$$y = a \cos \omega t (1 + \cos \phi) - a \sin \phi \sin \omega t$$

Let,

$$a(1 + \cos \phi) = A \cos \theta \quad \text{----- (i)}$$

$$a \sin \phi = A \sin \theta \quad \text{----- (ii)}$$

Squaring and adding equation (i) and (ii)

$$A^2 = a^2 (1 + \cos \phi)^2 + a^2 \sin^2 \phi$$

$$= a^2 (1 + \cos^2 \phi + 2 \cos \phi) + a^2 \sin^2 \phi$$

$$= 2a^2 (1 + \cos \phi)$$

$$= 4a^2 \cos^2 \phi / 2$$

$$I \propto A^2$$

$$I = kA^2$$

where k is constant

$$I = 4ka^2 \cos^2 \phi / 2$$

[Award full credit for this part for any other alternative methods]

$$(ii) \quad \phi_1 = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \pi/3$$

$$I_1 = 4I_0 \cos^2 \phi / 2$$

$$= 4I_0 \cos^2 (\pi/6)$$

$$I_1 = 3I_0$$

$$\phi_2 = \frac{2\pi}{\lambda} \times \frac{\lambda}{12} = \pi/6$$

$$I_2 = 4I_0 \cos^2 (\pi/12)$$

$$I_2 = 4I_0 \cos^2 15^\circ$$

$$\frac{I_1}{I_2} = \frac{3}{4 \cos^2 15^\circ}$$

1/2

1/2

1/2

1/2

1/2

1/2

1/2

1/2

1/2

1/2

19. Write two points of difference between interference and diffraction of light.

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19. Write two points of difference between interference and diffraction of light.

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2

Interference	Diffraction	1+1
(i) In interference pattern width of each maxima is same. (ii) In interference pattern intensity of all maxima is same.	(i) In diffraction pattern width of central maxima is twice the width of secondary maxima. (ii) In diffraction pattern intensity of maxima goes on decreasing as we move away from central maxima.	
[Award full credit if students write any other two differences]		

- 19.** (a) Two waves, each of amplitude 'a' and frequency ' ω ' emanating from two coherent sources of light superpose at a point. If the phase difference between the two waves is ϕ , obtain an expression for the resultant intensity at that point.

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19. (a) Two waves, each of amplitude 'a' and frequency 'ω' emanating from two coherent sources of light superpose at a point. If the phase difference between the two waves is ϕ , obtain an expression for the resultant intensity at that point.

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$$x_1 = a \cos \omega t$$

$$x_2 = a \cos(\omega t + \phi)$$

$$x = x_1 + x_2$$

$$= a(\cos \omega t + \cos(\omega t + \phi))$$

$$= a(2 \cos(\omega t + \frac{\phi}{2}) \cos \frac{\phi}{2})$$

$$= 2a \cos \frac{\phi}{2} \cos(\omega t + \frac{\phi}{2})$$

Intensity

$$I = K (\text{amplitude})^2 \quad \text{where K is a constant.}$$

$$= K(2a \cos \frac{\phi}{2})^2$$

$$= 4I_0 \cos^2 \frac{\phi}{2}$$

$$I_0 = Ka^2 = \text{intensity of each incident wave.}$$

(Award full credit of this part for all other alternative correct methods)

1/2

1/2

1/2

1/2

- (b) What is the effect on the interference pattern in Young's double-slit experiment when (i) the source slit is moved closer to the plane of the slits, and (ii) the separation between the two slits is increased ? Justify your answers.

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- (b) What is the effect on the interference pattern in Young's double-slit experiment when (i) the source slit is moved closer to the plane of the slits, and (ii) the separation between the two slits is increased ? Justify your answers.

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2

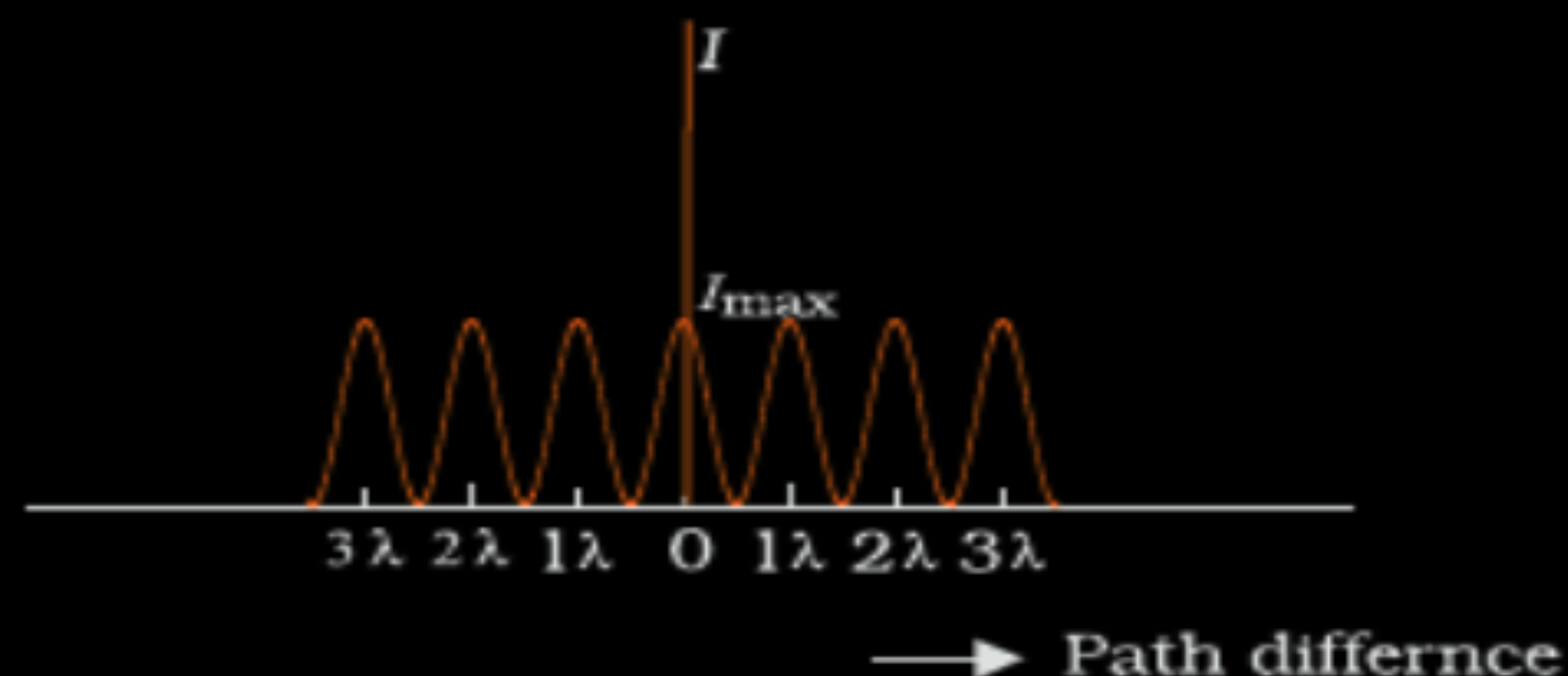
(i) Sharpness of interference pattern decreases $\frac{s}{S} < \frac{\lambda}{d}$ As S decreases, interference patterns produced by different parts of the source overlap and finally fringes disappear. Alternatively As the source slit is brought closer to the plane of the slits, the screen gets illuminated uniformly and fringes disappear. Alternatively Interference pattern is not formed. (Note : Award full credit of this part if a student merely attempts this part.)	1
(ii) $\beta = \frac{\lambda D}{d}$	$\frac{1}{2}$
As d increases, β decreases and fringes disappear.	$\frac{1}{2}$

- 31.** (a) (i) Give any two differences between the interference pattern obtained in Young's double-slit experiment and a diffraction pattern due to a single slit.
- (ii) Draw an intensity distribution graph in case of a double-slit interference pattern.
- (iii) In Young's double-slit experiment using monochromatic light of wavelength λ , the intensity of light at a point on the screen, where path difference is λ , is K units. Find the intensity of light at a point on the screen where the path difference is $\frac{\lambda}{6}$.

(i)

	Interference	Diffraction
1	Bands are equally spaced	Bands are not equally spaced.
2	Intensity of bright bands is same.	Intensity of maxima decreases on either side of central maxima.
3	First maxima is at an angle λ / a	First minima is at an angle λ / a

(ii)

(iii) Path difference $(\Delta) = \lambda$ **1 + 1****1**

$$\phi = \frac{2\pi\Delta}{\lambda}$$

$$\phi = 2\pi$$

$$I = 4I_0 \cos^2 \frac{\phi}{2}$$

$$K = 4I_0 \cos^2 \pi = 4I_0$$

$$\text{Path difference} = \frac{\lambda}{6}$$

$$\phi = \pi / 3$$

$$I = 4I_0 \cos^2 \frac{\pi}{6}$$

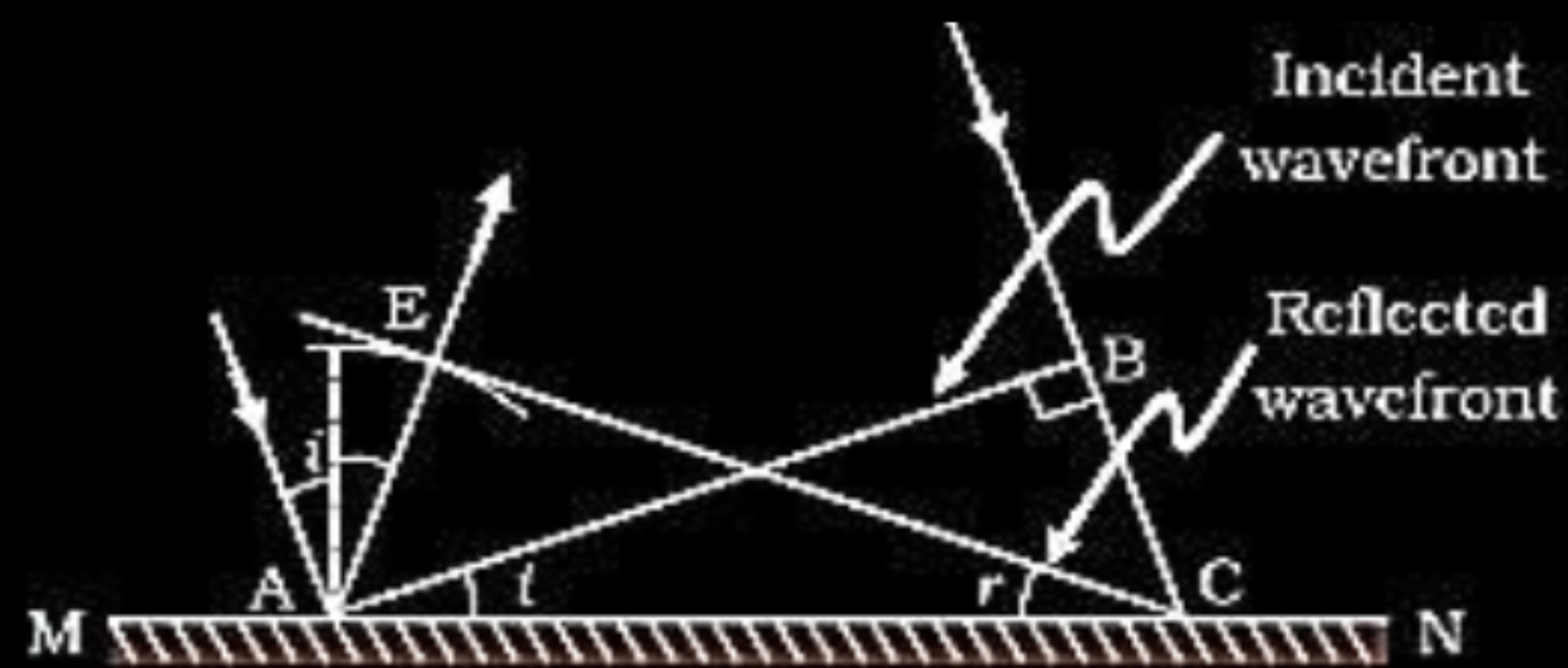
$$= 4I_0 \times \frac{3}{4}$$

$$= \frac{3}{4} K$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- (b) (i) State Huygens' principle. A plane wave is incident at an angle i on a reflecting surface. Construct the corresponding reflected wavefront. Using this diagram, prove that the angle of reflection is equal to the angle of incidence.
- (ii) What are the coherent sources of light ? Can two independent sodium lamps act like coherent sources ? Explain.
- (iii) A beam of light consisting of a known wavelength 520 nm and an unknown wavelength λ , used in Young's double slit experiment produces two interference patterns such that the fourth bright fringe of unknown wavelength coincides with the fifth bright fringe of known wavelength. Find the value of λ .

(i) Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the spread of the wave. Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets emanating from the wavefront are usually referred to as secondary wavelets and if we draw a common tangent to all these spheres, we obtain the new position of the wavefront at a later time.



ΔEAC is congruent to ΔBAC ; so $\angle i = \angle r$

(ii) Two sources are said to be coherent if the phase difference between them does not change with time.

No, two independent sodium lamps cannot be coherent.
Two independent sodium lamps cannot be coherent as the phase between them does not remain constant with time.

(iii)

$$4\beta_2 = 5\beta_1$$

$$4 \times \frac{\lambda D}{d} = 5 \times \frac{\lambda_{known} D}{d}$$

$$\Rightarrow \lambda = \frac{5}{4} \times \lambda_{known}$$

$$= \frac{5}{4} \times 520$$

$$= 650 \text{ nm}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) How will the fringe width be affected in a Young's double slit experiment with light of wavelength λ if :
- (i) the separation between the slits is reduced to half, keeping the distance of the screen from the slits constant ?
 - (ii) the distance of the screen from the slits is reduced to half, keeping the separation between the slits constant ?

(b) How will the fringe width be affected in a Young's double slit experiment with light of wavelength λ if :

- (i) the separation between the slits is reduced to half, keeping the distance of the screen from the slits constant ?
- (ii) the distance of the screen from the slits is reduced to half, keeping the separation between the slits constant ?

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2

i. Fringe width doubles / increases	1
ii. Fringe width becomes half / decreases	1

33. (a) (i) What are coherent sources of light ? Briefly explain how a pair of coherent sources are obtained in a Young's double-slit experiment. Mention the conditions for constructive and destructive interferences in this experiment.

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Coherent light sources are those sources which emit light waves with the same frequency/wavelength and have a constant phase difference.

1

A pair of coherent sources is obtained in Young's Double Slit Experiment by illuminating two identical narrow slits by a monochromatic source of light equidistant from the slits.

1

For Constructive Interference:

Phase difference = $2n\pi$

$n = 0, 1, 2, \dots$

Alternatively,

Path difference = $n\lambda$

$n = 0, 1, 2, \dots$

$\frac{1}{2}$

For Destructive Interference:

Phase difference = $(2n+1)\pi$

$n = 0, 1, 2, \dots$

Alternatively,

Path difference = $\left(n + \frac{1}{2}\right)\lambda$

$n = 0, 1, 2, \dots$

$\frac{1}{2}$

- (ii) Two students are separated by a 6 m partition wall in a room 9 m high. If both light and sound waves can bend around obstacles, how is it that the students are unable to see each other, though they can talk easily ?

(ii) Two students are separated by a 6 m partition wall in a room 9 m high. If both light and sound waves can bend around obstacles, how is it that the students are unable to see each other, though they can talk easily ?

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(ii) For diffraction or bending of a wave, size of the aperture should be comparable to the wavelength.

1

The wavelength of sound is comparable to the size of the aperture whereas wavelength of light is much smaller than size of the aperture so bending of sound will take place but bending of light will not occur. Therefore students are unable to see each other though they can talk easily.

1

9. In Young's double-slit experiment, the slit separation is made three times and the distance of the screen from the slits is doubled. The fringe width becomes :

- | | |
|-------------------------|-------------------------|
| (a) 6 times | (b) $\frac{3}{2}$ times |
| (c) $\frac{1}{6}$ times | (d) $\frac{2}{3}$ times |

9. In Young's double-slit experiment, the slit separation is made three times and the distance of the screen from the slits is doubled. The fringe width becomes :

- | | |
|-------------------------|-------------------------|
| (a) 6 times | (b) $\frac{3}{2}$ times |
| (c) $\frac{1}{6}$ times | (d) $\frac{2}{3}$ times |

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(d) $\frac{2}{3}$ times

- 21.** (a) The intensities of the principal maximum and secondary maxima in a single slit diffraction pattern are not equal. Explain.

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- 21.** (a) The intensities of the principal maximum and secondary maxima in a single slit diffraction pattern are not equal. Explain.

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2

The intensity of light of a secondary maxima goes on decreasing with the order of maxima because the intensity of central maxima is due to wavelets from all part of slit. First secondary maxima is due to wavelets from one third part of the slit and second secondary maxima is due to wavelets from one fifth part of slit and so on. Thus intensity of secondary maxima goes on decreasing and is lesser than central maxima.

- (b) In a Young's double-slit experiment, set-up in air, the fringe width is found to be β . If the entire set-up is immersed in water, how would β be affected ? Give reason for your answer.

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2

- (b) In a Young's double-slit experiment, set-up in air, the fringe width is found to be β . If the entire set-up is immersed in water, how would β be affected ? Give reason for your answer.

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2

$$\beta' = \frac{\beta}{\mu} \quad \text{Fringe width will decrease}$$

$$\beta = \frac{\lambda D}{d}$$

$$\lambda' = \frac{\lambda}{\mu}$$

$$\Rightarrow \beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{d \mu} = \frac{\beta}{\mu}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

- 33.** (a) (i) What is a wavefront ? What is the direction of a beam of light with respect to the wavefront associated with it ?
- (ii) State Huygens' principle. Using it, explain the propagation of light.
- (iii) State two conditions necessary for light from two sources to be coherent.

(i) A wavefront is defined as a surface of constant phase. Perpendicular	$\frac{1}{2}$ $\frac{1}{2}$
(ii) Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets emanating from the wavefront are usually referred to as secondary wavelets and if we draw a common tangent to all these spheres, we obtain the new position of the wavefront at a later time.	1
All the secondary sources emit wavelets and tangent drawn to all the wavelets gives the position of new wavefront at any later time.	1
(iii) (i) Two sources must emit light of same frequency. (ii) Phase difference between two light sources either should be zero or having a constant phase difference i.e. should not change with time.	1 1

6. In Young's double-slit experiment, the intensity on the screen is I_0 at a point where path difference is λ . The intensity at the point where path difference is $\frac{\lambda}{4}$ is :

(a) $\frac{I_0}{4}$

(b) $\frac{I_0}{2}$

(c) I_0

(d) zero

6. In Young's double-slit experiment, the intensity on the screen is I_0 at a point where path difference is λ . The intensity at the point where path difference is $\frac{\lambda}{4}$ is :

(a) $\frac{I_0}{4}$

(b) $\frac{I_0}{2}$

(c) I_0

(d) zero

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(b) $\frac{I_0}{2}$

- 28.** (a) (i) In which case is diffraction effect more dominant — slit formed by 2 blades or slit formed by two fingers ?
- (ii) Yellow light ($\lambda = 6000 \text{ \AA}$) illuminates a single slit of width $1 \times 10^{-4} \text{ m}$. Calculate (i) the distance between two dark lines on either side of central maximum, in the diffraction pattern observed on a screen kept 1.5 m away from the slit, and (ii) the angular spread of the first minimum.

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3

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- (ii) Yellow light ($\lambda = 6000 \text{ \AA}$) illuminates a single slit of width $1 \times 10^{-4} \text{ m}$. Calculate (i) the distance between two dark lines on either side of central maximum, in the diffraction pattern observed on a screen kept 1.5 m away from the slit, and (ii) the angular spread of the first minimum.

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3

(i) slit formed by two blades	1
(ii) (i) separation between two dark lines	$\frac{1}{2}$
$= \frac{2\lambda D}{d}$	
$= \frac{2 \times 6000 \times 10^{-10} \times 1.5}{1 \times 10^{-4}} = 18 \times 10^{-3} \text{ m}$	$\frac{1}{2}$
(ii) $\phi = \frac{\lambda}{a}$	$\frac{1}{2}$
$= \frac{6000 \times 10^{-10}}{1 \times 10^{-4}}$	
$= 6 \times 10^{-3} \text{ rad}$	$\frac{1}{2}$

- (b) (i) What will be the colour of the central bright fringe in Young's double slit experiment if the monochromatic source is replaced by a source of white light ? Give reason for your answer.
- (ii) In Young's double slit experiment, the slits are separated by 0.3 mm and the screen is placed 1.5 m away from the slits. The distance between the central bright fringe and the sixth bright fringe is found to be 1.8 cm. Find the wavelength of light used in the experiment.

- (b) (i) What will be the colour of the central bright fringe in Young's double slit experiment if the monochromatic source is replaced by a source of white light ? Give reason for your answer.
- (ii) In Young's double slit experiment, the slits are separated by 0.3 mm and the screen is placed 1.5 m away from the slits. The distance between the central bright fringe and the sixth bright fringe is found to be 1.8 cm. Find the wavelength of light used in the experiment.

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(b)

(i) White.

The interference pattern due to different component colors of white light overlap (incoherently). The central bright fringes for different colors are at the same position. Therefore the central fringe is white.

$$(ii) \lambda_n = \frac{n\lambda D}{d}$$

$$\lambda = \frac{\lambda_n d}{nD}$$

$$\lambda = \frac{1.8 \times 10^{-2} \times 0.3 \times 10^{-3}}{6 \times 1.5}$$

$$\lambda = 6000 \text{ \AA}$$

1/2

1/2

1/2

1

1/2

- (ii) In Young's double-slit experiment, the two slits are 1.0 mm apart. They are illuminated with light of wavelength 600 nm. Find the fringe width on a screen 2.0 m away from the slits.

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- (ii) In Young's double-slit experiment, the two slits are 1.0 mm apart. They are illuminated with light of wavelength 600 nm. Find the fringe width on a screen 2.0 m away from the slits.

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(ii) Fringe width

$$\beta = \frac{\lambda D}{d}$$

$$d = 1.0 \times 10^{-3} \text{ mm}$$

$$\lambda = 600 \times 10^{-9} \text{ m}$$

$$D = 2 \text{ m}$$

$$\beta = \frac{600 \times 10^{-9} \times 2}{1 \times 10^{-3}}$$

$$= 1200 \times 10^{-6}$$

$$= 1.2 \text{ mm}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

35. Diffraction of light : According to rectilinear propagation of light, light travels in a straight line. But Italian scientist Grimaldi discovered that light bends near the edges of the aperture of the obstacles whose size is comparable with the wavelength of light. This phenomenon of bending of light around the corners of an obstacle is called diffraction of light. If we place a screen in front of the slit, the diffraction pattern observed on it is of unequal widths and unequal intensities. The diffraction phenomenon can be explained on the basis of wave nature of light.

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Based on the above facts, answer the following questions :

- (i) What is the most essential condition for observing diffraction ?

- (ii) Which characteristic of light is used in X-ray crystallography ? 1
- (iii) How does the wave theory of light provide an explanation for diffraction ? Explain. 2

OR

- (iii) In a single slit diffraction, if red light is replaced by blue light, then how does the diffraction pattern change ? 2

(i) The most essential condition for observing diffraction pattern is that the slit width has to be comparable to the wavelength of light.	1
(ii) Wave nature characteristic of light is used in X-ray crystallography	1
(iii) Light bends near the edges of an aperture. All points of the aperture act as new sources of light. These are known as secondary wavelets. The overlapping of the secondary wavelets results in diffraction pattern.	2
OR	
(iii) Since wavelength of blue light is lesser than that of red light, so angular width of diffraction pattern is less when blue light is used. So the diffraction pattern becomes narrower.	2

1. In a Young's double-slit experiment, the fringe width is found to be β . If the entire apparatus is immersed in a liquid of refractive index μ , the new fringe width will be :

- (a) β (b) $\mu\beta$ (c) $\frac{\beta}{\mu}$ (d) $\frac{\beta}{\mu^2}$

1. In a Young's double-slit experiment, the fringe width is found to be β . If the entire apparatus is immersed in a liquid of refractive index μ , the new fringe width will be :

- (a) β (b) $\mu\beta$ (c) $\frac{\beta}{\mu}$ (d) $\frac{\beta}{\mu^2}$

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(c) $\frac{\beta}{\mu}$

9. A plane wavefront is incident on a concave mirror of radius of curvature R . The radius of the refracted wavefront will be :

- (a) $2R$ (b) R (c) $\frac{R}{2}$ (d) $\frac{R}{4}$

9. A plane wavefront is incident on a concave mirror of radius of curvature R. The radius of the refracted wavefront will be :
- (a) $2R$ (b) R (c) $\frac{R}{2}$ (d) $\frac{R}{4}$

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(c) $\frac{R}{2}$ for students who have opted to answer the question in

Hindi medium only.

English medium students-There is misprint in the English version of the question as the word ‘**reflected**’ appear as ‘**refracted**’. **Therefore full mark to be awarded to each student who have opted to answer the question in English medium.**

- 28.** (a) (i) In diffraction due to a single slit, the phase difference between light waves reaching a point on the screen is 5π . Explain whether a bright or a dark fringe will be formed at the point.
- (ii) What should the width (a) of each slit be to obtain eight maxima of two double-slit patterns (slit separation d) within the central maximum of the single slit pattern ?
- (iii) Draw the plot of intensity distribution in a diffraction pattern due to a single slit.

28. (a) (i) In diffraction due to a single slit, the phase difference between light waves reaching a point on the screen is 5π . Explain whether a bright or a dark fringe will be formed at the point.
- (ii) What should the width (a) of each slit be to obtain eight maxima of two double-slit patterns (slit separation d) within the central maximum of the single slit pattern ?
- (iii) Draw the plot of intensity distribution in a diffraction pattern due to a single slit.

CBSE 23

3

(i) For Bright fringe , $\phi = (2n+1)\pi = 5\pi$ for $n=2$

Alternatively:

$$\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x$$

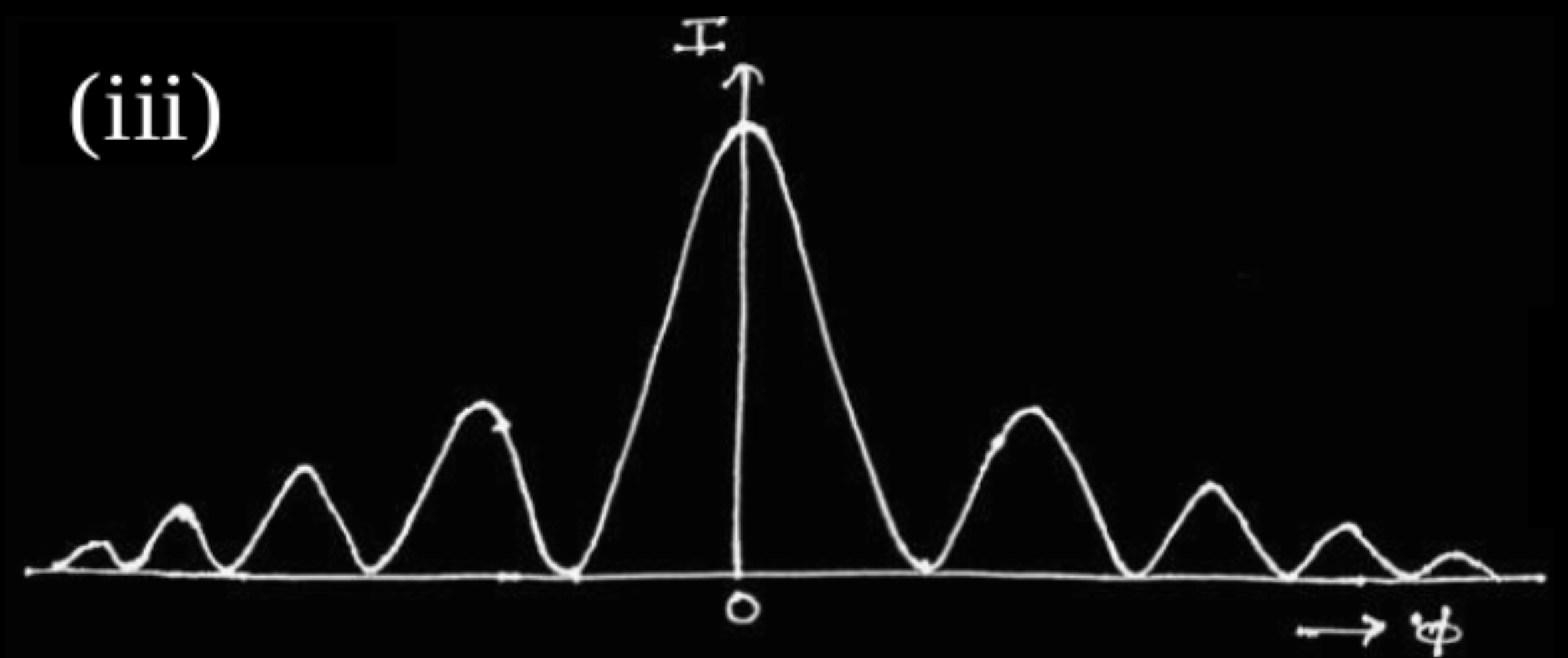
$$\Delta x = \frac{5}{2} \lambda$$

(ii) We want , $a\theta = \lambda$, $\theta = \lambda/a$

$$8 \frac{\lambda}{d} = 2 \frac{\lambda}{a} \Rightarrow a = \frac{d}{4}$$

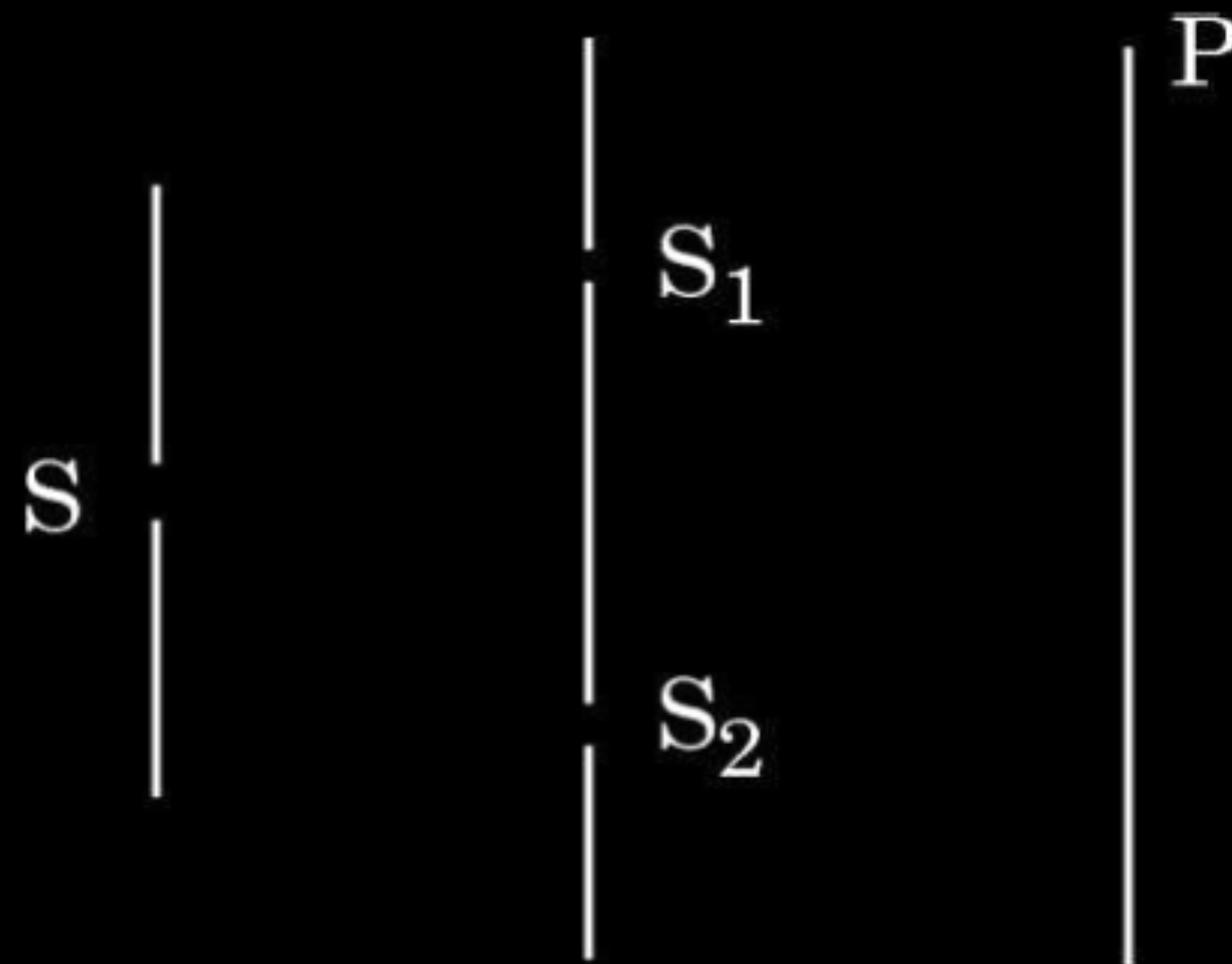
1

1



1

- (b) (i) In a Young's double-slit experiment $SS_2 - SS_1 = \frac{\lambda}{4}$, where S_1 and S_2 are the two slits as shown in the figure. Find the path difference $(S_2P - S_1P)$ for constructive and destructive interference at P.



CBSE 23

- (ii) What is the effect on the interference fringes in a Young's double-slit experiment, if the monochromatic source S is replaced by a source of white light ?

(i) Net path difference between two waves reaching on the screen through two slits is given as

$$SS_2P - SS_1P = (SS_2 - SS_1) + (S_2P - S_1P)$$

$$\Delta x = \frac{\lambda}{4} + (S_2P - S_1P)$$

For constructive interference $\Delta x = n\lambda$

$$\therefore (S_2P - S_1P) = n\lambda - \frac{\lambda}{4} = \lambda \left(n - \frac{1}{4} \right)$$

For destructive interference

$$\Delta x = (2n - 1) \frac{\lambda}{2}$$

$$\Delta x = \frac{\lambda}{4} + (S_2P - S_1P)$$

$$S_2P - S_1P = (2n - 1) \frac{\lambda}{2} - \frac{\lambda}{4}$$

$$= \frac{\lambda}{4} [4n - 3]$$

(ii) If monochromatic source is replaced by a source of white light the centre bright fringe is white.

1

1

1

6. In the wave picture of light, the intensity I of light is related to the amplitude A of the wave as :

(a) $I \propto \sqrt{A}$

(b) $I \propto A$

(c) $I \propto A^2$

(d) $I \propto \frac{1}{A^2}$

6. In the wave picture of light, the intensity I of light is related to the amplitude A of the wave as :

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CBSE 23

(c) $I \propto A^2$

- 12.** In a single-slit diffraction experiment, the width of the slit is halved. The width of the central maximum, in the diffraction pattern, will become :
- (a) half
 - (b) twice
 - (c) four times
 - (d) one-fourth

- 12.** In a single-slit diffraction experiment, the width of the slit is halved. The width of the central maximum, in the diffraction pattern, will become :
- (a) half
 - (b) twice
 - (c) four times
 - (d) one-fourth

CBSE 23

(b) twice

23. (a) State Huygens' principle. How did Huygens explain the absence of the backwave ?

CBSE 23

2

23. (a) State Huygens' principle. How did Huygens explain the absence of the backwave ?

CBSE 23

2

Statement: Each point of the wavefront is the source of secondary disturbance in all directions.

$\frac{1}{2}$

Common tangent to all the secondary wavelets gives new position of the wavefront.

$\frac{1}{2}$

Explanation: Light energy cannot travel in backward direction.

1

Alternatively:

It was an adhoc assumption .

Alternatively:

For back wave: $I = \frac{1}{2} (1 + \cos \theta)$

at $\theta = 180^\circ$; contribution is zero.

Alternatively:

Amplitude of secondary wavelets is maximum in forward direction and zero in backward in direction.

Note: If any other relevant explanation given, give full credit.

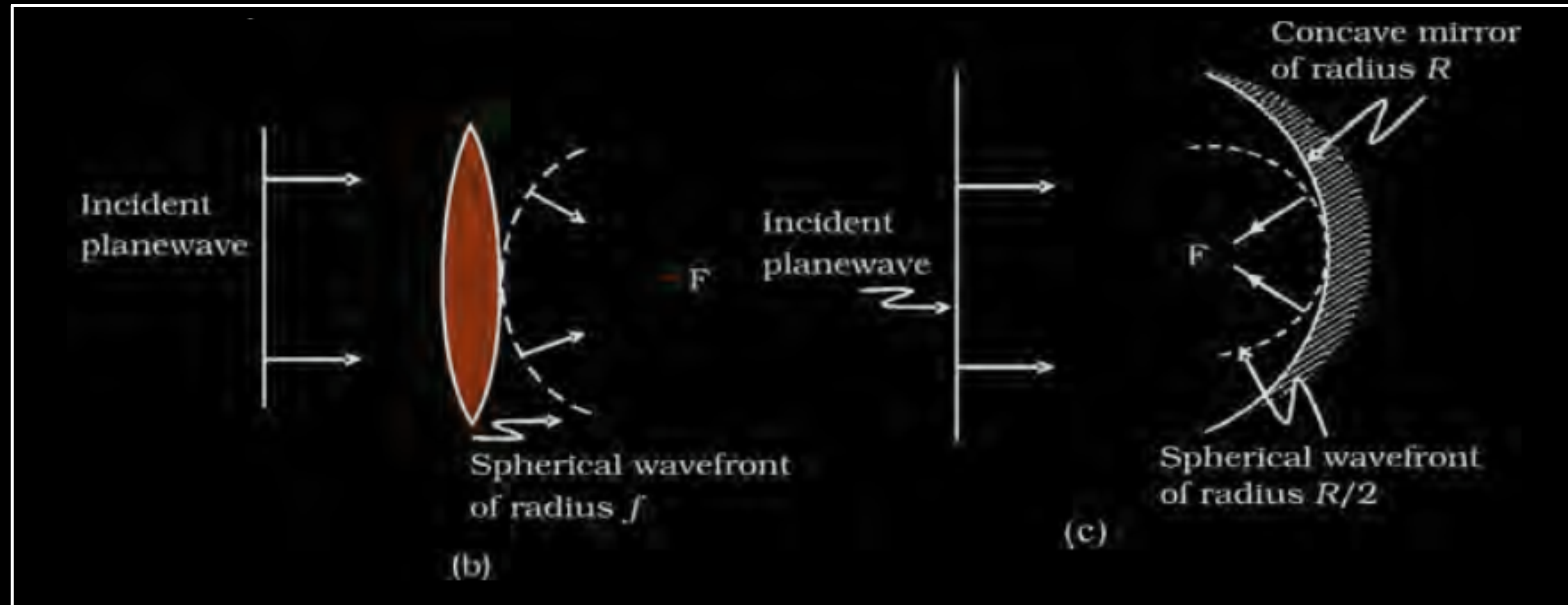
- (b) Use Huygens' principle to show reflection/refraction of a plane wave by (i) concave mirror, and (ii) a convex lens.

CBSE 23

(b) Use Huygens' principle to show reflection/refraction of a plane wave by (i) concave mirror, and (ii) a convex lens.

CBSE 23

2



- 35.** Diffraction of light is bending of light around the corners of an object whose size is comparable with the wavelength of light. Diffraction actually defines the limits of ray optics. This limit for optical instruments is set by the wavelength of light. An experimental arrangement is set up to observe the diffraction pattern due to a single slit.

Answer the following questions based on the above :

- (a) How will the width of central maximum be affected if the wavelength of light is increased ? 1
- (b) Under what condition is the first minimum obtained ? 1
- (c) Write two points of difference between interference and diffraction patterns. 2

OR

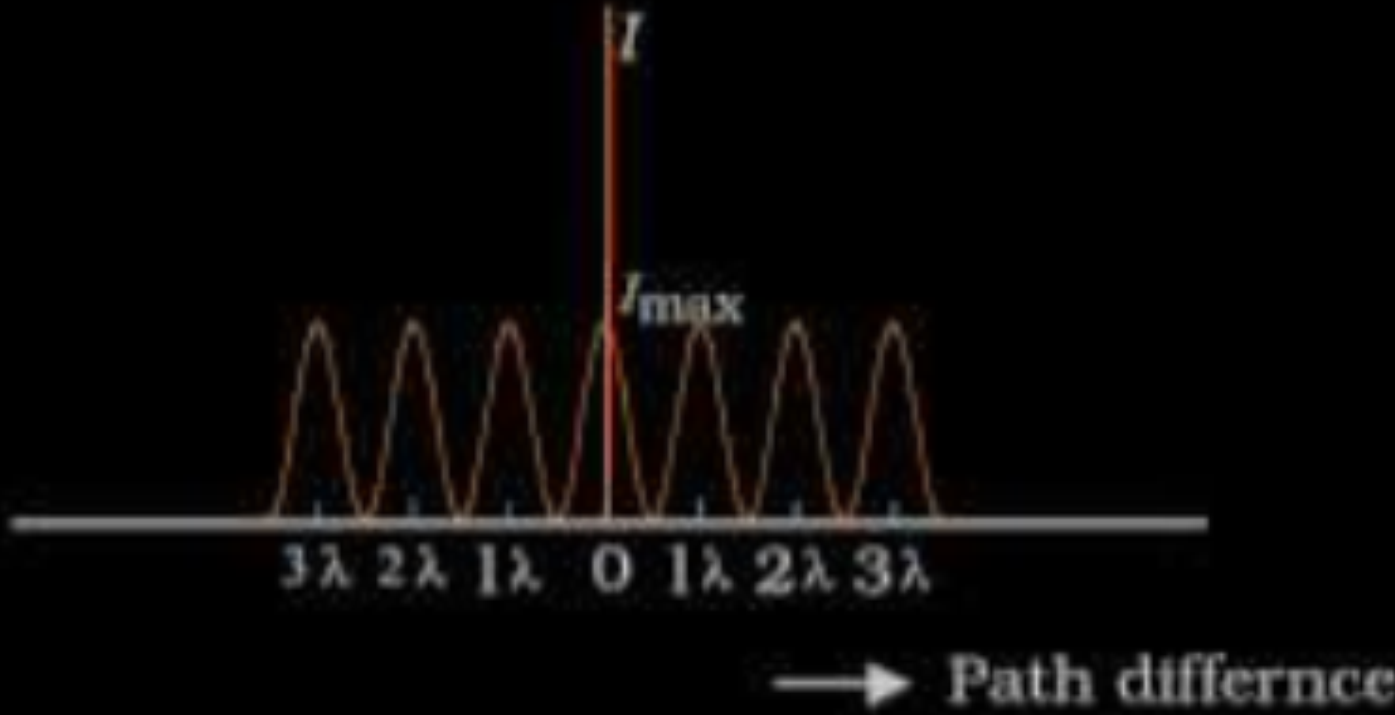
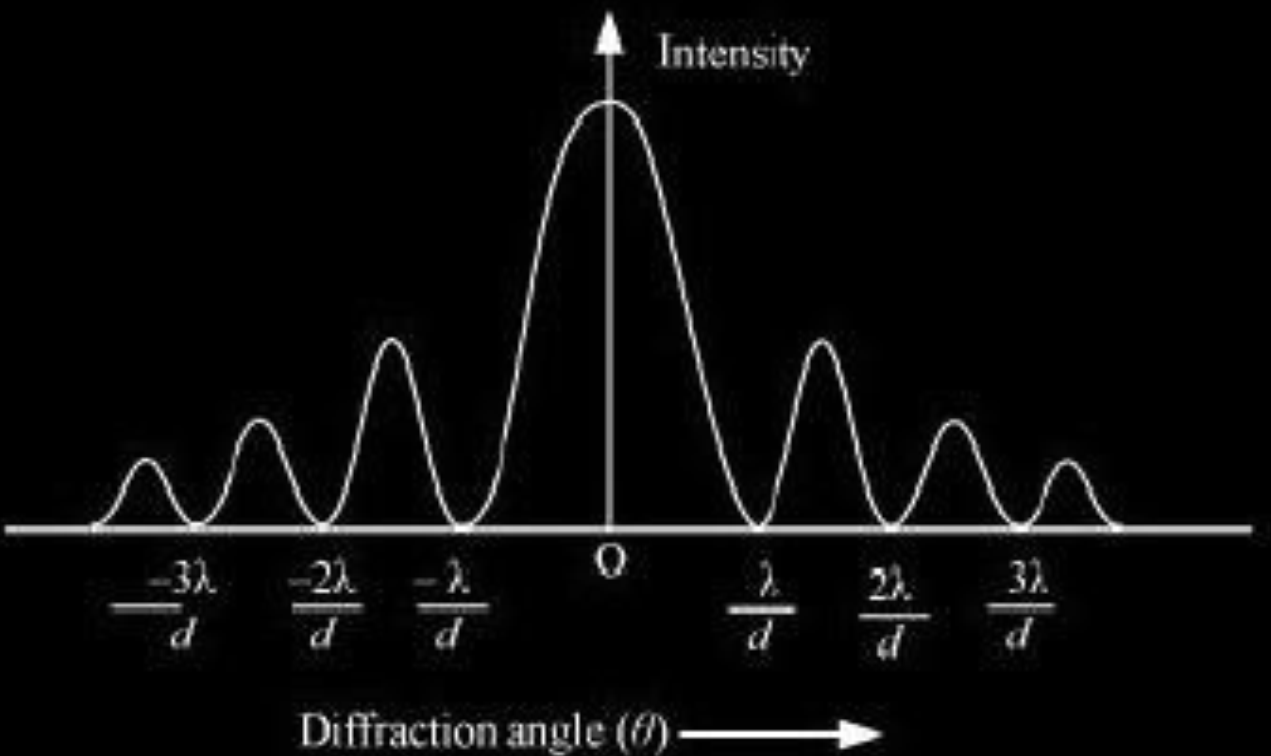
- (c) Two students are separated by a 7 m partition wall in a room 10 m high. If both light and sound waves can bend around obstacles, how is it that the students are unable to see each other even though they can converse easily ? 2

(a) $\beta_o \propto \lambda$

β_o will increase with increase in wavelength.

(b) When path difference $a \theta = \lambda$ or at an angle ; $\theta \approx \lambda / a$

(c) Differences

Interference Pattern	Diffraction Pattern
(i) All the maxima are equally spaced. (ii) The dark fringe is having zero intensity. (iii) All the maxima are of the same intensity. (iv)  Alternatively:- -It is obtained by the superposition of two waves originating from two sources/slits.	(i) Width of Central bright maxima is twice the width of the other maxima. (ii) The dark fringe is not completely dark. (iii) There is a sharp decrease in the intensity of maxima after the central bright maxima. (iv)  It is obtained by the superposition of waves from points on a single slit.

1

1

1+1

18. *Assertion (A) :* The phase difference between any two points on a wavefront is zero.

Reason (R) : All points on a wavefront are at the same distance from the source and thus oscillate in the same phase.

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Reason (R) : All points on a wavefront are at the same distance from the source and thus oscillate in the same phase.

CBSE 23

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correctly explanation of the Assertion (A).

- 23.** (a) Using Huygens' principle, draw a ray diagram showing the propagation of a plane wave refracting at a plane surface separating two media. Also verify Snell's law of refraction.

2

CBSE 23

23. (a) Using Huygens' principle, draw a ray diagram showing the propagation of a plane wave refracting at a plane surface separating two media. Also verify Snell's law of refraction.

2

CBSE 23

AB is incident wave front, incident at an angle i . Let τ be time taken by the wave front to travel distance BC.

$BC = v_1 \tau$ where v_1 is speed of wave in medium 1.

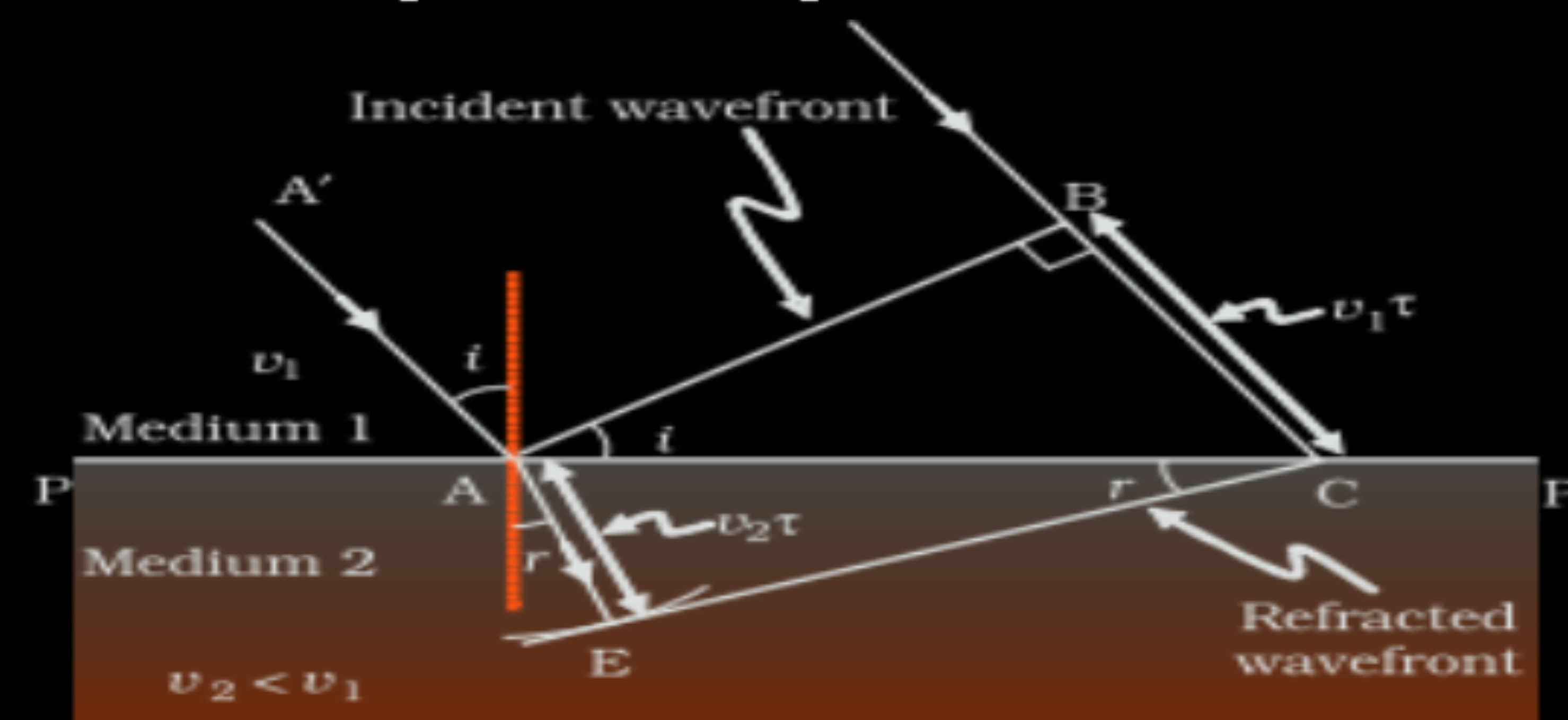
To determine shape of refracted wave front, we draw a sphere of radius $v_2 \tau$, where v_2 is speed of wave in medium 2.

CE represents a tangent drawn from point C on sphere, CE is the refracted wave front.

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$$

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = n_{21}$$



$\frac{1}{2}$

$\frac{1}{2}$

1

Note: Give full credit if student derives Snell's law by taking incident wave front in denser medium.

- 31.** (a) (i) (1) Write two points of difference between an interference pattern and a diffraction pattern.
- (2) Name any two factors on which the fringe width in a Young's double-slit experiment depends.
- (ii) In Young's double-slit experiment, the two slits are separated by a distance equal to 100 times the wavelength of light that passes through the slits. Calculate :
- (1) the angular separation in radians between the central maximum and the adjacent maximum.
- (2) the distance between these two maxima on a screen 50 cm from the slits.

(i) (1) (a) The interference pattern has a number of equally spaced bright and dark bands while diffraction pattern has a central bright maximum which is twice as wide as the other maxima. (b) Interference pattern is obtained by superposing two waves originating from two narrow slits, while diffraction pattern is a superposition of a continuous family of waves originating from each point on a single slit. (c) The maxima in interference pattern is obtained at angle λ / a, while the first minima is obtained at same angle λ / a for diffraction pattern. (d) in interference pattern the intensity of bright fringes remain same while in diffraction the intensity falls as we go to successive maxima away from the center on either side. (any two)	1+1	(ii) (1) $d \sin \theta = n \lambda$ $n=1$ $\sin \theta = \frac{\lambda}{d}$ For small angle $\sin \theta \approx \theta = \frac{\lambda}{100\lambda} = \frac{1}{100}$ radian. (2) $\beta = \frac{\lambda D}{d} = \theta D$ $= \frac{1}{100} \times 50 \times 10^{-2}$ $= 50 \times 10^{-4} m$ $= 5 \text{ mm}$
(2) Factors affecting fringes width Wave length (λ) / distance of screen from slits (D) / separation between slits (d). (any two)		

16. **Assertion (A)** : In Young's double slit experiment all fringes are of equal width.

Reason (R) : The fringe width depends upon wavelength of light (λ) used, distance of screen from plane of slits (D) and slits separation (d).

1

16. **Assertion (A)** : In Young's double slit experiment all fringes are of equal width.

Reason (R) : The fringe width depends upon wavelength of light (λ) used, distance of screen from plane of slits (D) and slits separation (d).

1

CBSE 23

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

19. What happens to the interference pattern when two coherent sources are

(a) infinitely close, and

(b) far apart from each other

19. What happens to the interference pattern when two coherent sources are
- (a) infinitely close, and
 - (b) far apart from each other

CBSE 23

2

(a) When ' d ' is very small, $\beta \propto \frac{1}{d}$, β will be very large and a single patch will occupy the whole field of view hence pattern cannot be observed.

Alternatively

Give full credit if a candidate writes that the fringe width will increase or the fringes will not be observed.

(b) When sources are far apart, i.e. d is very large, then fringe width will be so small that the fringes are not resolved and cannot be seen separately.

Alternatively

Give full credit if a candidate writes that the fringe width will decrease or the fringes may not be observed.

1

1

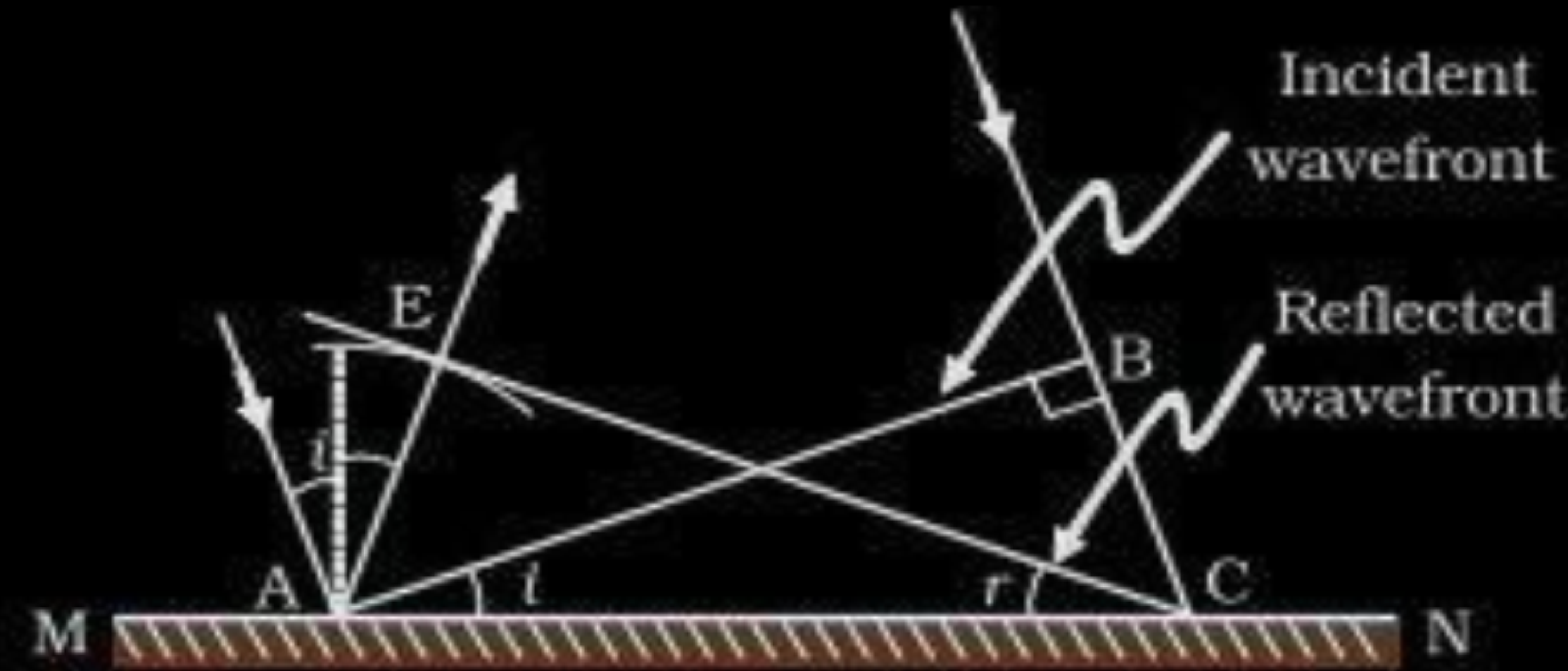
32. (a) (i) State Huygen's principle. With the help of a diagram, show how a plane wave is reflected from a surface. Hence verify the law of reflection.

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CBSE 23

(i) Huygen's principle

Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets emanating from the wavefront are usually referred to as secondary wavelets, a common tangent to all these spheres gives the new position of the wavefront at a later time.



Verification of law of reflection

In $\triangle AEC$ & $\triangle CBA$

$$EC = AB \quad (\text{c x t each})$$

$$\angle AEC = \angle CBA \quad (90^\circ \text{ each})$$

$$AC = AC \quad (\text{common side})$$

By RHS congruency $\triangle AEC \cong \triangle CBA$

$$\Rightarrow \angle i = \angle r$$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

19. In a Young's double slit experiment, the separation between the two slits is d and distance of the screen from the slits is $1000 d$. If the first minima falls at a distance d from the central maximum, obtain the relation between d and λ .

CBSE 23

2

19. In a Young's double slit experiment, the separation between the two slits is d and distance of the screen from the slits is $1000 d$. If the first minima falls at a distance d from the central maximum, obtain the relation between d and λ .

CBSE 23

2

$$D=1000 d$$

$$x_{1d} = d$$

Position of 1st minima.

$$x_{1d} = \frac{3}{2} \cdot \frac{\lambda D}{d}$$

$$d = \frac{3}{2} \cdot \frac{\lambda \times 1000d}{d}$$

$$\therefore d = 1500\lambda$$

$\frac{1}{2}$

1

$\frac{1}{2}$

6. In a Young's double-slit experiment, the screen is moved away from the plane of the slits. What will be its effect on the following ?

1

- (i) Angular separation of the fringes.
- (ii) Fringe-width.
- (A) Both (i) and (ii) remain constant.
- (B) (i) remains constant, but (ii) decreases.
- (C) (i) remains constant, but (ii) increases.
- (D) Both (i) and (ii) increase.

CBSE 23

14. In an interference experiment, third bright fringe is obtained at a point on the screen with a light of 700 nm. What should be the wavelength of the light source in order to obtain the fifth bright fringe at the same point ?

1

(A) 420 nm

(B) 750 nm

(C) 630 nm

(D) 500 nm

CBSE 23

26. (a) A plane wave-front propagating in a medium of refractive index ' μ_1 ' is incident on a plane surface making an angle of incidence (i). It enters into a medium of refractive index μ_2 ($\mu_2 > \mu_1$).

Use Huygen's construction of secondary wavelets to trace the refracted wave-front. Hence verify Snell's law of refraction.

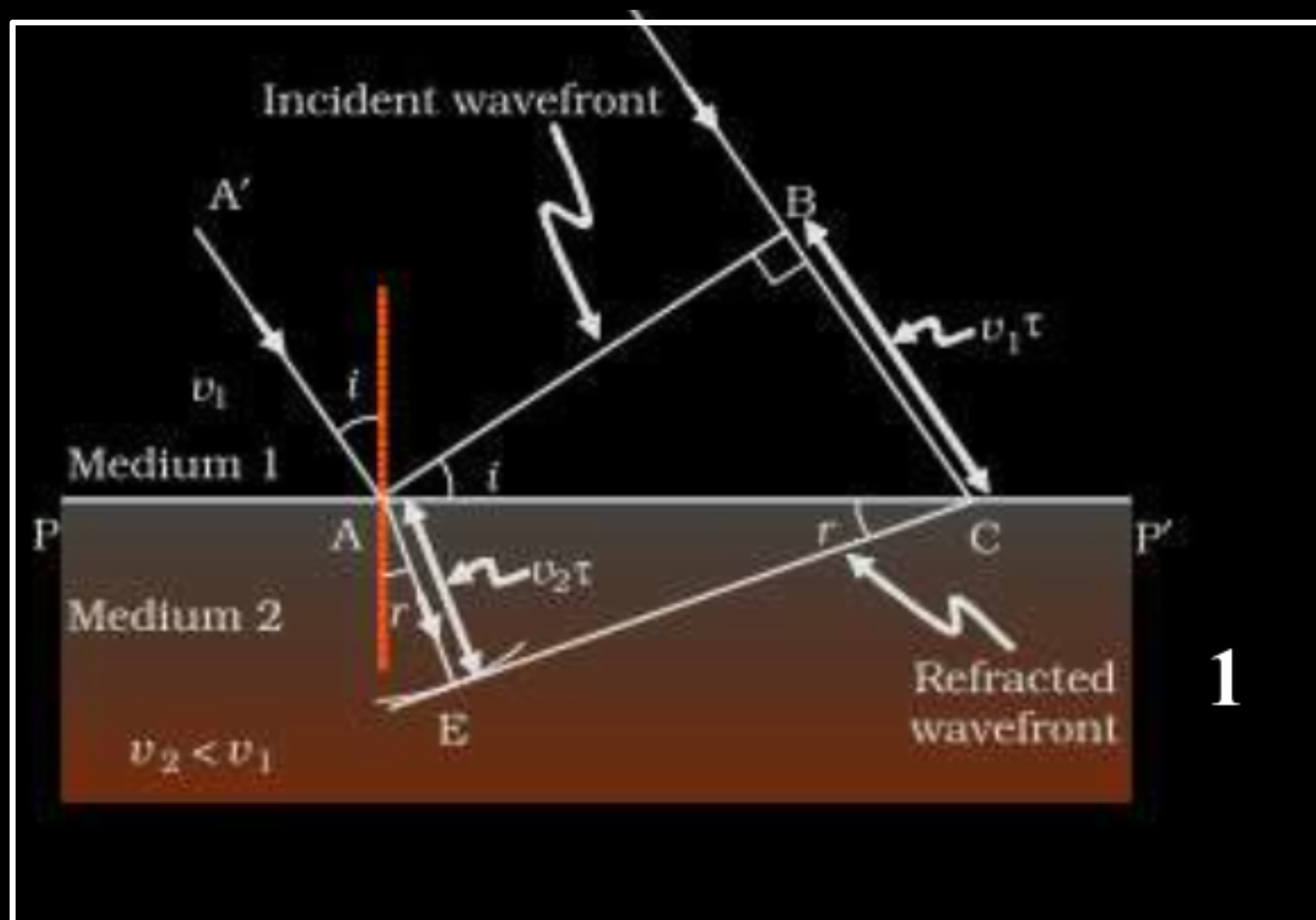
CBSE 23

3

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Use Huygen's construction of secondary wavelets to trace the refracted wave-front. Hence verify Snell's law of refraction. CBSE 23

3



AB is incident wavefront, incident at an angle i . let τ be the time taken by wavefront to travel distance BC.

$BC = v_1 \tau$ where v_1 is speed of wave in medium 1

To determine shape of refracted wavefront, we draw a sphere of radius $v_2 \tau$, where v_2 is speed of wave in medium 2.

CE represents a tangent drawn from point C on sphere, CE is the refracted wavefront.

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC} \quad \text{and}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$$

Where i and r are the angles of incidence and refraction, respectively.

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$$

$$\mu_1 \sin i = \mu_2 \sin r$$

This is the Snell's law of refraction.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b) Using Huygen's construction, show how a plane wave is reflected from a surface. Hence verify the law of reflection.

CBSE 23

3

(b) Using Huygen's construction, show how a plane wave is reflected from a surface. Hence verify the law of reflection.

CBSE 23

3

Consider a plane wave AB incident at an angle i on a reflecting surface MN. If v is the speed of the wave in the medium and if τ represents the time taken by the wavefront to advance from the point B to C then the distance

$$BC = v \tau$$

In order to construct the reflected wavefront, draw a sphere of radius $v\tau$ from the point A. Let CE represent the tangent plane, drawn from the point C to this sphere.

In $\triangle AEC$ and $\triangle ABC$

$$AE = BC = v \tau$$

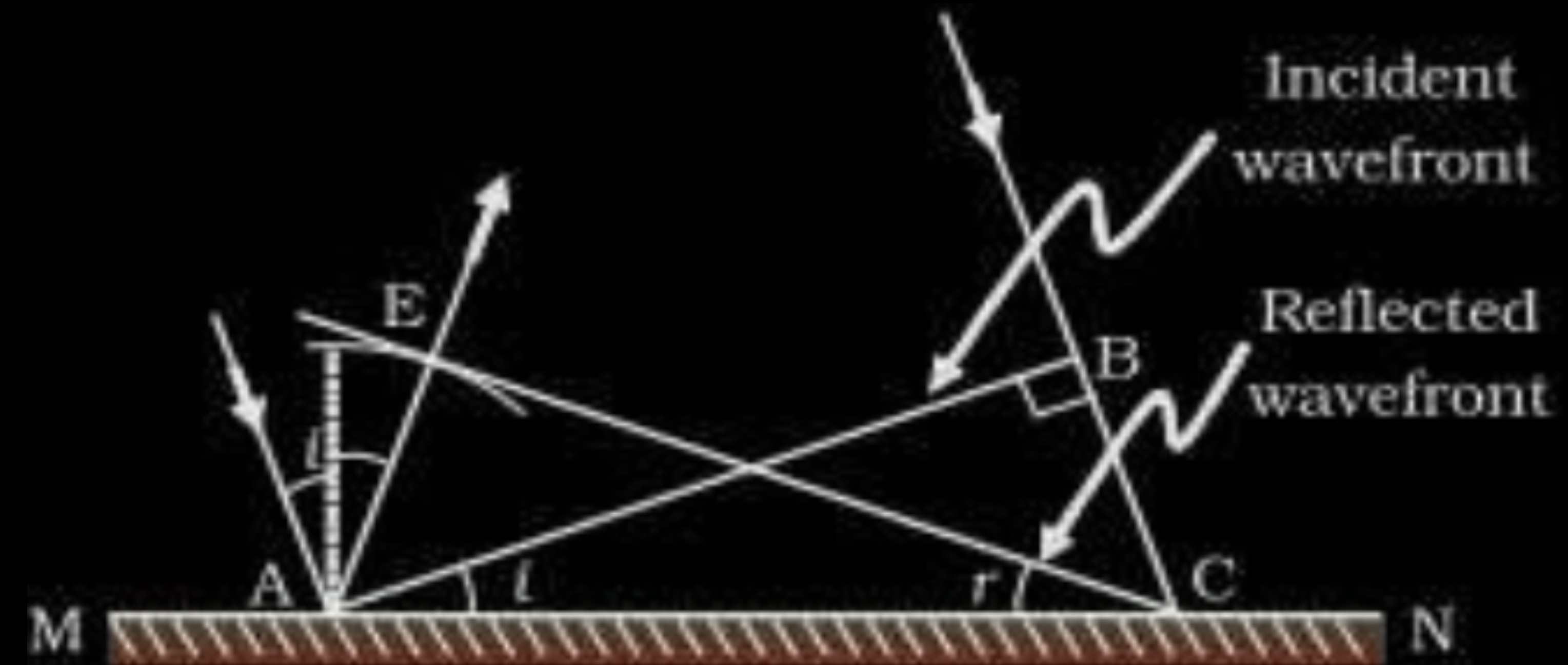
$$\angle CEA = \angle ABC \text{ (90}^\circ \text{ each)}$$

AC is common side

$$\triangle AEC \cong \triangle ABC$$

$$\therefore \angle i = \angle r$$

This is the law of reflection.



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

14. According to Huygens principle, the amplitude of secondary wavelets is **1**

- (A) equal in both the forward and the backward directions.
- (B) maximum in the forward direction and zero in the backward direction.
- (C) large in the forward direction and small in the backward direction.
- (D) small in the forward direction and large in the backward direction.

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- (C) large in the forward direction and small in the backward direction.
- (D) small in the forward direction and large in the backward direction.

CBSE 23

(B) Maximum in the forward direction and zero in the backward direction.

- (b) (i) How will the interference pattern in Young's double slit experiment be affected if the monochromatic source is replaced by a source of white light ?
- (ii) In Young's double slit experiment, the two slits are 0.18 mm apart and the screen is placed 1.4 m away from the plane of slits. If the distance between the central bright fringe and the third bright fringe is 1.4 cm, then calculate the wavelength of light used. 3

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CBSE 22 C

(i)	The central maxima will be white followed by coloured fringes.	$\frac{1}{2} + \frac{1}{2}$
(ii)	$x = \frac{n\lambda D}{d}$	1
	$1.4 \times 10^{-2} = \frac{3 \times \lambda \times 1.4}{0.18 \times 10^{-3}}$	$\frac{1}{2}$
	$\lambda = 6 \times 10^{-7} \text{ m} = 600 \text{ nm}$	$\frac{1}{2}$

(b) For diffraction due to a single slit :

- (i) Briefly explain the conditions for observing maximum and minimum intensity at a point on the screen.
- (ii) Why does the intensity of maxima in the diffraction pattern decrease sharply as their order (n) increases ?

CBSE 22 C

(i) When the path difference $(p) = d \sin \theta = (2n+1)\frac{\lambda}{2}$, then the secondary wavelets starting from consecutive parts of slit having a path difference of $\lambda/2$ except the last part, cancel each other contribution, leading to formation of maxima at a point on the screen.	1
When the path difference $(p) = d \sin \theta = n\lambda$, then secondary wavelets starting from consecutive parts of the slit having path difference of $\lambda/2$ cancel each other contribution leading to formation of minima at a point on the screen.	1
(ii) Maximum intensity is obtained at a point on the screen when the path difference is $(2n+1)\frac{\lambda}{2}$. With increasing 'n' only one- third, one-fifth, one-seventh etc of the slit contribute, hence intensity of maxima decreases sharply.	1
Alternatively With the increase of order (n) the number of secondary wavelets responsible for the formation of secondary maxima, decreases resulting in sharp decrease of intensity.	

12. The principle of superposition is an important principle in Physics. It helps us to understand many phenomena, for example interference of sound and light waves. According to this principle, the resultant displacement produced at a point, by a number of waves, is the vector sum of the displacements produced by each of the waves. Two light waves, originating from two coherent sources produce a stable interference, due to superposition. Thomas Young in 1801, used an innovative technique to obtain two coherent sources and observed interference of light waves coming from them, on a screen suitably placed away from the plane of the sources. The pattern consists of alternate bright and dark fringes. CBSE 22 C

(a) Two waves superimpose to produce stable interference.

The waves must have :

- (i) same amplitudes and varying phase difference.
- (ii) different amplitudes and varying phase difference.
- (iii) different wavelengths and constant phase difference.
- (iv) same wavelengths and constant phase difference.

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(b) Light added to light can produce darkness. This phenomenon is known as :

(i) Dispersion

(ii) Scattering

(iii) Interference

(iv) Polarization

(b) Light added to light can produce darkness. This phenomenon is known as :

(i) Dispersion

(ii) Scattering

(iii) Interference

(iv) Polarization

(c) Two waves of intensity I_0 each, superimpose and produce interference pattern. If the phase difference between them at a point is $\left(\frac{3\pi}{2}\right)$, then the resultant intensity at the point will be :

(i) $\frac{I_0}{4}$

(ii) $\frac{I_0}{2}$

(iii) $2I_0$

(iv) $4I_0$

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(i) $\frac{I_0}{4}$

(ii) $\frac{I_0}{2}$

(iii) $2I_0$

(iv) $4I_0$

(d) In Young's double slit experiment, with light of wavelength λ , the slit separation $d = n\lambda$. If the distance of the screen from the plane of the slits $D = n^3\lambda$, then the fringe width will be :

(i) $\frac{\lambda}{n^2}$

(ii) $\frac{\lambda}{n}$

(iii) $n\lambda$

(iv) $n^2\lambda$

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(i) $\frac{\lambda}{n^2}$

(ii) $\frac{\lambda}{n}$

(iii) $n\lambda$

(iv) $n^2\lambda$

(e) In Young's double slit experiment, the source used produces lights of wavelengths 450 nm and 600 nm. On the screen, the third maximum corresponding to 600 nm coincides with n^{th} maxima corresponding to 450 nm, where n equals :

(i) 1

(ii) 2

(iii) 3

(iv) 4

5×1=5

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(i) 1

(ii) 2

(iii) 3

(iv) 4

5×1=5

4. (a) In single slit diffraction, explain why the maxima at $\theta = \left(n + \frac{1}{2}\right) \left(\frac{\lambda}{a}\right)$ becomes weaker and weaker as n increases. State two important differences between interference and diffraction pattern.

Maximum intensity is obtained at a point on the screen when the path difference is

$(2n+1) \frac{\lambda}{2}$. With increasing 'n' only one- third, one-fifth, one-seventh etc

of the slit contribute, hence intensity of maxima decreases sharply.

Alternatively

With the increase of order (n) the number of secondary wavelets responsible for the formation of secondary maxima, decreases resulting in sharp decrease of intensity.

Differences (any two)

1. Diffraction is a pattern formed as a result of superposition of waves from different portions of the same wave front. Interference is a pattern on a screen a result of superposition of single slit diffraction from two slits.
2. In diffraction pattern width of central maximum is twice the width of secondary maxima. In interference pattern width of each maxima is same.
3. In diffraction pattern intensity of maxima goes on decreasing as we move away from centre maximum. In interference pattern intensity of all maxima is same.
4. In diffraction pattern there is no absolute minima. In interference pattern absolute minima depends on amplitude of waves superposing.

1

1+1

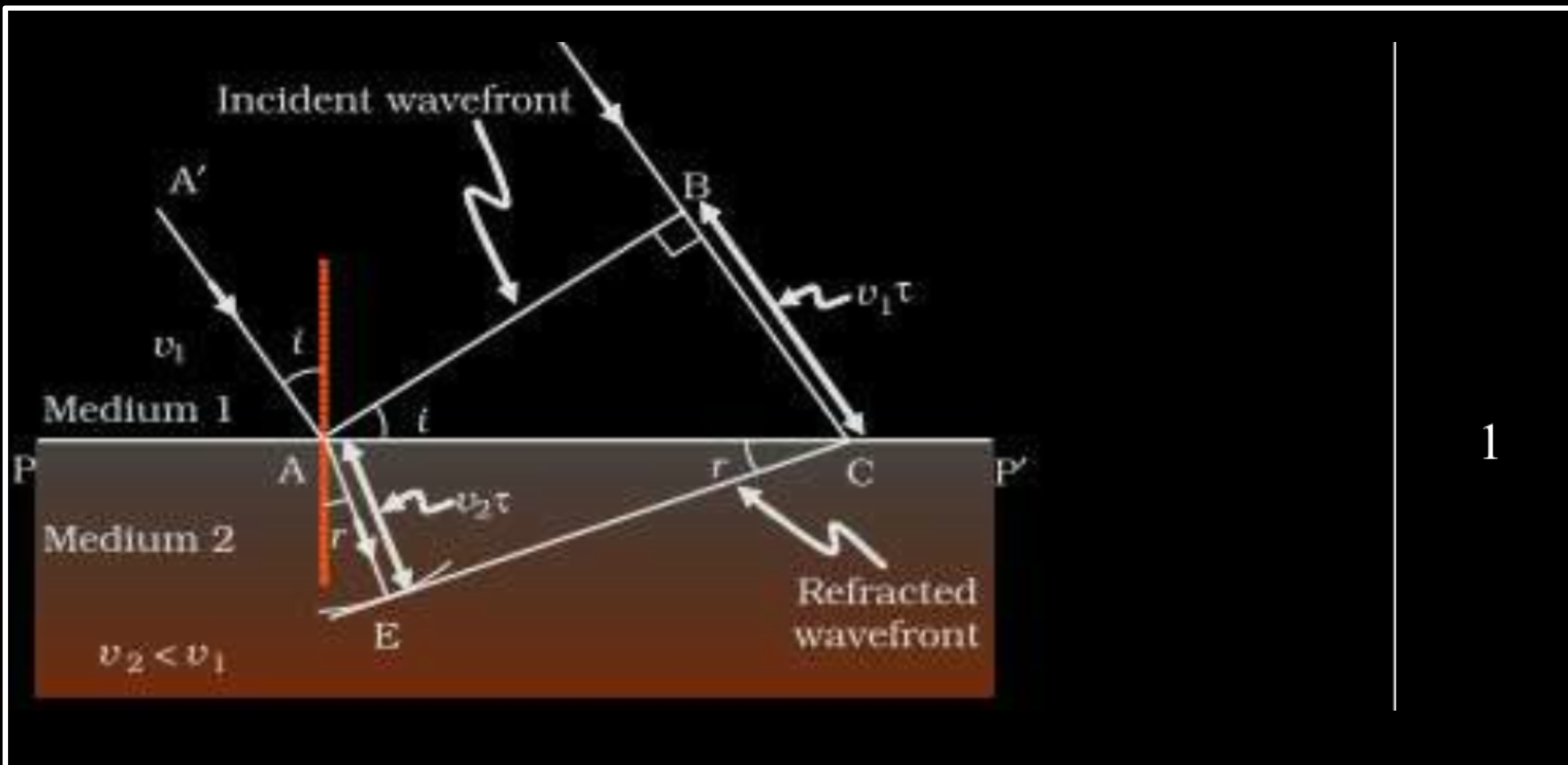
- (b) A plane wavefront is incident on a surface separating two media of refractive indices n_1 and n_2 ($> n_1$). With the help of a suitable diagram, explain its propagation from the rarer to denser medium. Hence, verify Snell's law.

CBSE 22 C

- (b) A plane wavefront is incident on a surface separating two media of refractive indices n_1 and n_2 ($> n_1$). With the help of a suitable diagram, explain its propagation from the rarer to denser medium. Hence, verify Snell's law.

CBSE 22 C

3



1

Consider the triangles ABC and AEC, we readily obtain

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$$

And

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$$

where i and r are the angles of incidence and refraction, respectively.

Thus we obtain

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} \quad \text{-----(1)}$$

If c represents the speed of light in vacuum, then,

$$n_1 = \frac{c}{v_1}$$

and

$$n_2 = \frac{c}{v_2}$$

are known as the refractive indices of medium 1 and medium 2, respectively. In terms of the refractive indices, eq. (1) can be written as

$$n_1 \sin i = n_2 \sin r$$

This is the Snell's law of refraction.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

8. (a) In a Young's double slit experimental set-up, the intensity of the light waves from two coherent sources are in the ratio of 4 : 9. Find the ratio of intensity of bright and dark fringes in the interference pattern. 2
- (b) Light travels from a rarer to a denser medium. How will
- (i) the energy carried, and
 - (ii) the wavelength of the light wave,
- be affected ? CBSE 22 C 1

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- (b) Light travels from a rarer to a denser medium. How will
- (i) the energy carried, and
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- be affected ? CBSE 22 C 1

(a) $r = \frac{A_1}{A_2} = \sqrt{\frac{I_1}{I_2}} = \frac{2}{3}$	1
$\frac{I_{\max}}{I_{\min}} = \left(\frac{r+1}{r-1} \right)^2 = \frac{25}{1}$	1
(b) (i) No effect	$\frac{1}{2}$
(c) (ii) Wavelength decreases	$\frac{1}{2}$

5. In Young's double slit experiment, the slit separation is 1 mm and the screen is 1 m away from the slits. For a monochromatic light of wavelength 500 nm, find
- (a) the distance of third minima, and
 - (b) the distance of second maxima,
- from the central maxima.

5. In Young's double slit experiment, the slit separation is 1 mm and the screen is 1 m away from the slits. For a monochromatic light of wavelength 500 nm, find

(a) the distance of third minima, and

(b) the distance of second maxima,

from the central maxima.

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3

a) $x = (2n + 1) \frac{\lambda D}{2d}$	$\frac{1}{2}$
$= \frac{(2 \times 2 + 1) \times 500 \times 1 \times 10^{-9}}{2 \times 10^{-3}}$	$\frac{1}{2}$
$= \frac{5 \times 5 \times 10^{-4}}{2} = 12.5 \times 10^{-4} \text{ m} = 1.25 \text{ mm}$	$\frac{1}{2}$
b) $x = \frac{n\lambda D}{d}$	$\frac{1}{2}$
$x = \frac{2 \times 500 \times 1 \times 10^{-9}}{10^{-3}}$	$\frac{1}{2}$
$x = 10^{-3} \text{ m} = 1 \text{ mm}$	$\frac{1}{2}$

5. (a) In a Young's double slit experimental set-up, the intensity of the light waves from two coherent sources are in the ratio of 9 : 1. Find the ratio of intensity of bright and dark fringes in the interference pattern. 2
- (b) Monochromatic light of wavelength 600 nm is incident from air on a water surface. If the refractive index of water is $\frac{4}{3}$, then find the wavelength of the refracted light. 1

CBSE 22 C

5. (a) In a Young's double slit experimental set-up, the intensity of the light waves from two coherent sources are in the ratio of 9 : 1. Find the ratio of intensity of bright and dark fringes in the interference pattern. 2
- (b) Monochromatic light of wavelength 600 nm is incident from air on a water surface. If the refractive index of water is $\frac{4}{3}$, then find the wavelength of the refracted light. 1

CBSE 22 C

a)	
$r = \frac{A_1}{A_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{9}{1}} = \frac{3}{1}$	1
$\frac{I_{max}}{I_{min}} = \frac{(r + 1)^2}{(r - 1)^2} = \left(\frac{4}{2}\right)^2 = \frac{4}{1}$	1
b) Wavelength of refracted light	
$\lambda = \frac{\lambda_{air}}{n} = \frac{3 \times 600}{4} = \frac{1800}{4}$	$\frac{1}{2}$
$= 450 \text{ nm}$	$\frac{1}{2}$

- (b) A slit of width 0.6 mm is illuminated by a beam of light consisting of two wavelengths 600 nm and 480 nm. The diffraction pattern is observed on a screen 1.0 m from the slit. Find :
- (i) The distance of the second bright fringe from the central maximum pertaining to light of 600 nm.
 - (ii) The least distance from the central maximum at which bright fringes due to both the wavelengths coincide.

$$(i) \quad \sin \theta \simeq \theta = \frac{x}{D} = \frac{5\lambda}{2a}$$

$\frac{1}{2}$

$$\therefore x = \frac{5\lambda D}{2a} = \frac{5 \times 600 \times 10^{-9} \times 1}{2 \times 0.6 \times 10^{-3}}$$

$\frac{1}{2}$

$$= 2.5 \text{ mm}$$

$\frac{1}{2}$

$$(ii) \quad (2n+1) \frac{\lambda}{2} = \frac{(2(n+1)+1)\lambda'}{2}$$

$\frac{1}{2}$

$$(2n+1) \times 600 = (2n+3) \times 480$$

$$\therefore n = 3.5$$

$$x_n = \frac{(2n+1)\lambda D}{2d} = \frac{(2 \times 3.5 + 1) \times 600 \times 10^{-9} \times 1}{2 \times 0.6 \times 10^{-3}}$$

$\frac{1}{2}$

$$= 4 \text{ mm}$$

$\frac{1}{2}$

(Note: Full credit for finding the position by taking $n = 3$ or $n = 4$

for $n = 3$, $x_n = 3.5 \text{ mm}$ and for $n = 4$, $x_n = 4.5 \text{ mm}$)

4. How will the interference pattern in Young's double-slit experiment be affected if : 3

- (i) The screen is moved away from the plane of the slits.
- (ii) The source slit is moved away from the plane of the slits.
- (iii) The phase difference between the light waves emanating from the two slits S_1 and S_2 changes from 0 to π and remains constant.

i) $\beta = \frac{\lambda D}{d}$ As D increases fringe width increases (Note: Give full credit, if students does not write the formula)	$\frac{1}{2}$ $\frac{1}{2}$
ii) Fringe width remains same but intensity of fringes decreases.	1
<u>Alternatively</u> s=size of source S=distance between source and slit for interference to be seen $\frac{s}{S} < \frac{\lambda}{D}$ remains satisfied Sharpness of fringes increases/interference pattern may disappear. There will be no distinct pattern.	
iii) There will be a dark fringe at the centre instead of a bright fringe. <u>Alternatively</u> As the phase difference changes from 0 to π there will be a shift in pattern on the screen till the central maximum turns to dark fringe.	1

7. How can you differentiate whether a pattern is produced by a single slit or double slits ? Derive the expression for the angular position of (i) bright and (ii) dark fringes produced in a single slit diffraction.

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7. How can you differentiate whether a pattern is produced by a single slit or double slits ? Derive the expression for the angular position of (i) bright and (ii) dark fringes produced in a single slit diffraction.

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In the pattern of fringes produced by a single slit, the central fringe (band) is brighter as compared to other fringes i.e intensity goes on decreasing as the order (n) of the maxima increases, while in the fringe pattern produced by double slits all bright fringes including central fringe are of same intensity.

Alternatively:

In the fringe pattern produced by single slit, the fringe at the centre is wider as compared to the width of other bright fringes, while in the fringe pattern produced by double slits all bright fringes are of equal width.

(Note : Give full credit, for the differentiation by drawing intensity distribution curves of the patterns produced by single slit and double slits)

Calculation of angular position

For the slit of width 'a ' and angle of diffraction 'θ'

Path difference (Δp) = $a \sin \theta$

(i) Condition for Bright Fringe, $\Delta p = (2n+1)\frac{\lambda}{2}$

$\therefore a \sin \theta = (2n+1)\frac{\lambda}{2}$

1

$$\sin \theta = \frac{(2n+1)\lambda}{2a} \quad (n = 1, 2, \dots)$$

for small angle, $\sin \theta \approx \theta$

$$\theta = \frac{(2n+1)\lambda}{2a}$$

 $\frac{1}{2}$

(ii) Condition for dark fringe , $\Delta p = n \lambda$

 $\frac{1}{2}$

$$a \sin \theta = n \lambda \quad (n = 1, 2, \dots)$$

$$\therefore \sin \theta = \frac{n\lambda}{2a}$$

for small angle, $\sin \theta \approx \theta$

$$\theta = \frac{n\lambda}{a}$$

 $\frac{1}{2}$ $\frac{1}{2}$

6. A beam of light consisting of two wavelengths 600 nm and 500 nm is used in a Young's double slit experiment. The slit separation is 1.0 mm and the screen is kept 0.60 m away from the plane of the slits. Calculate :
- (i) the distance of the second bright fringe from the central maximum for wavelength 500 nm, and
 - (ii) the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.

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6. A beam of light consisting of two wavelengths 600 nm and 500 nm is used in a Young's double slit experiment. The slit separation is 1.0 mm and the screen is kept 0.60 m away from the plane of the slits. Calculate :
- the distance of the second bright fringe from the central maximum for wavelength 500 nm, and
 - the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.

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$$\text{i) } x_n = n \frac{\lambda D}{d}$$

$$x_2 = \frac{2 \times 500 \times 10^{-9} \times 0.60}{10^{-3}}$$

$$= 0.6 \text{ mm}$$

$$\text{ii) } n_1 \lambda_1 = n_2 \lambda_2$$

$$n \times 600 = (n + 1) 500$$

$$n = 5$$

$$x_5 = \frac{5 \lambda D}{d}$$

$$x_5 = \frac{5 \times 600 \times 10^{-9} \times 0.60}{10^{-3}}$$

$$= 1.8 \text{ mm}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (b)
 - (i) State two conditions for two light sources to be coherent.
 - (ii) Give two points of difference between an interference pattern due to a double – slit and a diffraction pattern due to a single slit.

8. In a diffraction pattern due to a single slit, how will the angular width of central maximum change, if
- (i) Orange light is used in place of green light,
 - (ii) the screen is moved closer to the slit,
 - (iii) the slit width is decreased ?
- Justify your answer in each case.

8. In a diffraction pattern due to a single slit, how will the angular width of central maximum change, if
- Orange light is used in place of green light,
 - the screen is moved closer to the slit,
 - the slit width is decreased ?
- Justify your answer in each case.

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3

In diffraction pattern the angular width of central maximum $= \frac{2\lambda}{a}$ where a is the slit width and λ is the wavelength.

(i) Increases

$$\text{As } \theta = \frac{2\lambda}{a} \text{ and } \lambda_{\text{orange}} > \lambda_{\text{green}}$$

$\frac{1}{2}$

$\frac{1}{2}$

(ii) No change / no effect

As θ does not depend upon the distance of the screen from the slit(D)

$\frac{1}{2}$

$\frac{1}{2}$

(iii) Increases

As θ is inversely proportional to the slit width(a).

$\frac{1}{2}$

$\frac{1}{2}$

(Note :- Give $\frac{1}{2}$ mark , if only the formula $\theta = \frac{2\lambda}{a}$ is given.)

10. How is the spacing between fringes in a double slit experiment affected if :
- (i) the slits separation is increased,
 - (ii) the colour of light used is changed from red to blue,
 - (iii) the whole apparatus is submerged in a oil of refractive index 1.2 ?
- Justify your answer in each case.

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10. How is the spacing between fringes in a double slit experiment affected if :
- the slits separation is increased,
 - the colour of light used is changed from red to blue,
 - the whole apparatus is submerged in a oil of refractive index 1.2 ?
- Justify your answer in each case.

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3

Spacing between fringes in Young's doubles slit experiment	
$\beta = \frac{\lambda D}{d}$	
1) β decreases as d increases as $\beta \propto \frac{1}{d}$	$\frac{1}{2} + \frac{1}{2}$
2) $\lambda_{\text{blue}} < \lambda_{\text{red}}$ as $\beta \propto \lambda \therefore$ fringe width decreases.	$\frac{1}{2} + \frac{1}{2}$
3) $\beta' = \frac{\beta}{\mu} \therefore \beta$ decreases as $\mu > 1$ [$\therefore \mu = 1.2$]	$\frac{1}{2} + \frac{1}{2}$

10. In a Young's double slit experiment using light of wavelength 600 nm, the slit separation is 0.8 mm and the screen is kept 1.6 m from the plane of the slits. Calculate :
- (i) the fringe width
 - (ii) the distance of (a) third minimum and (b) fifth maximum, from the central maximum.

10. In a Young's double slit experiment using light of wavelength 600 nm, the slit separation is 0.8 mm and the screen is kept 1.6 m from the plane of the slits. Calculate :
- the fringe width
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(i)	$\beta = \frac{\lambda D}{d} = \frac{600 \times 10^{-9} \times 1.2}{8 \times 10^{-4}} = 1.2 \text{ mm}$	$\frac{1}{2} + \frac{1}{2}$
(ii) (a)	$x_3 = \frac{5}{2} \frac{\lambda D}{d} = \frac{5}{2} \times 1.2 \text{ mm} = 3 \text{ mm}$	$\frac{1}{2} + \frac{1}{2}$
(iii) (b)	$x_5 = \frac{5\lambda D}{d} = 2 \times 3 \text{ mm} = 6 \text{ mm}$	$\frac{1}{2} + \frac{1}{2}$

4. (a) A parallel beam of light of wavelength 600 nm is incident normally on a slit of width 0.2 mm. If the resulting diffraction pattern is observed on a screen 1 m away, find the distance of 3
- (i) first minimum, and
- (ii) second maximum, from the central maximum.

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4. (a) A parallel beam of light of wavelength 600 nm is incident normally on a slit of width 0.2 mm. If the resulting diffraction pattern is observed on a screen 1 m away, find the distance of

(i) first minimum, and

(ii) second maximum, from the central maximum.

CBSE 22

$$\begin{aligned}
 \text{(i) } y &= \frac{\lambda D}{a} \\
 &= \frac{600 \times 10^{-9} \times 1}{0.2 \times 10^{-3}} \\
 &= 3 \times 10^{-3} \text{ m} = 3 \text{ mm}
 \end{aligned}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\text{(ii) } y = \left(n + \frac{1}{2}\right) \frac{\lambda D}{a}$$

$$\frac{1}{2}$$

$$y = \left(2 + \frac{1}{2}\right) \frac{\lambda D}{a}$$

$$y = \frac{5 \lambda D}{2 a}$$

$$\frac{1}{2}$$

$$y = \frac{5}{2} \times \frac{600 \times 10^{-9} \times 1}{0.2 \times 10^{-3}}$$

$$\frac{1}{2}$$

$$= 7.5 \times 10^{-3} = 7.5 \text{ mm}$$

6. Using Huygen's principle, for a plane wavefront incident on a plane reflecting surface, draw the reflected wavefront. Hence prove laws of reflection.

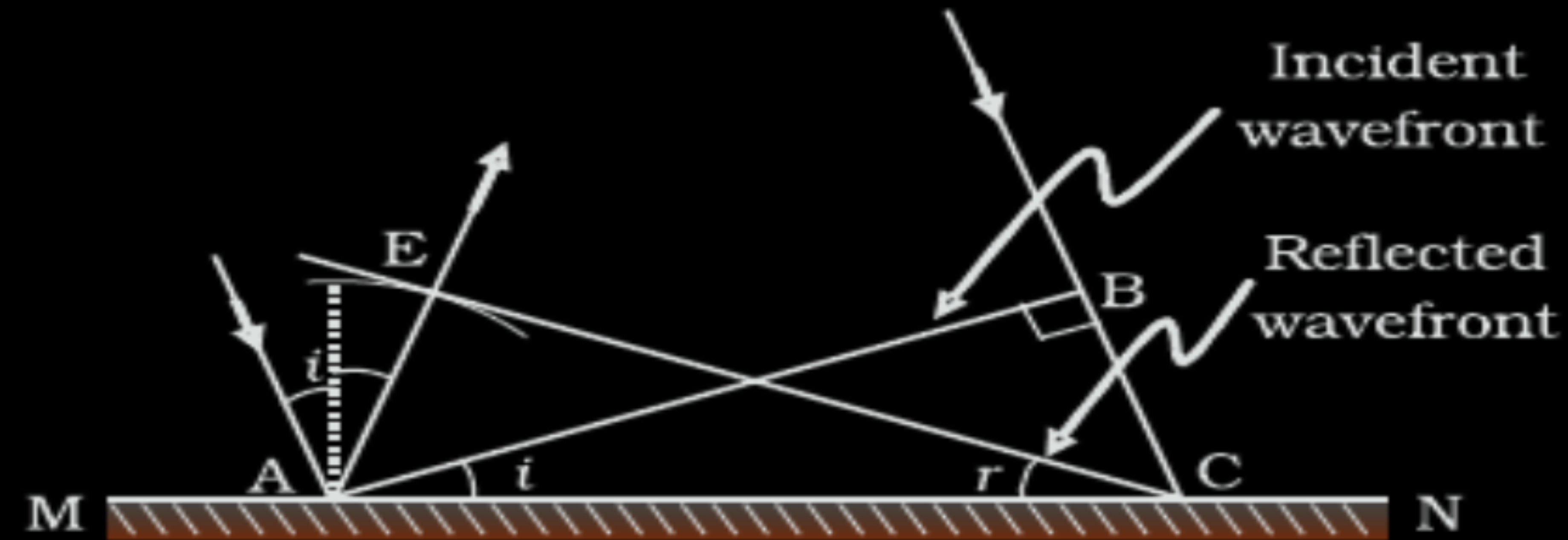
CBSE 22

3

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3



$\triangle EAC$ and $\triangle BAC$ are congruent

(RHS congruency)

$$\therefore \angle i = \angle r$$

The plane of mirror is $\perp$ to plane of paper hence the incident ray, reflected ray and normal ray lie in same plane.

1

1

$\frac{1}{2}$

$\frac{1}{2}$

9. (a) (i) Monochromatic light is incident on a surface separating two media. The frequency of the light after refraction remains unaffected but its wavelength changes. Why ?

CBSE 22

9. (a) (i) Monochromatic light is incident on a surface separating two media. The frequency of the light after refraction remains unaffected but its wavelength changes. Why ?

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i) Refraction arises through interaction of incident light with the atomic constituents of matter. Atoms may be viewed as oscillators which take up the frequency of the external agency causing forced oscillations. Thus the frequency of refracted light equals the frequency of incident light.

Alternatively

Frequency is the characteristic of the source of light. So it remains unaffected. But λ depends on refractive index (μ) of the medium as —

$$\lambda_m = \frac{\lambda_o}{\mu}$$

12. The principle of superposition is used to understand the phenomenon of interference of light waves. The principle states that at a particular point, the resultant displacement produced by a number of waves is the vector sum of the displacements produced by each wave. Light waves from two coherent sources produce interference pattern. Thomas Young devised a way to obtain two coherent sources using two identical pinholes (S_1 and S_2) illuminated by a single monochromatic pinhole source S . Using these sources in his experiment known as Young's double slit experiment, Young studied the interference pattern. The pattern consists of alternate bright and dark fringes. The distance between two successive bright or dark fringes depends on the distance between S_1 and S_2 , the distance of the screen from the plane of S_1S_2 and the wavelength of light used.

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CBSE 22

I. Consider the following waves :

(i) $y_1 = a \sin \omega t$

(ii) $y_2 = a \sin 2\omega t$

(iii) $y_3 = a \sin (2\omega t + \phi)$

(iv) $y_4 = a \sin \left(4\omega t + \frac{\pi}{2} \right)$

Which pair of the waves coming from two sources S_1 and S_2 will produce interference ?

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (iii) and (iv)

(D) (iv) and (i)

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CBSE 22

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(D) (iv) and (i)

I (B)

II. Two light waves of the same intensity I_0 each, having a path difference of $\lambda/4$, emanating from two coherent sources, meet at a point. The resultant intensity at the point will be

(A) Zero

(B) I_0

(C) $2 I_0$

(D) $4 I_0$

II. Two light waves of the same intensity I_0 each, having a path difference of $\lambda/4$, emanating from two coherent sources, meet at a point. The resultant intensity at the point will be

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II (C)

III. Vandana performs Young's double slit experiment by using orange, green and red lights successively. If the fringe widths measured in the three cases are ω_1 , ω_2 and ω_3 respectively, then which of the following is correct ?

(A) $\omega_2 > \omega_1 > \omega_3$

(B) $\omega_1 > \omega_2 > \omega_3$

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III (D)

IV. In a Young's double slit experiment, the slit separation is 0.8 mm and the interference pattern is obtained on a screen kept 50 cm from the plane of the slits S_1 and S_2 . If the first bright fringe is formed 0.4 mm from the central maximum, the wavelength of light used is

(A) 480 nm

(B) 560 nm

(C) 640 nm

(D) 680 nm

IV. In a Young's double slit experiment, the slit separation is 0.8 mm and the interference pattern is obtained on a screen kept 50 cm from the plane of the slits S_1 and S_2 . If the first bright fringe is formed 0.4 mm from the central maximum, the wavelength of light used is

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(C) 640 nm (D) 680 nm

IV (C)

- V. Consider the effect on the angular separation of the fringes in a Young's double slit experiment due to the following operations :
- (i) the screen is moved away from the plane of the slits,
 - (ii) the separation between the two slits is increased till fringes are observed.

Which of the following options is correct ?

- (A) It remains constant in both cases.
- (B) It decreases in both cases.
- (C) It remains constant in (i) but decreases in (ii).
- (D) It decreases in (i) but remains constant in (ii).

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8. (a) (i) Two independent monochromatic sources of light cannot constitute coherent sources. Explain.
- (ii) Two waves represented by

$$y_1 = a_1 \cos \omega t, \text{ and}$$

$$y_2 = a_2 \cos \omega t$$

produce an interference pattern in Young's double slit experiment. If the ratio of intensities at maximum and at minimum is 25 : 9, then calculate the ratio of intensities of light waves from the sources.

CBSE 22

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produce an interference pattern in Young's double slit experiment. If the ratio of intensities at maximum and at minimum is 25 : 9, then calculate the ratio of intensities of light waves from the

(i) Explanation: Two different monochromatic sources can not act as coherent sources because they can not emit light waves having the same phase or having a constant phase difference.

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$$(ii) \frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{25}{9}$$

$$\therefore \frac{a_1 + a_2}{a_1 - a_2} = \frac{5}{3}$$

$$\therefore \frac{a_1}{a_2} = \frac{4}{1}$$

$$\therefore \frac{I_1}{I_2} = \left(\frac{a_1}{a_2} \right)^2 = \frac{16}{1}$$

OR

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

- (b) (i) State Huygens' principle. How did Huygens explain the absence of secondary wavelets in the backward direction ?
- (ii) What is the shape of a wavefront emanating from (I) a point source, and (II) a linear source ?

3

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3

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• Each point on a wavefront acts as the new source of secondary wavelets which also travel with the speed of the original wavefront of the secondary wavelets	1
• The amplitude of the secondary wavelet, according to Huygen's argument, is maximum in the forward direction and zero in the backward direction.	1
• (i) spherical	$\frac{1}{2}$
(ii) cylindrical	$\frac{1}{2}$

12. When light passes through a narrow aperture or falls on an obstacle of very small dimensions, it bends into the shadow region. The bending of light waves around the corners and edges of the obstacles as well as aperture is called diffraction of light. For prominent diffraction effect, the size of the aperture or obstacle must be comparable to the wavelength of the light used. If we place a screen in front of the slit, a broad central maxima and a few alternate dark and bright bands of successively decreasing intensities on either sides of central bright band are obtained on the screen.

CBSE 22

(a) Light of wavelength ' λ ' falls normally on a slit of width ' a '. The diffraction due to the slit will be most prominently observed when :

(i) $a = 5\lambda$

(ii) $a = 100\lambda$

(iii) $a = 10^3\lambda$

(iv) $a = 10^5\lambda$

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(iii) $a = 10^3\lambda$

(iv) $a = 10^5\lambda$

a-(i)

- (b) When monochromatic light is replaced by white light, then :
- (i) the diffraction pattern disappears.
 - (ii) the diffraction band becomes circular.
 - (iii) a white band is formed at the centre and coloured bands are formed at both its sides.
 - (iv) only a white band is formed at the centre.

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 - (iv) only a white band is formed at the centre.

b-(iii)

- (c) If in a diffraction experiment, the width of the slit is slightly increased, then the central maximum of the diffraction pattern becomes :
- (i) broader and brighter.
 - (ii) broader and fainter.
 - (iii) narrow and brighter.
 - (iv) narrow and fainter.

- (c) If in a diffraction experiment, the width of the slit is slightly increased, then the central maximum of the diffraction pattern becomes :
- (i) broader and brighter.
 - (ii) broader and fainter.
 - (iii) narrow and brighter.
 - (iv) narrow and fainter.

(d) If a convex lens of focal length 'f' is placed very close to the slit of width 'a' and the screen is kept at the focus of the lens, the size of the central maximum in the diffraction pattern will be :

(i) $\frac{f \lambda}{a}$

(ii) $\frac{2f \lambda}{a}$

(iii) $\frac{f \lambda}{2a}$

(iv) $\frac{f \lambda}{4a}$

- (d) If a convex lens of focal length 'f' is placed very close to the slit of width 'a' and the screen is kept at the focus of the lens, the size of the central maximum in the diffraction pattern will be :

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(ii) $\frac{2f \lambda}{a}$

(iii) $\frac{f \lambda}{2a}$

(iv) $\frac{f \lambda}{4a}$

d-(i)

(e) A parallel beam of light of wavelength 500 nm falls on a narrow slit. The diffraction pattern is observed on a screen 1 m away. The first minimum is at a distance of 2.5 mm from the centre of the screen. The width of the slit is :

(i) 0.16 mm

(ii) 0.20 mm

(iii) 0.40 mm

(iv) 0.60 mm

5×1=5

(e) A parallel beam of light of wavelength 500 nm falls on a narrow slit. The diffraction pattern is observed on a screen 1 m away. The first minimum is at a distance of 2.5 mm from the centre of the screen. The width of the slit is :

(i) 0.16 mm

(ii) 0.20 mm

(iii) 0.40 mm

(iv) 0.60 mm

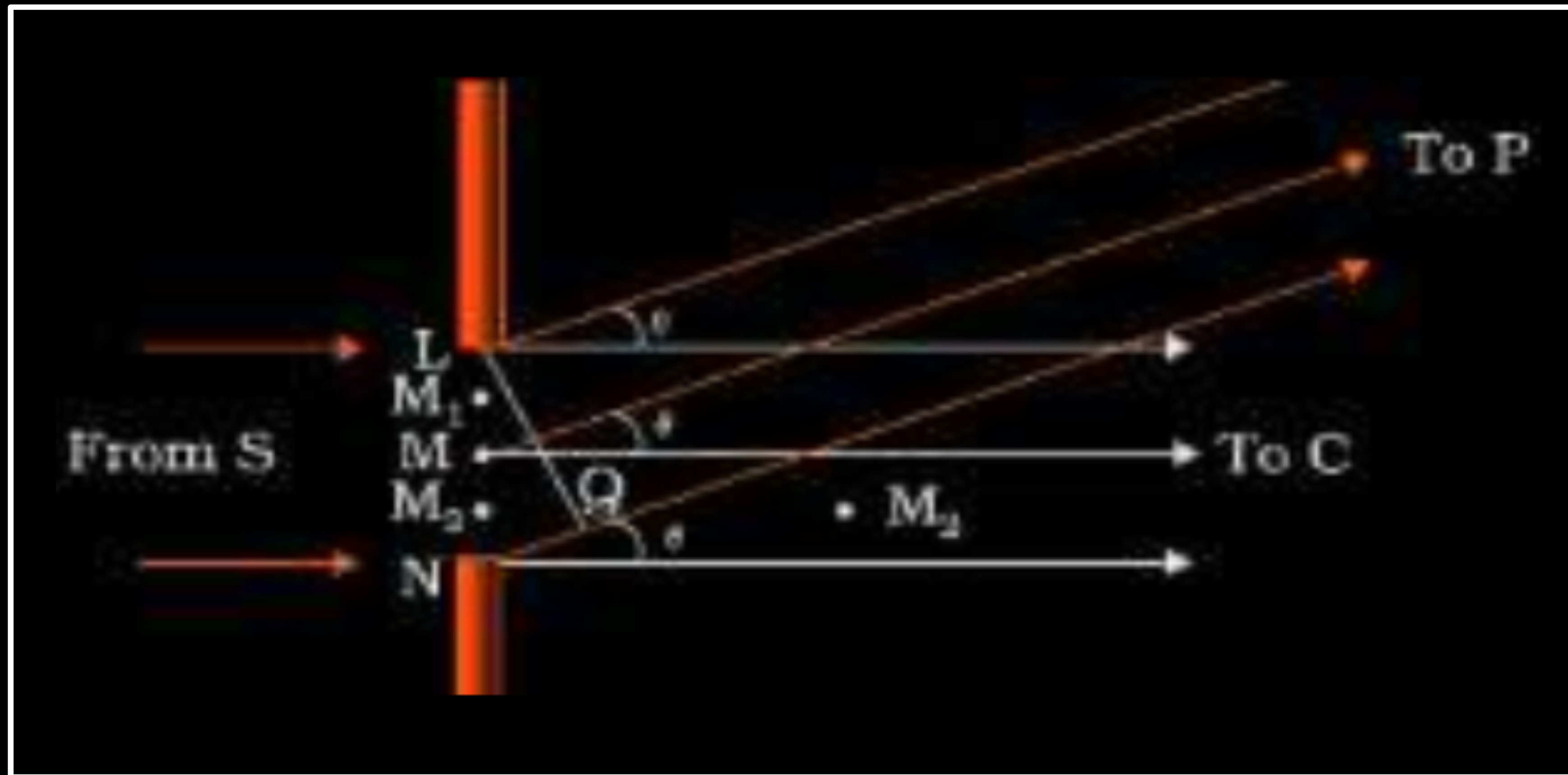
$5 \times 1 = 5$

e-(ii)

- 6.** Briefly explain how bright and dark fringes are formed on a screen due to the diffraction at a single slit. Hence, explain why the intensity at the bright fringes decreases sharply as their order (n) increases.

CBSE 22

6. Briefly explain how bright and dark fringes are formed on a screen due to the diffraction at a single slit. Hence, explain why the intensity at the bright fringes decreases sharply as their order (n) increases. CBSE 22 3



Dividing the slit into smaller parts, and adding their contributions at P with the proper phase differences. The path difference $NP - LP$ between the two edges of the slit, $NP - LP = NQ = a \sin \theta \approx a\theta$

$\frac{1}{2}$

At the central point C on the screen, the angle θ is zero. All path differences are zero and hence all the parts of the slit contribute in phase. This gives maximum intensity at C,

$\frac{1}{2}$

Similarly other secondary maxima will be formed at $\theta = (n+1/2) \lambda/a$,

$\frac{1}{2}$

For minima (zero intensity) the contributions from M_1 and M_2 are 180° out of phase and cancel in the direction $\theta = \lambda/a$. where λ is the path difference. Similarly other minima are formed due to cancellation of contributions. At an angle $\theta = n\lambda/a$, where $n = \pm 1, \pm 2, \pm 3, \dots$

$\frac{1}{2}$

Consider an angle $\theta = 3\lambda/2a$ which is midway between two of the dark fringes. Divide the slit into three equal parts. If we take the first two thirds of the slit, the path difference between the two ends would be λ .

$\frac{1}{2}$

The first two-thirds of the slit can be divided into two halves which have a $\lambda/2$ path difference. The contributions of these two halves cancel, only the remaining one-third of the slit contributes to the intensity at a point between the two minima. Which will be much weaker than the central maximum (where the entire slit contributes in phase). Similarly there are maxima at $(n + 1/2) \lambda/a$ with $n = 2, 3$, etc. These become weaker with increasing n , since only one-fifth, one-seventh, etc., of the slit contributes in these cases.

$\frac{1}{2}$

- (b) A plane wavefront is propagating from a rarer into a denser medium. Use Huygens principle to show the refracted wavefront and verify Snell's law.

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3

We consider, refracted wavefront CE and triangles ABC & AEC. From the triangles we obtained

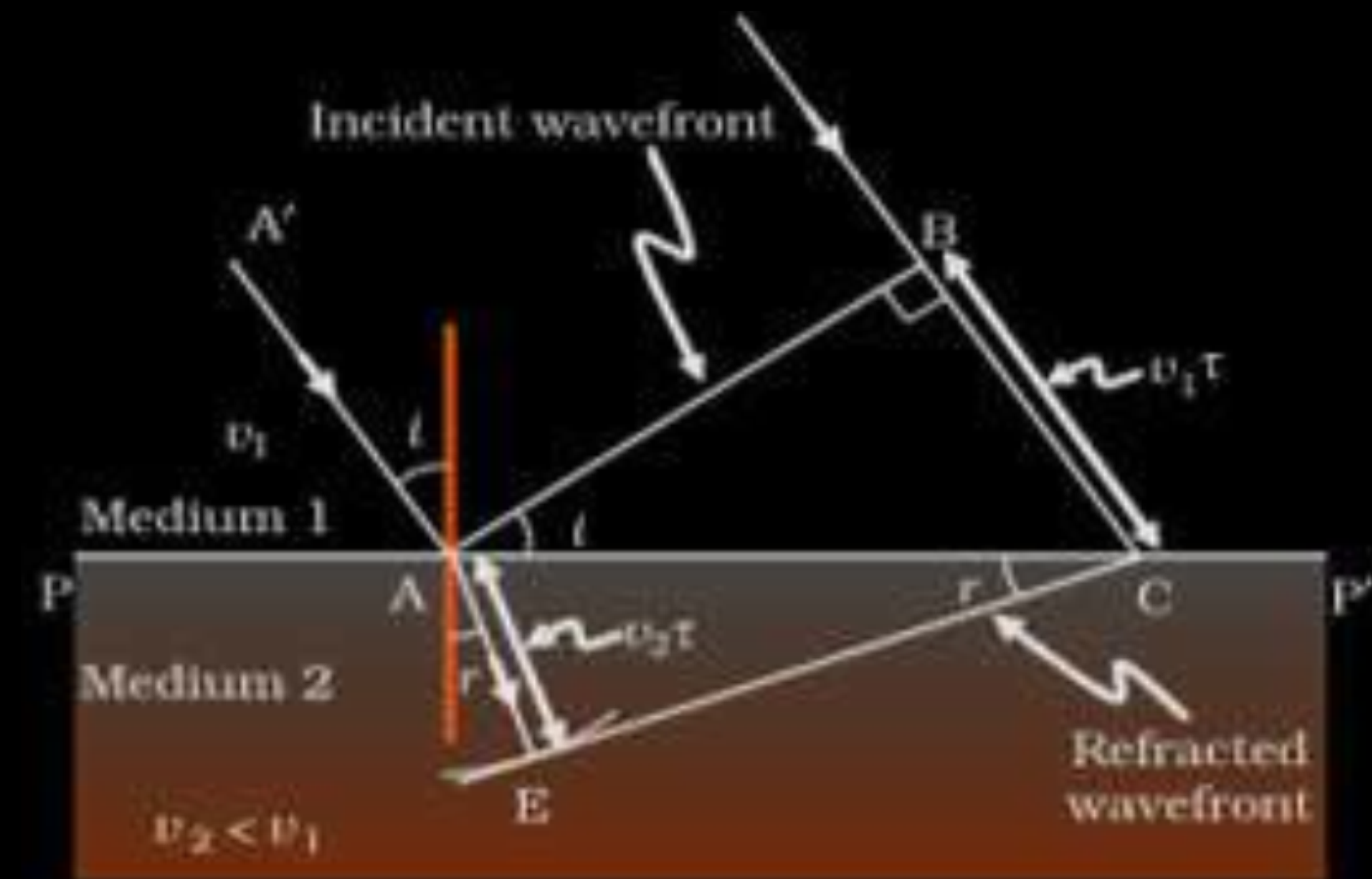
$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$$

$$\text{Thus } \frac{\sin i}{\sin r} = \frac{v_2}{v_1}$$

$$\text{We know } n = \frac{c}{v}, \text{ So } \frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$

which is the Snell's law.



1

1/2

1/2

1/2

1/2

7. (a) The interference pattern is not observed in Young's double slit experiment when the two sources S_1 and S_2 are far apart. Explain.
- (b) Mention the conditions for the two sources to be coherent.
- (c) What is the effect on the interference pattern in a Young's double slit experiment, if the source of wavelength λ is replaced by another source of wavelength 1.5λ , with the interference pattern still observable ?

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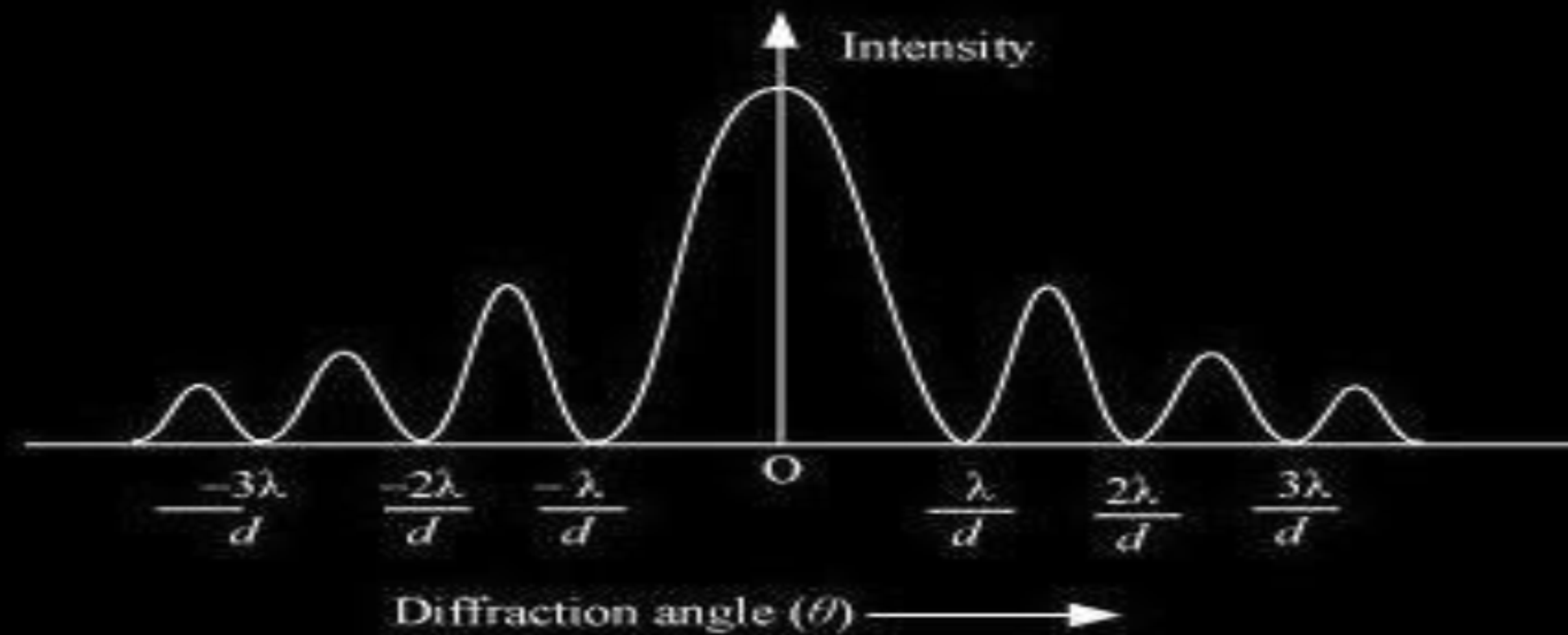
CBSE 22

3

a) $\beta = \frac{\lambda D}{d}$, When space between point source increases, fringe width decrease. Interference pattern gets less and less sharp/ Interference pattern will not be observable. Alternatively On increasing separation between the slits the condition $\frac{s}{S} < \frac{\lambda}{d}$, will not be satisfied and interference pattern disappears.	1
b) Two sources must have same frequency and constant or zero phase difference / constant or zero path difference.	1
c) $\beta \propto \lambda$ fringe width will increase & intensity (Brightness) of fringe will decrease.	$\frac{1}{2} + \frac{1}{2}$

- (b) A plane wavefront of light of wavelength ' λ ' is incident normally on a narrow slit of width ' a ' and a diffraction pattern is observed on a screen at a distance ' D ' from the slit.
- (i) Depict the intensity distribution in the pattern observed.
 - (ii) Obtain the expression for the first maximum from the central maximum.

(i) Intensity distribution curve



For maximum

$$a \sin \theta = (n + \frac{1}{2})\lambda$$

For first Maximum ; $n = 1$

$$\text{For small } \theta; a \theta = \frac{3\lambda}{2} \Rightarrow \theta = \frac{3\lambda}{2a}$$

$$\therefore \frac{x}{D} = \frac{3\lambda}{2a}$$

$$\therefore x = \frac{3\lambda D}{2a}$$

1½

½

½

½

12. The British physicist Thomas Young explained the interference of light using the principle of superposition of waves. He observed the interference pattern on the screen, in his experimental set-up, known now as Young's double slit experiment. The two slits S_1 and S_2 were illuminated by light from a slit S . The interference pattern consists of dark and bright bands of light. Such bands are called fringes. The distance between two consecutive bright and dark fringes is called fringe width.

- (a) If the screen is moved closer to the plane of slits S_1 and S_2 , then the fringe width :
- (i) will decrease, but the intensity of bright fringe remains the same.
 - (ii) will increase, but the intensity of bright fringe decreases.
 - (iii) will decrease, but the intensity of bright fringe increases.
 - (iv) and the intensity both remain the same.

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CBSE 22

a) (iii)

- (b) What will happen to the pattern on the screen, when the two slits S_1 and S_2 are replaced by two independent but identical sources ?
- (i) The intensity of pattern will increase
 - (ii) The intensity of pattern will decrease
 - (iii) The number of fringes will become double
 - (iv) No pattern will be observed on the screen

- (b) What will happen to the pattern on the screen, when the two slits S_1 and S_2 are replaced by two independent but identical sources ?
- (i) The intensity of pattern will increase
 - (ii) The intensity of pattern will decrease
 - (iii) The number of fringes will become double
 - (iv) No pattern will be observed on the screen

b) (iv)

- (c) Two sources of light are said to be coherent, when both emit light waves of :
- (i) same amplitude and have a varying phase difference.
 - (ii) same wavelength and a constant phase difference.
 - (iii) different wavelengths and same intensity.
 - (iv) different wavelengths and a constant phase difference.

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 - (iii) different wavelengths and same intensity.
 - (iv) different wavelengths and a constant phase difference.

c) (ii)

(d) The fringe width in a Young's double slit experiment is β . If the whole set-up is immersed in a liquid of refractive index ' μ ', then the new fringe width will be :

(i) β

(ii) $\beta\mu$

(iii) $\frac{\beta}{\mu}$

(iv) $\frac{\beta}{\mu^2}$

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(iv) $\frac{\beta}{\mu^2}$

d) (iii)

(e) The total path difference between two waves meeting at points P_1 and P_2 on the screen are $\left(\frac{3\lambda}{2}\right)$ and 2λ respectively. Then :

- (i) bright fringes are formed at both points.
- (ii) dark fringes are formed at both points.
- (iii) a bright fringe is formed at P_1 and a dark fringe is formed at P_2 .
- (iv) a bright fringe is formed at P_2 and a dark fringe is formed at P_1 .

5×1=5

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5×1=5

e) (iv)

18. Write the conditions on path difference under which (i) constructive
(ii) destructive interference occur in Young's double slit experiment.

1

CBSE 20

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1

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(i) for constructive interference path difference, $\Delta p = n \lambda$

$\frac{1}{2}$

(ii) for destructive interference path difference, $\Delta p = (2n+1) \frac{\lambda}{2}, n = 0, 1, 2, 3, \dots$

$\frac{1}{2}$

Alternatively

$\Delta p = (2n-1) \frac{\lambda}{2}, n = 1, 2, 3, \dots$

27. In a single slit diffraction experiment, the width of the slit is increased. How will the (i) size and (ii) intensity of central bright band be affected ? Justify your answer.

CBSE 20

2

27. In a single slit diffraction experiment, the width of the slit is increased. How will the (i) size and (ii) intensity of central bright band be affected ? Justify your answer.

CBSE 20

2

(i) On increasing the width of the slit, the size of the central bright band will decrease	$\frac{1}{2}$
(ii) Justification: Angular width = $\frac{2\lambda}{a}$, i.e. angular width is inversely proportional to the width of the slit	$\frac{1}{2}$
(iii) The intensity of central bright band will increase	$\frac{1}{2}$
Justification: The amplitude/intensity of light passing through slit has increased.	$\frac{1}{2}$

8. When a wave undergoes reflection at an interface from rarer to denser medium, adhoc change in its phase is :

(a) $\frac{\pi}{2}$

(b) 0

(c) π

(d) $\frac{\pi}{4}$

1

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(a) $\frac{\pi}{2}$

(b) 0

(c) π

(d) $\frac{\pi}{4}$

1

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(c)

π

33. Two coherent light waves of intensity $5 \times 10^{-2} \text{ Wm}^{-2}$ each super-impose and produce the interference pattern on a screen. At a point where the path difference between the waves is $\lambda/6$, λ being wavelength of the wave, find the

CBSE 20

- (a) phase difference between the waves.
- (b) resultant intensity at the point.
- (c) resultant intensity in terms of the intensity at the maximum.

3

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CBSE 20

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3

(a) Phase difference $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$

(b)
$$I_1 = I_2 + I_3 + 2\sqrt{I_3 I_2} \cos \phi$$

$$= I + I + 2I \times \frac{1}{2} = 3I$$

$$= 15 \times 10^{-2} \text{ Wm}^{-2}$$

(c) $I_{\max} = 4I$

$$I_1 = \frac{3I}{4I} \times 4I = \frac{3}{4} I_{\max}$$

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

- 13.** In Young's double slit experiment, the separation between the two slits is halved. The new fringe width will be _____ times its initial value. 1

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CBSE 20

- (a) Briefly describe the Young's double slit experiment of interference of light. Drive the expression for fringe width in the pattern.
- (b) Monochromatic light of wavelength 588 nm is incident from air to water interface. Find the wavelength and speed of the refracted light. The refractive index of water is $\frac{4}{3}$.

CBSE 20

S is a monochromatic source of light. S_1 and S_2 are two pinholes separated by a distance d . GG' is the screen placed at the distance D from the pinholes.

P is a general point on the screen.

Derivation

$$\begin{aligned}(S_2P)^2 - (S_1P)^2 &= \left[D^2 + \left(x + \frac{d}{2} \right)^2 \right] - \left[D^2 + \left(x - \frac{d}{2} \right)^2 \right] \\ &= D^2 + x^2 + \frac{d^2}{4} + xd - D^2 - x^2 - \frac{d^2}{4} + xd \\ &= 2xd\end{aligned}$$

$$\text{path difference} = S_2P - S_1P = \frac{2xd}{S_2P + S_1P} \approx \frac{2xd}{2D}$$

$$\text{Path difference} = \frac{xd}{D}$$

For maxima

$$\frac{xd}{D} = n\lambda, \quad n = 0, 1, 2, \dots$$

$$\text{or } x_n = \frac{n\lambda D}{d}$$

$$x_{n+1} = \frac{(n+1)\lambda D}{d}$$

$$\beta = x_{n+1} - x_n$$

$$\beta = \frac{\lambda D}{d}$$

b)

$$\mu_w = \frac{c_0}{c_w} = \frac{v\lambda_0}{v\lambda_w} = \frac{\lambda_0}{\lambda_w}$$

$$\lambda_w = \frac{\lambda_0}{\mu_w} = \frac{588 \times 3}{4} = 441 \text{ nm}$$

$$c_w = \frac{c_0}{\mu_w} = \frac{3 \times 10^8 \times 3}{4} = 2.25 \times 10^8 \text{ m/s}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

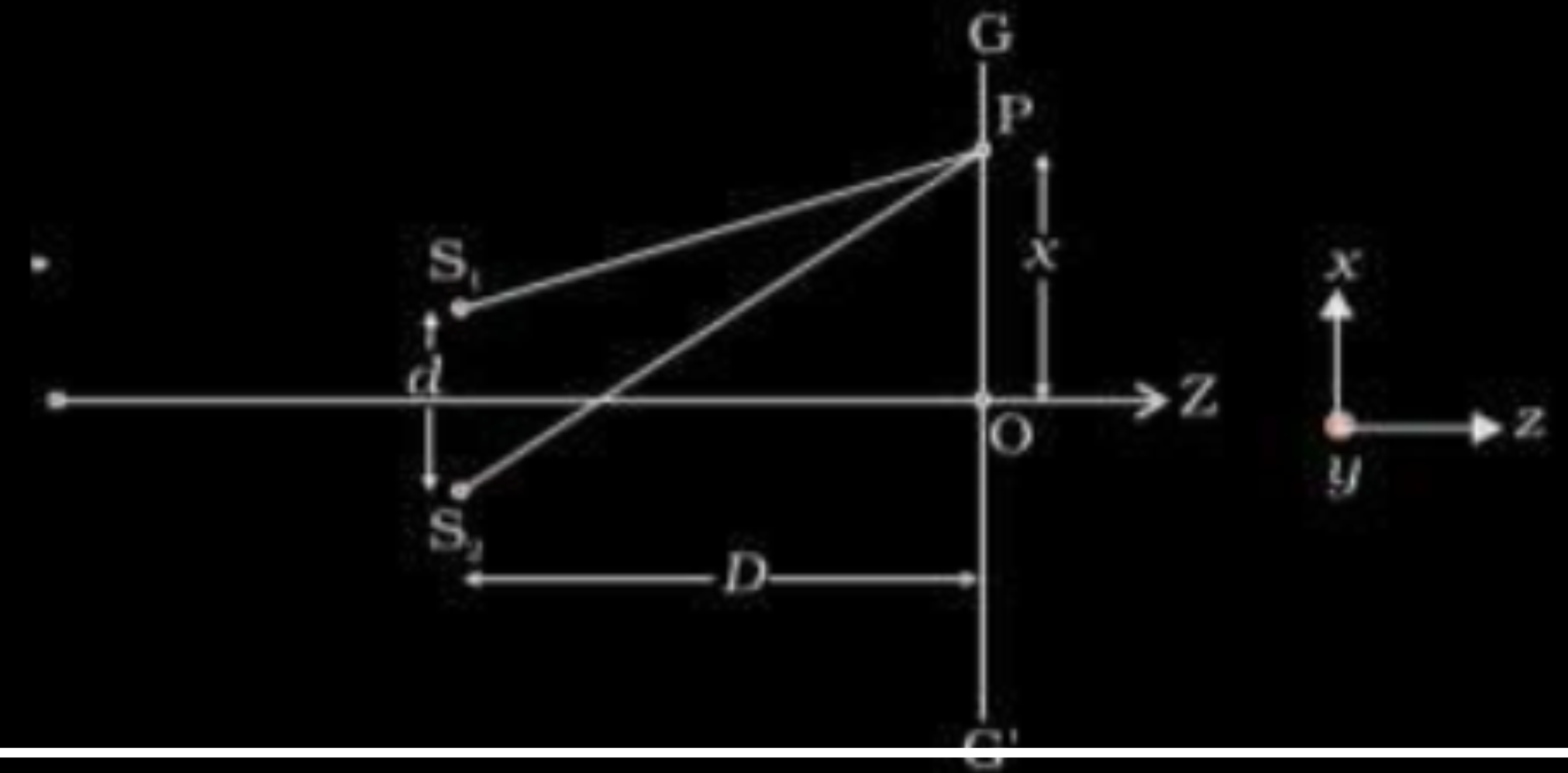
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$



12. In Young's double slit experiment, the path difference between two interfering waves at a point on the screen is $\frac{5\lambda}{2}$, λ being wavelength of the light used. The _____ dark fringe will lie at this point. 1

12. In Young's double slit experiment, the path difference between two interfering waves at a point on the screen is $\frac{5\lambda}{2}$, λ being wavelength of the light used. The Third dark fringe will lie at this point.

1

If one of the slits in Young's double slit experiment is fully closed, the new pattern has _____ central maximum in angular size. *1*

If one of the slits in Young's double slit experiment is fully closed, the new pattern has _____ central maximum in angular size. 1

CBSE 20

$$\frac{2\lambda}{a}$$

[Alternatively, broader]

- (a) What is a wavefront ? How does it propagate ? Using Huygens' principle, explain reflection of a plane wavefront from a surface and verify the laws of reflection.
- (b) A parallel beam of light of wavelength 500 nm falls on a narrow slit and the resulting diffraction pattern is obtained on a screen 1 m away. If the first minimum is formed at a distance of 2.5 mm from the centre of the screen, find the (i) width of the slit, and (ii) distance of first secondary maximum from the centre of the screen.

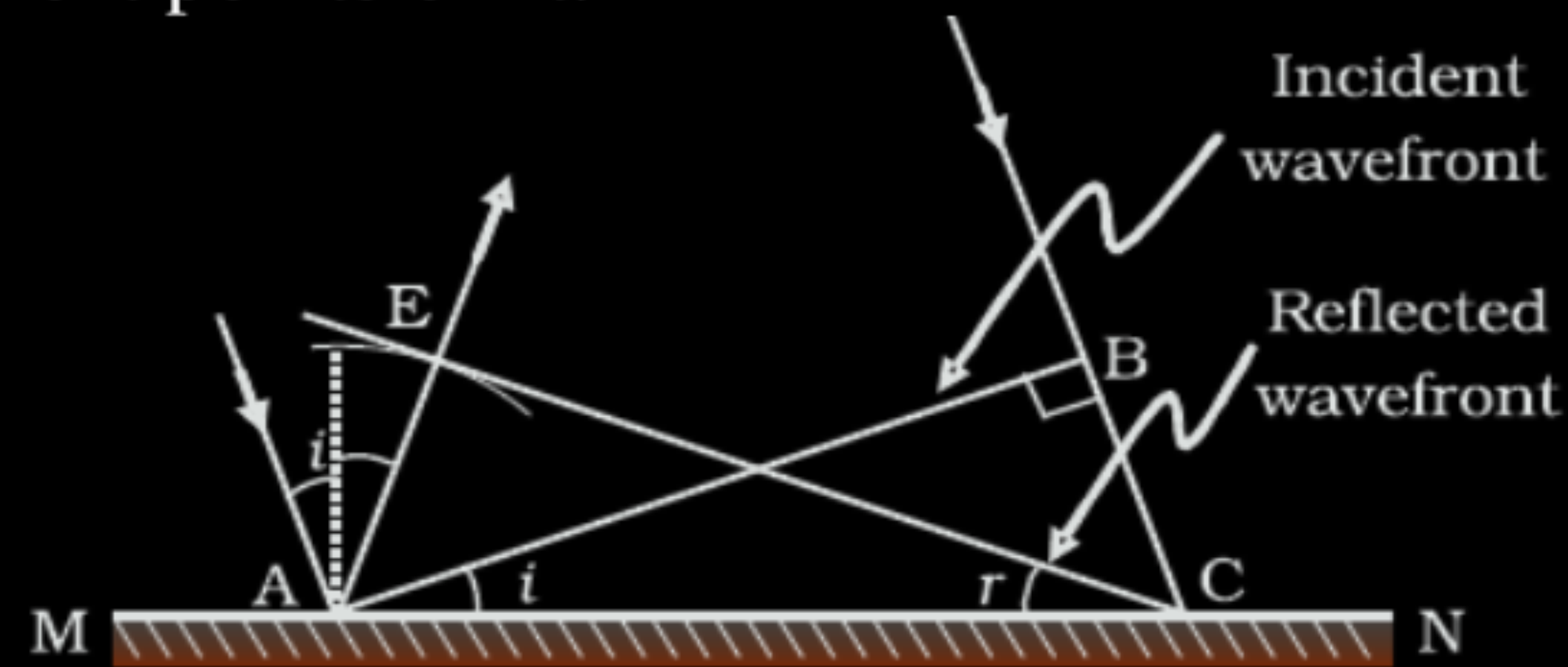
5

(a)

Wavefront is a surface of constant phase.

Alternatively, It is the locus of all those points which are in the same phase of disturbance.

The wave propagates in a direction perpendicular to the wavefront through secondary wavelets originating from different points on it.



Consider a plane wave AB incident at an angle i with speed v on the surface MN in time τ

Therefore

$$BC = v\tau$$

Using Huygen's principle, a sphere of radius $v\tau$ which has tangent plane CE is reflected at an angle r

$$\therefore AE = BC = v\tau$$

$\therefore \triangle EAC$ and $\triangle BAC$ are congruent

$$\therefore \angle i = \angle r$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b)

(i)

$$x = \frac{\lambda D}{d}$$

$$\Rightarrow d = \frac{\lambda D}{x} = \frac{500 \times 10^{-9} \times 1}{2.5 \times 10^{-3}} = 2 \times 10^{-4} \text{ m}$$

$\frac{1}{2}$

$\frac{1}{2}$

(ii) For the first Secondary maxima

$$x = \frac{3\lambda D}{2d}$$

$$= \frac{3 \times 500 \times 10^{-9} \times 1}{2 \times 2 \times 10^{-4}} = 3.75 \text{ mm}$$

$\frac{1}{2}$

$\frac{1}{2}$

[Even if a student finds location of first secondary maxima by $(2.5) + (\frac{1}{2} \times 2.5) = 3.75 \text{ mm}$, award full 1 mark for b(ii)]

- 24.** Define wavefront of a travelling wave. Using Huygens principle, obtain the law of refraction at a plane interface when light passes from a denser to rarer medium.

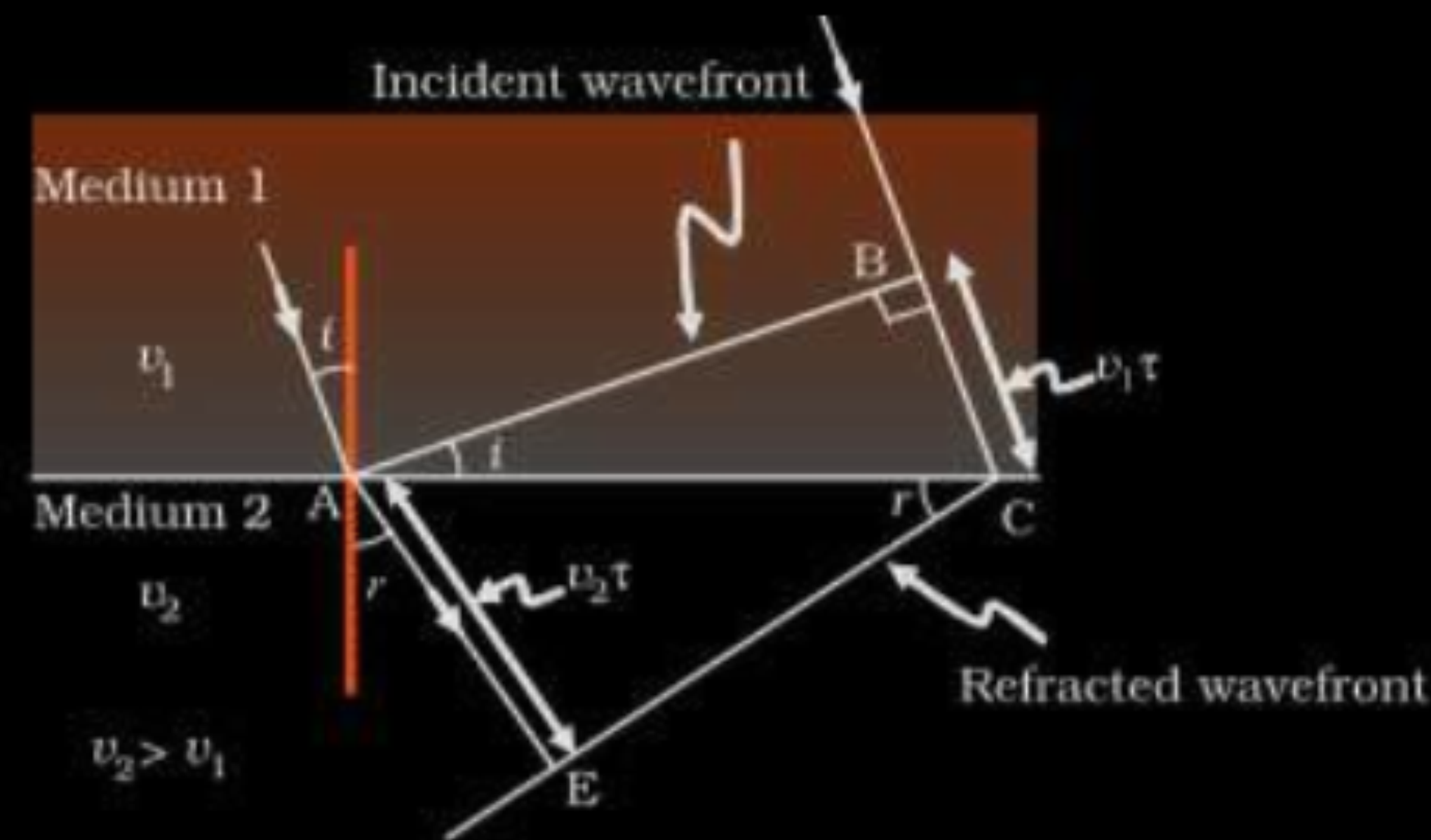
CBSE 20

24. Define wavefront of a travelling wave. Using Huygens principle, obtain the law of refraction at a plane interface when light passes from a denser to rarer medium.

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2

Wavefront is defined as the surface of constant phase;
Alternatively
It is a locus of all the points in the same phase of disturbance



$$\sin i = \frac{BC}{AC} = \frac{v_1 t}{AC}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 t}{AC}$$

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

1/2

1/2

1/2

1/2

31. What is the effect on the interference fringes in Young's double slit experiment due to each of the following operations ? Justify your answers.

3

- (a) The screen is moved away from the plane of the slits.
- (b) The separation between slits is increased.
- (c) The source slit is moved closer to the plane of double slit.

CBSE 20

31. What is the effect on the interference fringes in Young's double slit experiment due to each of the following operations ? Justify your answers.

3

- (a) The screen is moved away from the plane of the slits.
- (b) The separation between slits is increased.
- (c) The source slit is moved closer to the plane of double slit.

CBSE 20

- (a) Fringe width increases, $\beta \propto D$
- (b) Fringe width decreases, $\beta \propto \frac{1}{d}$
- (c) Fringes disappear because $\frac{s}{S} < \frac{\lambda}{d}$ not satisfied

$\frac{1}{2} + \frac{1}{2}$
 $\frac{1}{2} + \frac{1}{2}$
 $\frac{1}{2} + \frac{1}{2}$

- (a) Define a wavefront. Using Huygens' geometrical construction, explain with the help of a diagram how the plane wavefront travels from the instant t_1 to t_2 in air. CBSE 19
- (b) A plane wavefront is incident on a convex lens. Explain, with the help of the diagram, the shape of the refracted wavefront formed.

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(b) A plane wavefront is incident on a convex lens. Explain, with the help of the diagram, the shape of the refracted wavefront formed.

The wave front is defined as a surface of constant phase

1

Alternatively: The wave front is a locus of points which oscillate in phase

Consider a plane wave AB incident at an angle 'i' on a reflecting surface MN
let t = time taken by the wave front to advance from B to C.

$$\therefore BC = vt$$

Let CE represent the tangent plane drawn from the point C to the sphere of radius 'vt' having A as its center.

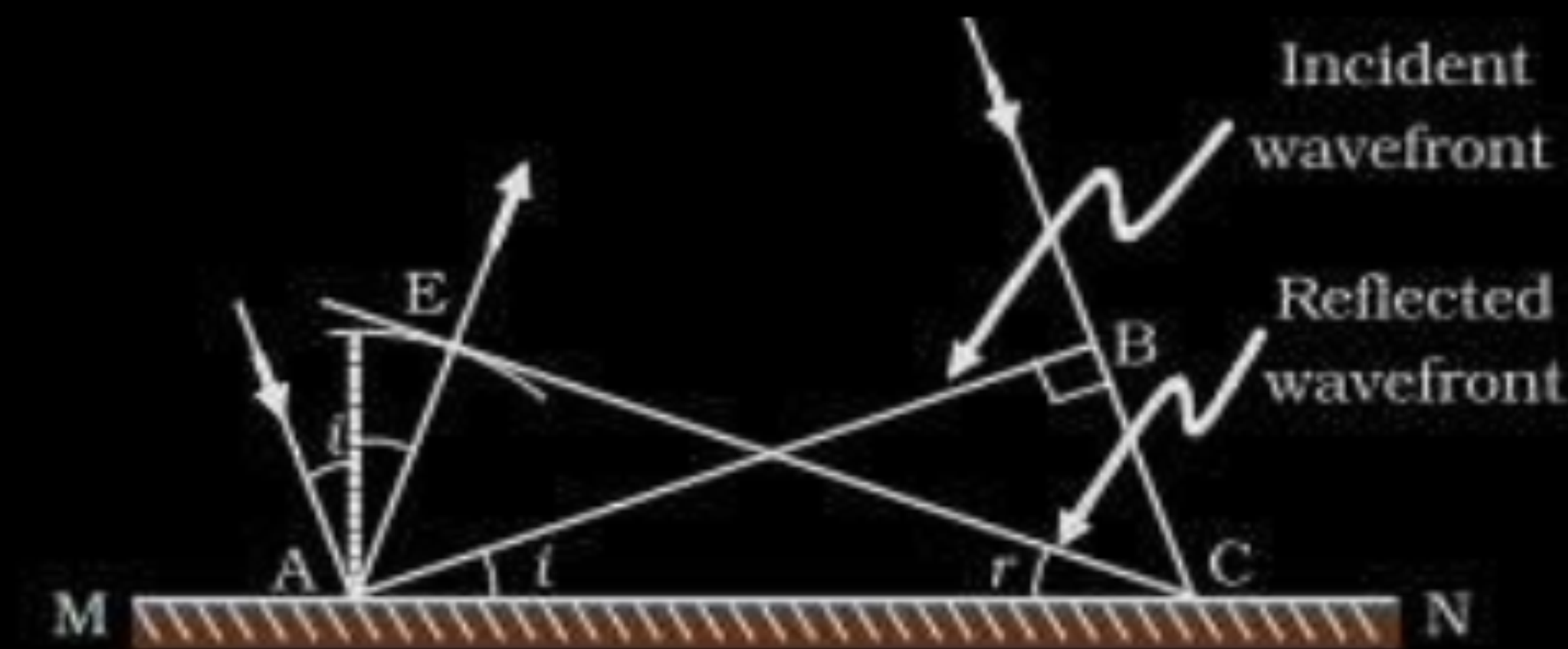
$$\text{then } AE = BC = vt$$

it follows that

$$\Delta EAC \cong \Delta BAC$$

$$\text{Hence } \angle i = \angle r$$

$$\therefore \text{Angle of incidence} = \text{angle of reflection}$$



$\frac{1}{2}$

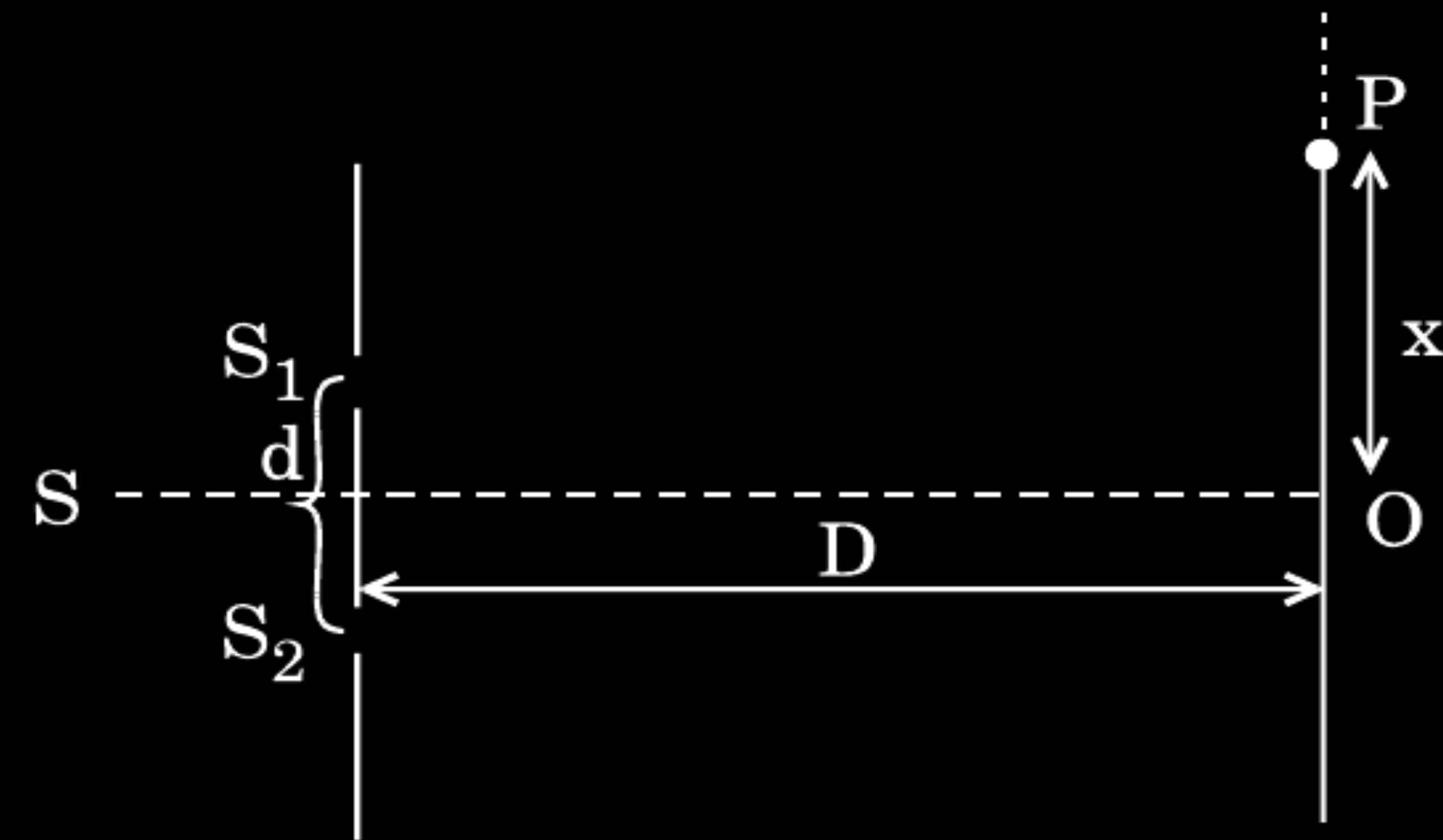
$\frac{1}{2}$

- (a) Describe any two characteristic features which distinguish between interference and diffraction phenomena. Derive the expression for the intensity at a point of the interference pattern in Young's double slit experiment.
- (b) In the diffraction due to a single slit experiment, the aperture of the slit is 3 mm. If monochromatic light of wavelength 620 nm is incident normally on the slit, calculate the separation between the first order minima and the 3rd order maxima on one side of the screen. The distance between the slit and the screen is 1.5 m.

(a) (Any two of the following) (i) Interference pattern has number of equally spaced bright and dark bands while diffraction pattern has central bright maximum which is twice as wide as the other maxima. (ii) Interference is obtained by the superposing two waves originating from two narrow slits. The diffraction pattern is the superposition of the continuous family of waves originating from each point on a single slit. (iii) In interference pattern, the intensity of all bright fringes is same, while in diffraction pattern intensity of bright fringes go on decreasing with the increasing order of the maxima	$\frac{1}{2} + \frac{1}{2}$
(a) Distance of First order minima from centre of the central maxima = $x_{D1} = \frac{\lambda D}{a}$ Distance of third order maxima from centre of the central maxima $X_{B3} = \frac{7D\lambda}{2a}$	$\frac{1}{2}$
$\therefore$ Distance between first order minima and third order maxima = $x_{B3} - x_{d1}$ $= \frac{7D\lambda}{2a} - \frac{\lambda D}{a}$ $= \frac{5D\lambda}{2a}$ $= \frac{5 \times 620 \times 10^{-9} \times 1.5}{2 \times 3 \times 10^{-3}}$ $= 775 \times 10^{-6} \text{ m}$ $= 7.75 \times 10^{-4} \text{ m}$	$\frac{1}{2}$

- (a) Can the interference pattern be produced by two independent monochromatic sources of light ? Explain.
- (b) The intensity at the central maximum (O) in a Young's double slit experimental set-up shown in the figure is I_0 . If the distance OP equals one-third of the fringe width of the pattern, show that the intensity at point P, would equal $\frac{I_0}{4}$.

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- (c) In Young's double slit experiment, the slits are separated by 0.5 mm and screen is placed 1.0 m away from the slit. It is found that the 5th bright fringe is at a distance of 4.13 mm from the 2nd dark fringe. Find the wavelength of light used.

a) No. Sustained interference pattern cannot be obtained Light waves emitted from a source undergoes abrupt phase changes in times of the order of 10^{-10} s. So light from two independent sources will not have fixed phase relationship and will be incoherent.	1 $\frac{1}{2} + \frac{1}{2}$
b) $x = \frac{\beta}{3}, \text{ path difference} = \frac{\lambda}{3}$ $\text{phase diff} = \frac{2\pi}{3}$ $I = I_0 \cos^2 \frac{\phi}{2}$ $I = I_0 \cos^2 \left(\frac{2\pi}{3 \times 2} \right) = I_0 \cos^2 \left(\frac{\pi}{3} \right)$ $I = I_0 \left(\frac{1}{4} \right) = \frac{I_0}{4}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
c) Distance of 5th bright fringe from 2nd dark fringe $x = \frac{5 \lambda D}{d} - \frac{3 \lambda D}{2 d} = \frac{7}{2} \frac{\lambda D}{d}$ $\lambda = \frac{2xd}{7D} = \frac{2 \times 4.13 \times 10^{-3} \times 0.5 \times 10^{-3}}{7 \times 1}$ $\lambda = 0.59 \times 10^{-6} \text{ m} = 5900 \text{ \AA}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- (a) Derive the relation $a \sin \theta = \lambda$ for the first minimum of the diffraction pattern produced due to a single slit of width 'a' using light of wavelength λ .
- (b) State with reason, how the linear width of central maximum will be affected if (i) monochromatic yellow light is replaced with red light, and (ii) distance between the slit and the screen is increased.
- (c) Using the monochromatic light of same wavelength in the experimental set-up of the diffraction pattern as well as in the interference pattern where the slit separation is 1 mm, 10 interference fringes are found to be within the central maximum of the diffraction pattern. Determine the width of the single slit, if the screen is kept at the same distance from the slit in the two cases.

From diagram path difference between the waves from L and N
 $= a \sin \theta$

$\frac{1}{2}$

When first minimum is obtained at P then path difference $= \lambda$

$\frac{1}{2}$

[imagine the slit be divided into two halves, for each wavelets from first half of the slit has a corresponding wavelet from second half of the slit differing by a path of $\frac{\lambda}{2}$ and cancel each other]

Condition for first minimum

$$\therefore \lambda = a \sin \theta$$

$$\text{b.) } \beta_{\text{cm}} = \frac{2 \lambda D}{d}$$

(i) increases

$\frac{1}{2}$

(ii) increases

$\frac{1}{2}$

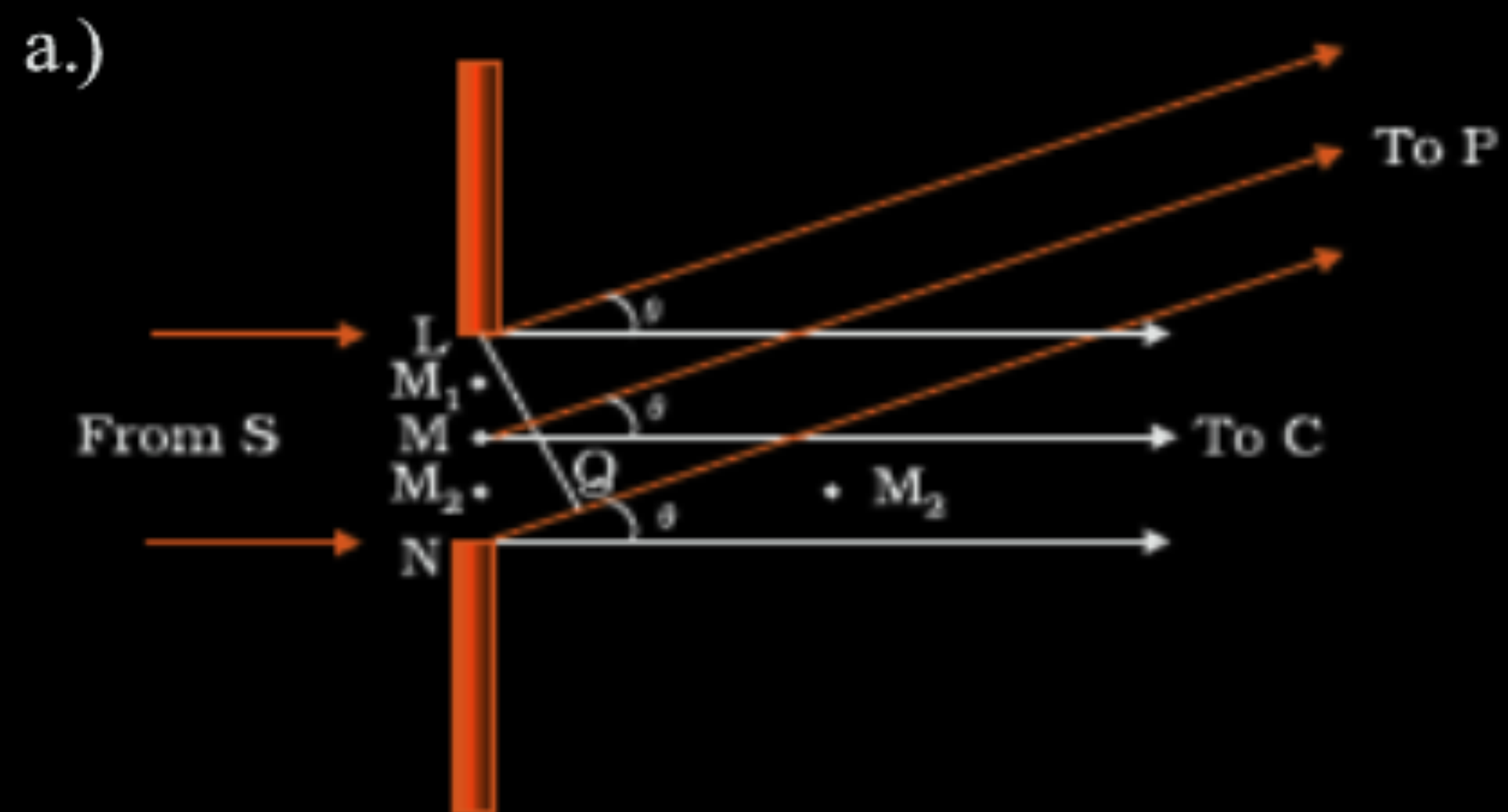
$\frac{1}{2}$

$$\text{c.) } 10 \frac{\lambda}{d} = 2 \frac{\lambda}{a}$$

1

$$a = \frac{d}{5} = 0.2 \text{mm}$$

1



- (a) Why cannot the phenomenon of interference be observed by illuminating two pin holes with two sodium lamps ?
- (b) Two monochromatic waves having displacements $y_1 = a \cos \omega t$ and $y_2 = a \cos (\omega t + \phi)$ from two coherent sources interfere to produce an interference pattern. Derive the expression for the resultant intensity and obtain the conditions for constructive and destructive interference.
- (c) Two wavelengths of sodium light of 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture 2×10^{-6} m. If the distance between the slit and the screen is 1.5 m, calculate the separation between the positions of the second maxima of diffraction pattern obtained in the two cases.

(a) Because two independent sources cannot be coherent OR they are not coherent

(b) $y_1 = a \cos \omega t$

$$y_2 = a \cos(\omega t + \phi)$$

So resultant displacement is give by

$$y = y_1 + y_2$$

$$y = a \cos \omega t + a \cos(\omega t + \phi)$$

$$y = 2a \cos(\phi / 2) \cos(\omega t + \phi / 2)$$

The amplitude of the resultant displacement is $2a \cos(\phi / 2)$ and therefore intensity at that point will be $I = 4I_0 \cos^2(\phi / 2)$

For constructive interference: $\phi = 0, \pm 2\pi, \pm 4\pi, \dots$

For destructive interference: $\phi = \pm \pi, \pm 3\pi, \pm 5\pi, \dots$

(c) Position of second maxima $y_2 = \frac{5 \lambda D}{2 a}$

Separation between the positions of the second maxima with λ_1 and λ_2 is:

$$\Delta y = \frac{5D(\lambda_2 - \lambda_1)}{2a} = \frac{5 \times 1.5 \times (596 - 590) \times 10^{-9}}{2 \times 2 \times 10^{-6}} = 11.25 \times 10^{-3} m$$

Draw a graph showing the intensity distribution of fringes due to diffraction at single slit.

- (i) Write two points to distinguish between interference and diffraction fringes.
- (ii) In a Young's double slit experiment, fringes are obtained on a screen placed a certain distance away from the slits. If the screen is moved by 5 cm towards the slits, the fringe width changes by $30\text{ }\mu\text{m}$. Given that the slits are 1 mm apart, calculate the wavelength of the light used.

- (a) Explain two features to distinguish between the interference pattern in Young's double slit experiment with the diffraction pattern obtained due to a single slit.
- (b) A monochromatic light of wavelength 500 nm is incident normally on a single slit of width 0.2 mm to produce a diffraction pattern. Find the angular width of the central maximum obtained on the screen.

Estimate the number of fringes obtained in Young's double slit experiment with fringe width 0.5 mm, which can be accommodated within the region of total angular spread of the central maximum due to single slit.

a)

Interference Pattern	Diffraction pattern
1) All fringes are of equal width.	1) Width of central maxima is twice the width of higher order bands.
2) Intensity of all bright bands is equal.	2) Intensity goes on decreasing for higher order of diffraction bands.

[Note: Also accept any other two correct distinguishing features.]

b) Angular width of central maximum

$$= \frac{\lambda}{a} \text{ radian}$$

$$= \frac{\lambda}{a} \text{ radian}$$

Linear width of central maxima in the diffraction pattern

Let 'n' be the number of interference fringes which can be accommodated in the central maxima

[Award the last $\frac{1}{2}$ mark if the student writes the answers as 2 (taking $d=a$), or just attempts to do these calculation.]

1+1

$\frac{1}{2}$
 $\frac{1}{2}$

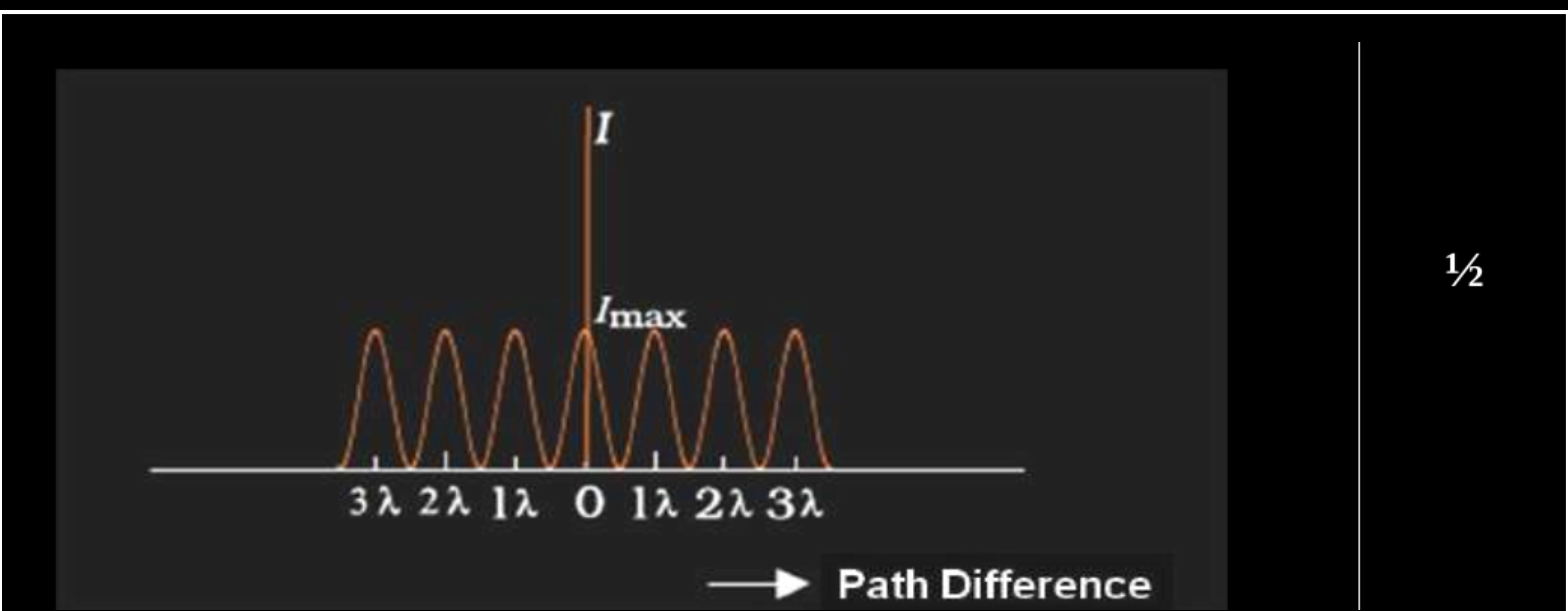
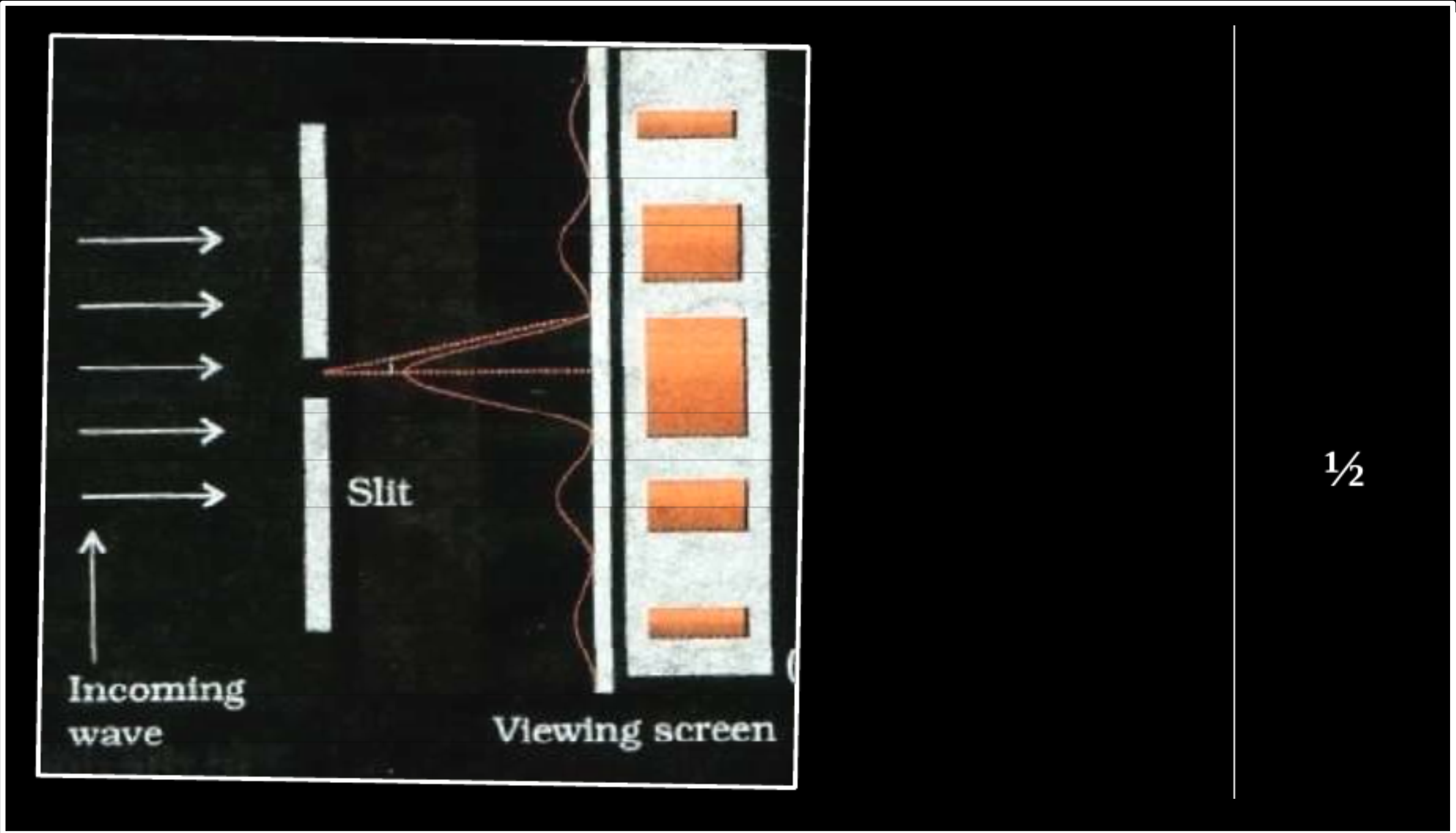
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$\frac{1}{2}$

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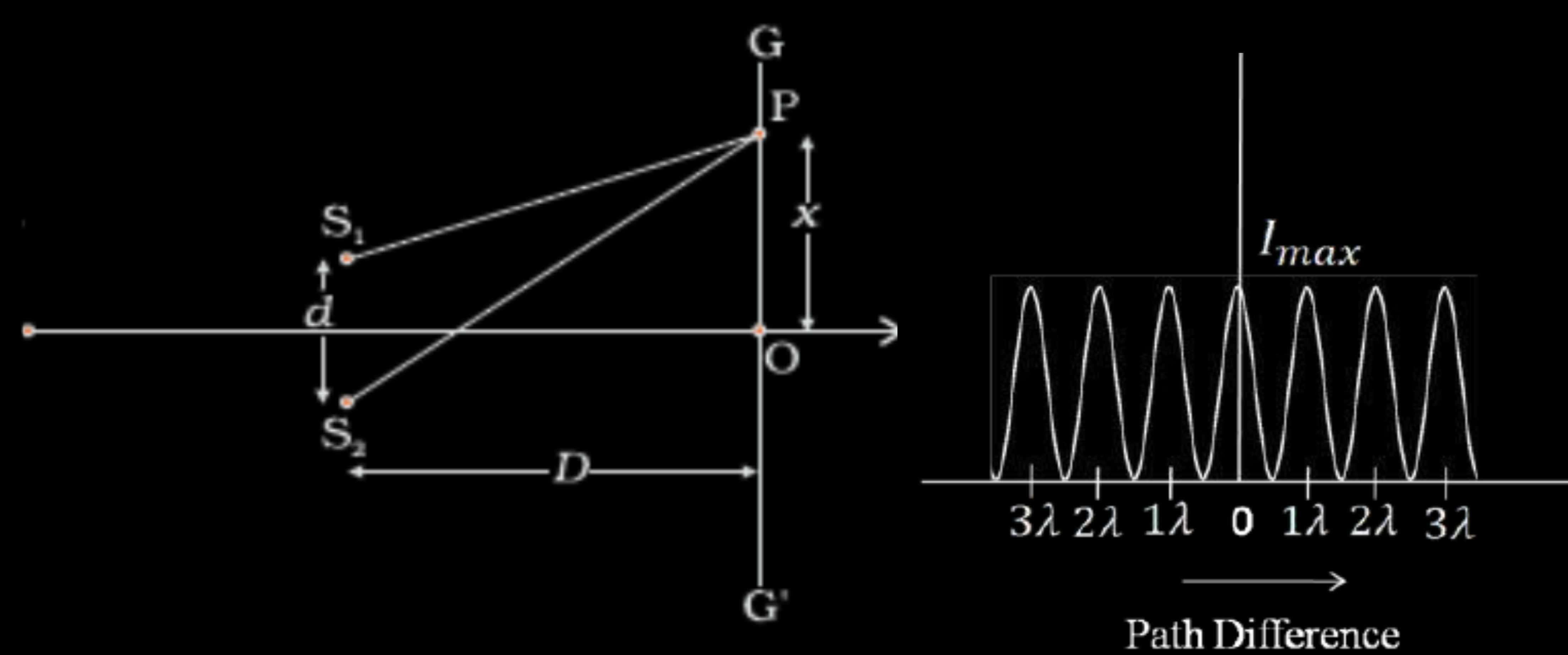
Draw the intensity pattern for single slit diffraction and double slit interference. Hence, state two differences between interference and diffraction patterns.

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Differences		$\frac{1}{2} + \frac{1}{2}$
Interference	Diffraction	
All maxima have equal intensity	Maxima have different (/rapidly decreasing) intensity	
All fringes have equal width.	Different (/changing) width.	
Superposition of two wavefronts	Superposition of wavelets from the same wavefront	
(Any two)		

- (i) In Young's double slit experiment, deduce the condition for (a) constructive, and (b) destructive interference at a point on the screen. Draw a graph showing variation of intensity in the interference pattern against position 'x' on the screen.
- (ii) Compare the interference pattern observed in Young's double slit experiment with single slit diffraction pattern, pointing out three distinguishing features.



From figure

$$\text{Path difference} = (S_2P - S_1P)$$

$$(S_2P)^2 - (S_1P)^2 = [D^2 + (x + \frac{d}{2})^2] - [D^2 + (x - \frac{d}{2})^2]$$

$$(S_2P + S_1P)(S_2P - S_1P) = 2xd$$

$$S_2P - S_1P = \frac{2xd}{S_2P + S_1P}$$

For $x, d \ll D$

$$S_2P + S_1P = 2D$$

$$\therefore S_2P - S_1P = \frac{2xd}{2D} = \frac{xd}{D}$$

For constructive interference $S_2P - S_1P = n\lambda, n=0,1,2,\dots$

$$\Rightarrow \frac{xd}{D} = n\lambda$$

$$\Rightarrow x = \frac{n\lambda D}{d}$$

For destructive interference $S_2P - S_1P = (2n + 1)\frac{\lambda}{2}$

$$n = 0, 1, 2, \dots$$

$$\frac{xd}{D} = (2n + 1)\frac{\lambda}{2}$$

$$\Rightarrow x = (2n + 1)\frac{\lambda D}{2d}$$

ii)

- (a) The Interference pattern has number of equally spaced bright and dark bands, while in the diffraction pattern the width of the central maximum is twice the width of other maxima.
- (b) In Interference all bright fringes are of equal intensity, whereas in the diffraction pattern the intensity falls as order of maxima increases.
- (c) In Interference pattern, maxima occurs at an angle $\frac{\lambda}{a}$, where a is the slit width, whereas in diffraction pattern, at the same angle, first minimum occurs. (Here ' a ' is the size of the slit)

(Any other distinguishing feature)

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

Two harmonic waves of monochromatic light

$$y_1 = a \cos \omega t \text{ and } y_2 = a \cos(\omega t + \phi)$$

are superimposed on each other. Show that maximum intensity in interference pattern is four times the intensity due to each slit. Hence write the conditions for constructive and destructive interference in terms of the phase angle ϕ .

Two harmonic waves of monochromatic light

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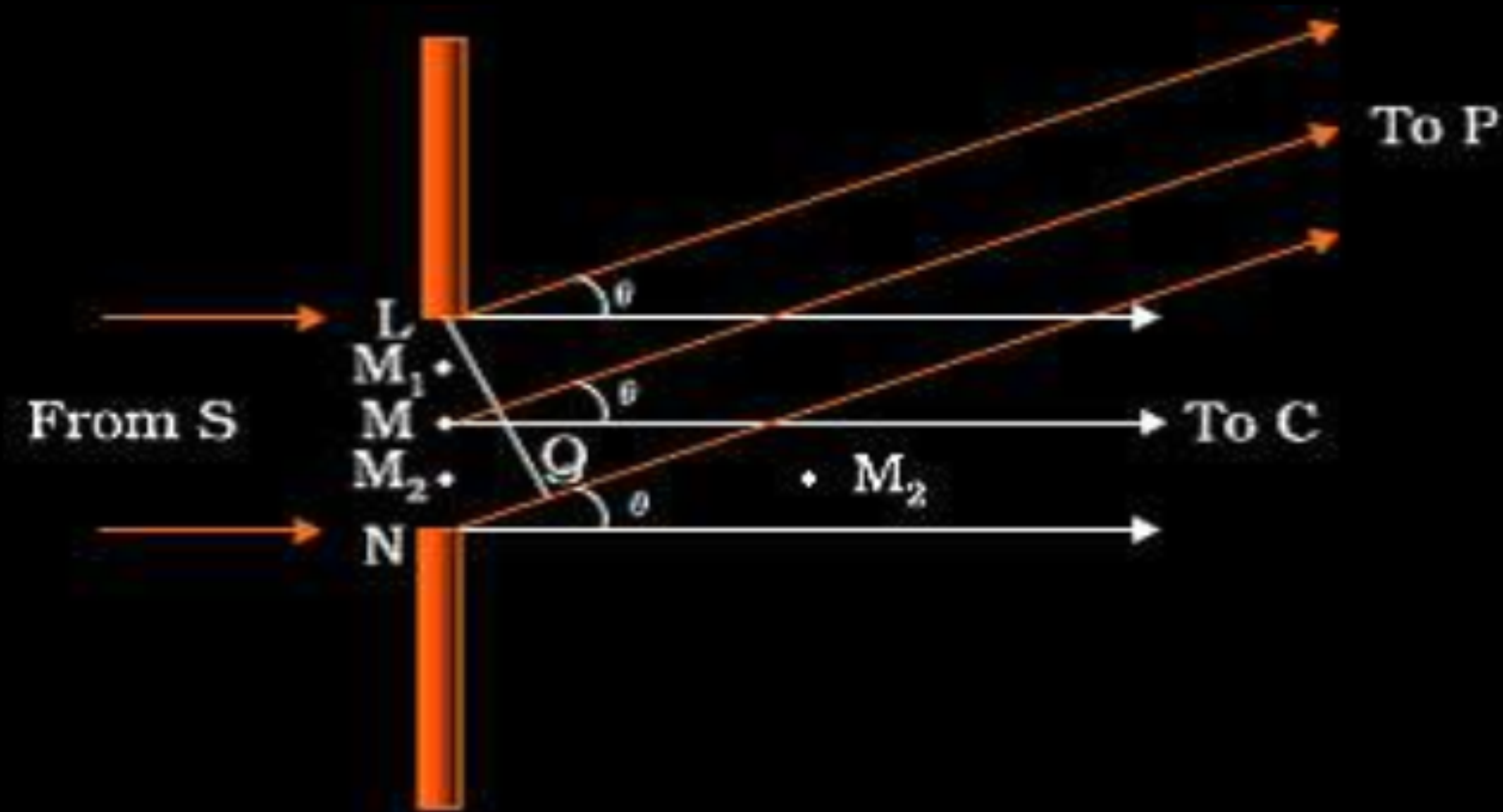
are superimposed on each other. Show that maximum intensity in interference pattern is four times the intensity due to each slit. Hence write the conditions for constructive and destructive interference in terms of the phase angle ϕ .

Resultant displacement	
$y = y_1 + y_2$ $= a[\cos(\omega t) + \cos(\omega t + \phi)]$ $= 2a \cos\left(\frac{\phi}{2}\right) \cos\left(\omega t + \frac{\phi}{2}\right)$	$\frac{1}{2}$
$\therefore$ amplitude of resultant wave = $2a \cos\left(\frac{\phi}{2}\right)$	$\frac{1}{2}$
$\therefore$ Intensity = $4I_o \cos^2\left(\frac{\phi}{2}\right)$, where $I_o = a^2$ is the intensity of each harmonic wave	$\frac{1}{2}$
At the maxima, $\phi = \pm 2n\pi \therefore \cos^2 \frac{\phi}{2} = 1$	$\frac{1}{2}$
At the maxima, $I = 4I_o = 4 \times$ intensity due to one slit	
$I = 4I_o \cos^2\left(\frac{\phi}{2}\right)$	$\frac{1}{2}$
For constructive interference, I is maximum	
It is possible when $\cos^2\left(\frac{\phi}{2}\right) = 1; \frac{\phi}{2} = n\pi; \phi = 2n\pi$	
For destructive interference, I is minimum, i.e, $I=0$	
It is possible when $\cos^2\left(\frac{\phi}{2}\right) = 0; \frac{\phi}{2} = \frac{(2n-1)\pi}{2}; \phi = (2n \pm 1) \frac{\pi}{2}$	$\frac{1}{2}$

When a parallel beam of monochromatic source of light of wavelength λ is incident on a single slit of width a , show how the diffraction pattern is formed at the screen by the interference of the wavelets from the slit.

Show that, besides the central maximum at $\theta=0$, secondary maxima are observed at $\theta = n + \frac{1}{2} \lambda/a$ and the minima at $\theta = n\lambda/a$.

Why do secondary maxima get weaker in intensity with increasing n ? Explain.



The diffraction pattern formed can be understood by adding the contributions from the different wavelets of the incident wavefront, with their proper phase differences.

For the central point, we imagine the slit to be divided into two equal halves. The contribution of corresponding wavelets, in the two halves, are in phase with each other. Hence we get a maxima at the central point. The entire incident wavefront contributes to this maxima.

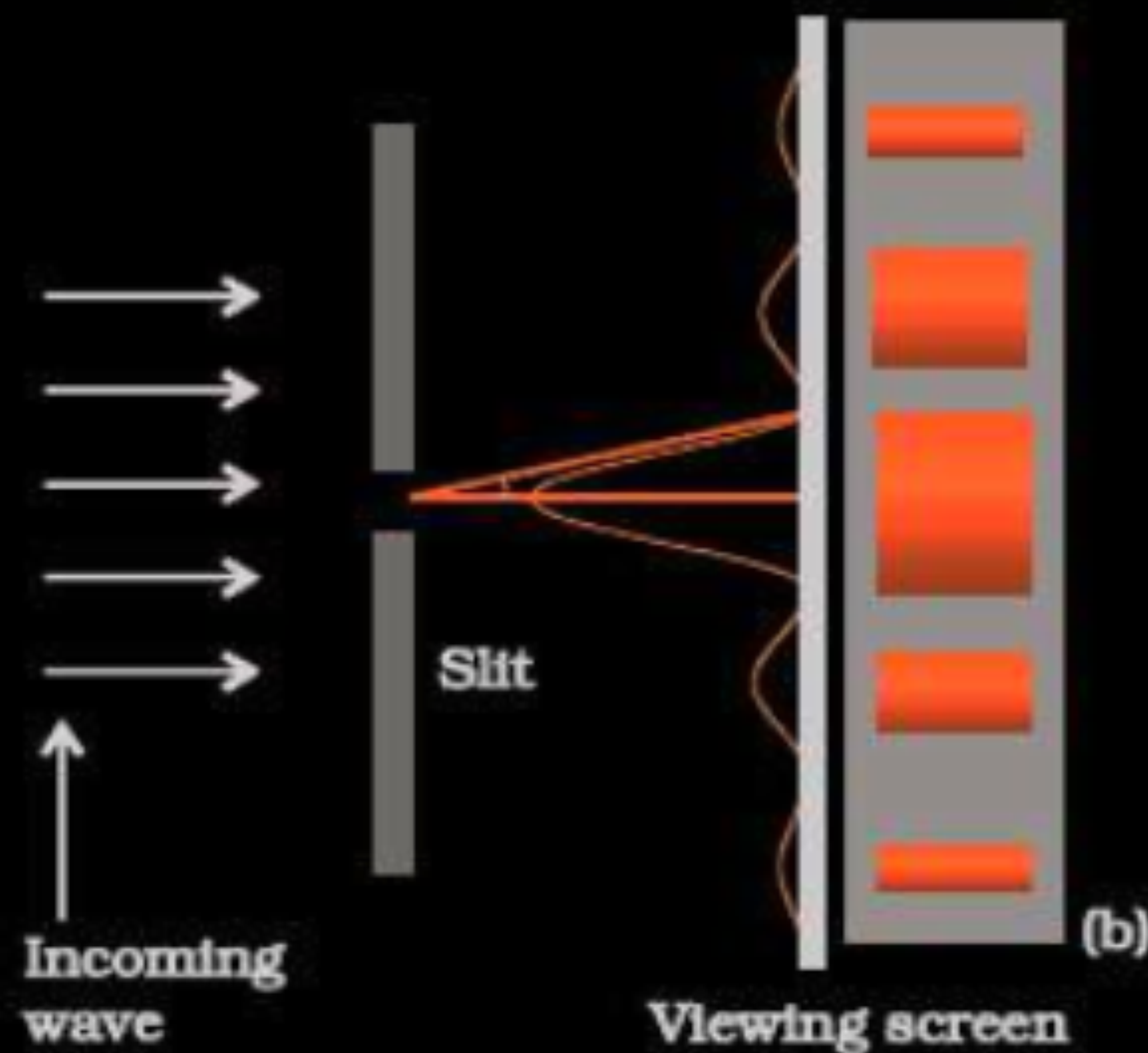
All other points, for which $\theta = (n + \frac{1}{2}) \frac{\lambda}{a}$, get a net non zero contribution from all the wavelets. Hence all such points are also points of maxima.

Points for which $\theta = \frac{n\lambda}{a}$, the net contribution, from all the wavelets, is zero. Hence these points are point of minima.

We thus get a diffraction pattern on the screen, made up of points of maxima and minima.

Secondary maxima keep on getting weaker in intensity, with increasing n . This is because, at the

- (i) First secondary maxima, the net contribution is only from (effectively) $1/3$ rd of the incident wavefront on the slit.
- (ii) Second secondary maxima, the net contribution is only from (effectively) $1/5^{\text{th}}$ of the incident wavefront on the slit. And so on.

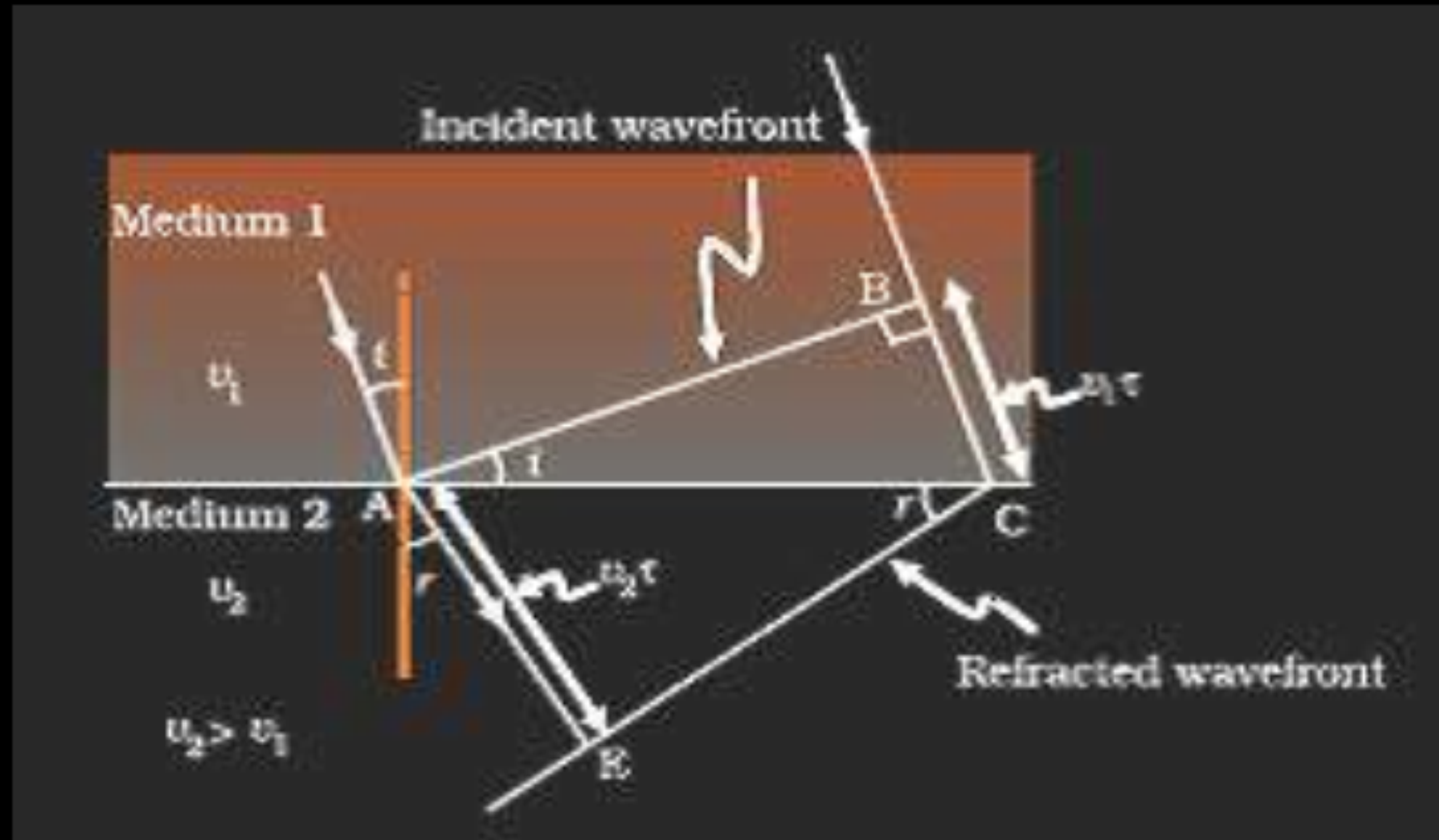


(a)

(b)

- (i) Derive Snell's law on the basis of Huygen's wave theory when light is travelling from a denser to a rarer medium.
- (ii) Draw the sketches to differentiate between plane wavefront and spherical wavefront.

(i)



We have $BC = v_1 \tau$ and $AE = v_2 \tau$

Also $\sin i = \frac{BC}{AC}$ and $\sin r = \frac{AE}{AC}$

$$\therefore \frac{\sin i}{\sin r} = \frac{BC}{AE} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

= a constant

This is Snell's law.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

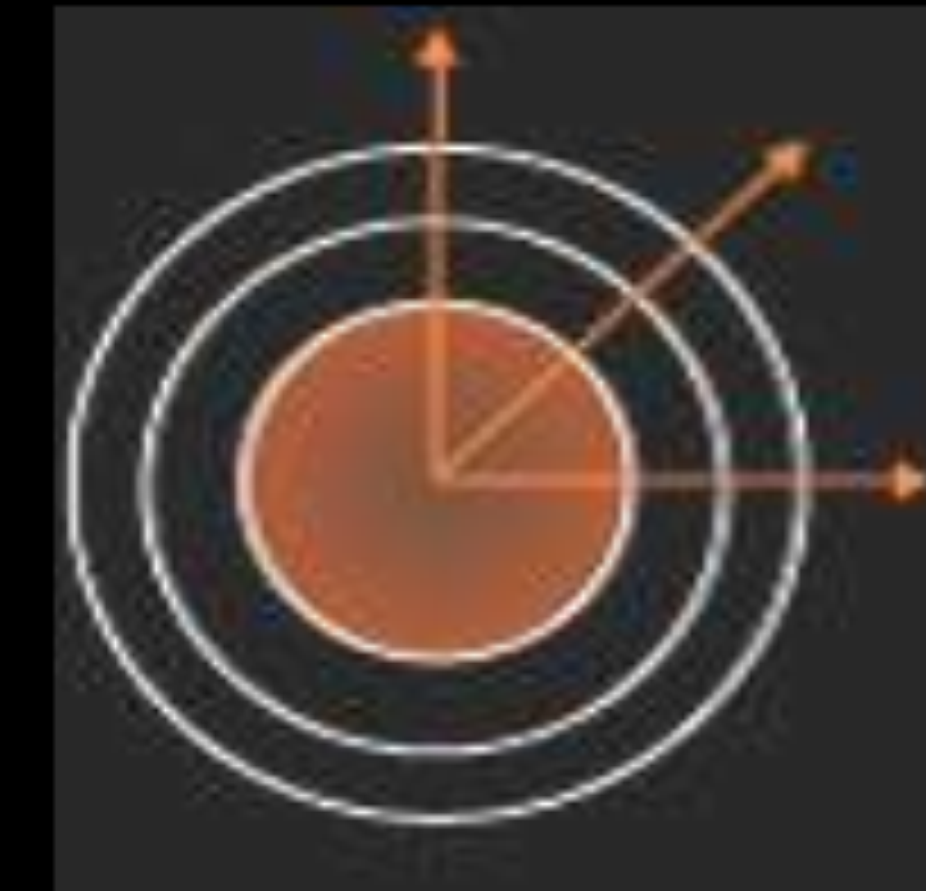
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(ii) Plane wavefront



$\frac{1}{2}$

Spherical wavefront



$\frac{1}{2}$

- (a) Derive an expression for path difference in Young's double slit experiment and obtain the conditions for constructive and destructive interference at a point on the screen.
- (b) The intensity at the central maxima in Young's double slit experiment is I_0 . Find out the intensity at a point where the path difference is $\frac{\lambda}{6}$, $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$.

- (a) Derive an expression for path difference in Young's double slit experiment and obtain the conditions for constructive and destructive interference at a point on the screen.
- (b) The intensity at the central maxima in Young's double slit experiment is I_0 . Find out the intensity at a point where the path difference is $\frac{\lambda}{6}$, $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$.

(b) The general expression, for the intensity, at a point is $I = I_0 \cos^2 \phi / 2$		
(i)	For path difference $= \lambda / 6$, $I = 3I_0 / 4$	$\frac{1}{2}$
(ii)	For path difference $= \lambda / 4$, $I = I_0 / 2$	$\frac{1}{2}$
(iii)	For path difference $= \lambda / 3$, $I = I_0 / 4$	$\frac{1}{2}$

- (a) In Young's double slit experiment, two slits are 1 mm apart and the screen is placed 1 m away from the slits. Calculate the fringe width when light of wavelength 500 nm is used.
- (b) What should be the width of each slit in order to obtain 10 maxima of the double slits pattern within the central maximum of the single slit pattern ?

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- (b) What should be the width of each slit in order to obtain 10 maxima of the double slits pattern within the central maximum of the single slit pattern ?

$$\begin{aligned}
 \text{(a) } \beta &= \frac{\lambda D}{d} \\
 &= \frac{500 \times 10^{-9} \times 1}{10^{-3}} \\
 &= 0.5 \text{ mm or } 5 \times 10^{-4} \text{ m}
 \end{aligned}$$

$$\text{(b) } \beta_0 = \frac{2\lambda D}{a} = 10 \beta$$

$$a = \frac{2 \times 500 \times 10^{-9} \times 1}{10 \times 5 \times 10^{-4}}$$

$$a = 2 \times 10^{-4} \text{ m or } 0.2 \text{ mm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Define the term wave front. State Huygen's principle.

Consider a plane wave front incident on a thin convex lens. Draw a proper diagram to show how the incident wave front traverses through the lens and after refraction focusses on the focal point of the lens, giving the shape of the emergent wave front.

Definition- Locus of all points which oscillate in phase.

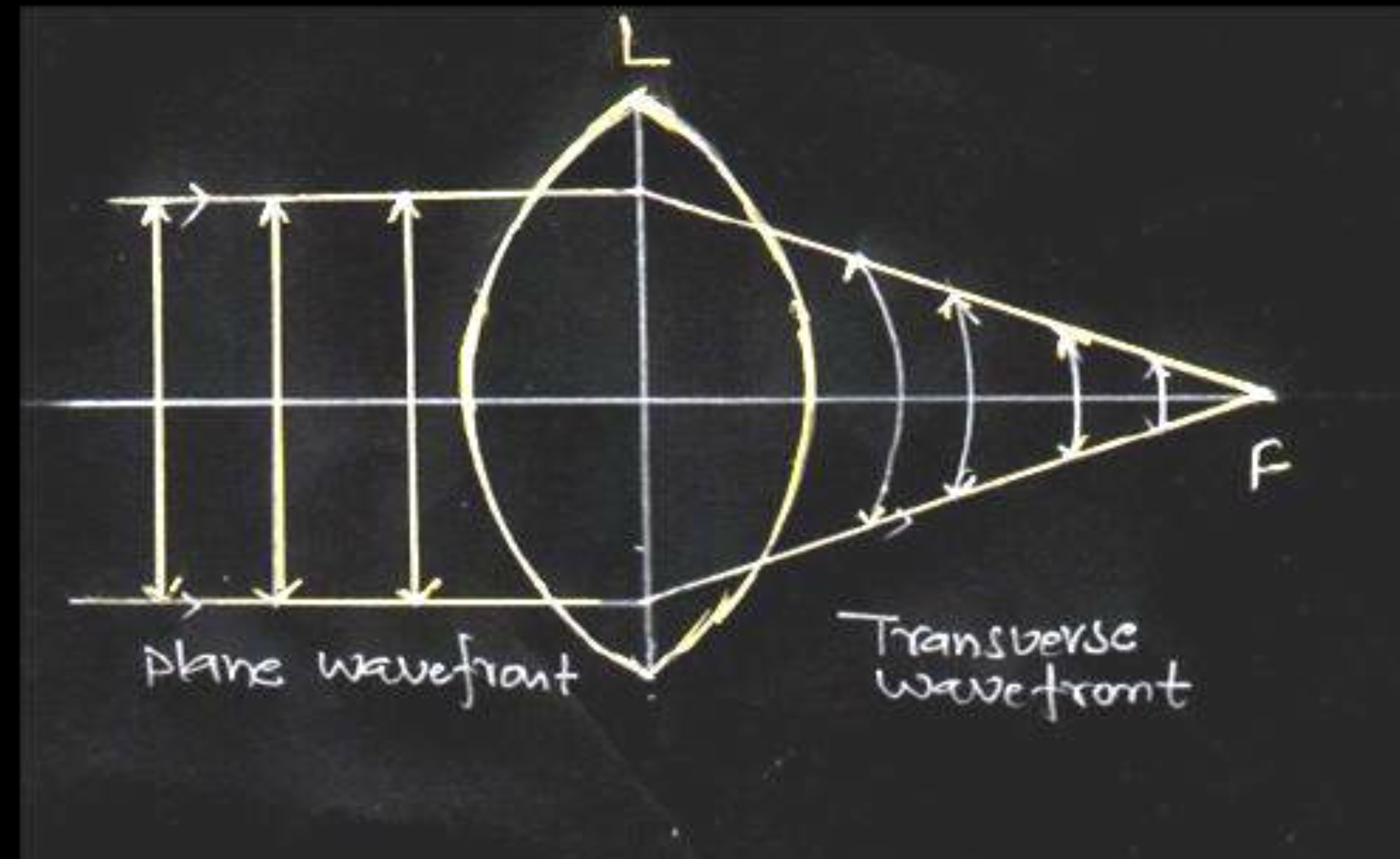
1

i. Huygen's Principle- Each point of the wave front is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions. These travel with the same velocity as that of the original wave front.

$\frac{1}{2}$

ii. The shape and position of the wave front, after time 't', is given by the tangential envelope to the secondary wavelets.

$\frac{1}{2}$



$\frac{1}{2} + \frac{1}{2}$

Explain the following, giving reasons :

- (i) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency.
- (ii) When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a reduction in the energy carried by the wave ?
- (iii) In the wave picture of light, intensity of light is determined by the square of the amplitude of the wave. What determines the intensity in the photon picture of light ?

i. Reflection and refraction arise through interaction of incident light with atomic constituents of matter which vibrate with the same frequency as that of the incident light. Hence frequency remains unchanged.

ii. No. [Energy carried by a wave depends on the amplitude of the wave, not on the speed of wave propagation].

iii. For a given frequency, intensity of light in the photon picture is determined by the number of photon incident normally on a crossing an unit area per unit time.